

# SA-AI (Spalart–Allmaras with Autogenous Inception): Technical Summary

Qiqi Wang (Flexcompute, Inc. / MIT)

July 29, 2026

## 1 Model

### 1.1 Transport equation and blended sources

$$\frac{D\tilde{\nu}}{Dt} = P - D + \frac{1}{\sigma} \nabla \cdot [(c_{\nu, \text{ai}} \nu + \tilde{\nu}) \nabla \tilde{\nu}] + \frac{c_{b2}}{\sigma} |\nabla \tilde{\nu}|^2, \quad (1)$$

$$P = \max[(1 - \sigma_P) a S \omega \tilde{\nu}, \sigma_P P_{\text{SA}}], \quad D = \sigma_D D_{\text{SA}}, \quad (2)$$

$\sigma_P \equiv 1$  (and with it  $\sigma_D = 1$ ),  $c_{\nu, \text{ai}} = 1$ , and  $b \equiv 0$  recover baseline SA exactly.

$$\begin{aligned} P_{\text{SA}} &= c_{b1} \tilde{S} \tilde{\nu}, & D_{\text{SA}} &= c_{w1} f_w \left( \frac{\tilde{\nu}}{d} \right)^2, & \tilde{S} &= \omega + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v2}, & f_{v2} &= 1 - \frac{\chi}{1 + \chi f_{v1}}, \\ f_{v1} &= \frac{\chi^3}{\chi^3 + c_{v1}^3}, & f_w &= g \left( \frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6}, & g &= r + c_{w2}(r^6 - r), & r &= \min\left(\frac{\tilde{\nu}}{\tilde{S} \kappa^2 d^2}, 10\right), \end{aligned} \quad (3)$$

The non-negative  $\tilde{S}$  modification and the negative- $\tilde{\nu}$  safeguards of Allmaras et al. (2012) are retained, with one consistency requirement: the negative-branch diffusion factor  $f_n$  is formed with the scaled coefficient  $c_{\nu, \text{ai}} \nu$ , which keeps the total diffusivity positive,  $c_{\nu, \text{ai}} \nu + f_n \tilde{\nu} > 0$ .

$$\sigma_P = \sigma_t(\chi; \tau), \quad \sigma_t(\chi; \tau) = \max\left[1 - e^{-(\chi-1)/\tau}, 0\right], \quad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}} (1 - \sigma_P). \quad (4)$$

### 1.2 Local indicators

In a thin quasi-two-dimensional layer the profile is captured to second order about any wall-normal point by three velocity scales formed from the wall distance  $d$ : the velocity  $u$ , the shear  $du'$ , and the curvature  $\frac{1}{2}d^2u''$ . Normalized by their common magnitude  $R = \sqrt{u^2 + (du')^2 + (\frac{1}{2}d^2u'')^2}$ , they place the local state on the unit sphere of directions

$$(X, Y, Z) = (u, du', \frac{1}{2}d^2u'')/R, \quad (5)$$

$$\hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}} \quad (\text{vorticity fraction}), \quad \hat{I} = Y - X - Z \quad (\text{accumulated inflection}). \quad (6)$$

A near-wall parabola  $u = By + Cy^2$  satisfies  $\hat{I} = 0$  identically.

$$I(d) \equiv R \hat{I} = d u' - u - \frac{1}{2}d^2 u'' = - \int_0^d \frac{1}{2} s^2 u'''(s) ds, \quad (7)$$

Every profile leaves the wall with  $\hat{I}$  third-order small ( $u_w''' = 0$  on a steady wall). The vorticity Reynolds number  $Re_\Omega = d^2 \omega / \nu$  is reserved for the onset gate. Fig. 1.

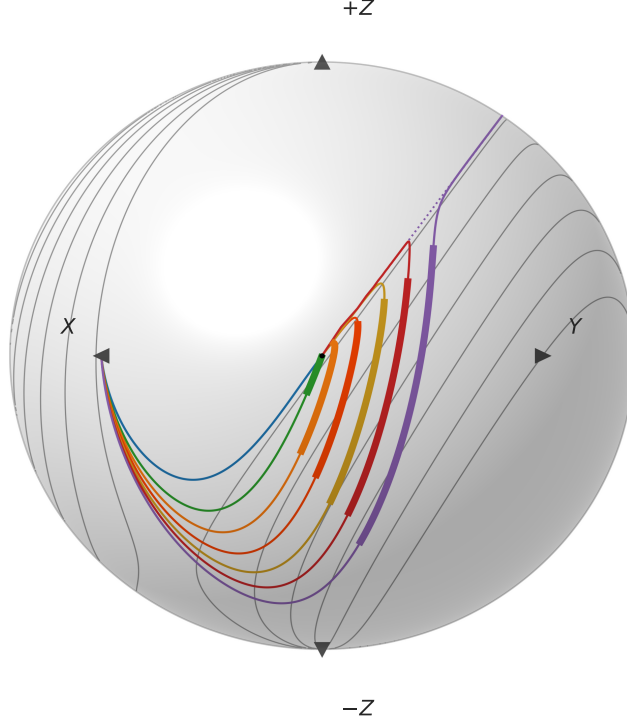


Figure 1: Indicator sphere. Falkner–Skan profiles  $\beta = +0.30, 0, -0.10, -0.16, -0.19, -0.1988$ ; separated Stewartson  $\beta = -0.19, H \approx 4.9$ . Attached profiles at  $Re_\theta = 500$ , reversed-flow at  $Re_\theta = 200$ ;  $\hat{\Omega}\hat{I}$  contours  $\{0.025, 0.2, 0.4, 0.6, 0.8, 1.0\}$ .

In the solver the triple is assembled from four local vector-calculus objects: the velocity  $\mathbf{u}$ , the vorticity  $\boldsymbol{\omega}$ , the wall-distance vector  $\mathbf{d} = d \nabla d$ , and the curvature vector  $\boldsymbol{\ell} = \frac{1}{2} \langle \mathbf{d}, \mathbf{d} \rangle \nabla^2 \mathbf{u}$ , with  $\nabla^2 \mathbf{u} = -\nabla \times \boldsymbol{\omega}$  supplying the second derivative.

$$(X, Y, Z) = \left( \langle \mathbf{u}, \mathbf{u} \rangle, \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{d}, \mathbf{d} \rangle \langle \boldsymbol{\omega}, \boldsymbol{\omega} \rangle}, \langle \boldsymbol{\ell}, \mathbf{u} \rangle \right), \quad \hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}}, \quad \hat{I} = \frac{Y - X - Z}{\sqrt{X^2 + Y^2 + Z^2}}, \quad (8)$$

The solver’s equivalent ratio realization ( $|\mathbf{u}|, d|\boldsymbol{\omega}|, \frac{1}{2}d^2(\nabla^2 \mathbf{u}) \cdot \hat{\mathbf{u}}$ ) coincides with it wherever  $\mathbf{u} \neq 0$ .

### 1.3 Amplification rate and onset threshold

The complete amplification source is

$$P_{\text{AI}} = a S \omega \tilde{\nu}, \quad a = a_{\text{max}} \text{clip} \langle \hat{\Omega} \hat{I} \rangle_0^1, \quad S = S(Re_\Omega / Re_\Omega^c(\hat{\Omega} \hat{I})), \quad S(\zeta) = \frac{1}{2} [1 + \tanh((\zeta - 1)/w)], \quad (9)$$

$$Re_\Omega^c(\hat{\Omega} \hat{I}) = k \text{softmin}_2 \left( C, A + B \langle \hat{\Omega} \hat{I} \rangle_+^{-2} \right), \quad \text{softmin}_2(x_1, x_2) = \frac{x_1 x_2}{\sqrt{x_1^2 + x_2^2}}, \quad \langle x \rangle_+ = \max(x, 0). \quad (10)$$

Fig. 2: graze ratios 0.93–1.14, root-mean-square  $\sim 6\%$ , from incipient separation through  $\beta = +0.15$ . The marched Blasius  $N=9$  crossing lands at  $Re_\theta = 1181$ , 7% past Drela’s 1108; across the attached family the  $N=1$  stations land on the Drela–Giles stations to 6% root-mean-square (adverse wedges within  $\pm 4\%$ , the favorable pair 11–12% early). By  $\beta = +0.35$  the rate is  $\approx 0.1 \times$  Drela and the onset  $\approx 3 \times$  its critical  $Re_\theta$ ; by  $\beta \approx 0.45$  transition has retreated beyond any station of practical relevance, and stronger favorable wedges are suppressed outright (Figs. 3–4).

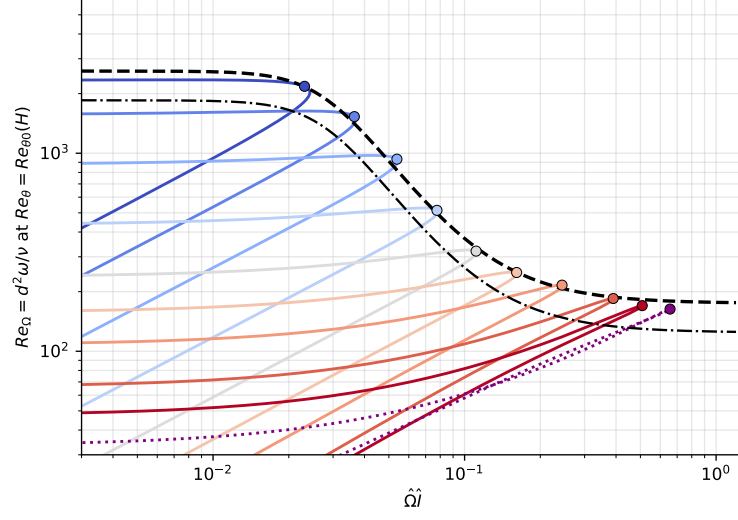


Figure 2: Onset-threshold graze. Attached family  $\beta = +0.15, +0.10, +0.05, 0, -0.05, -0.10, -0.15, -0.19, -0.1988$ , each at its Drela–Giles  $Re_{\theta 0}(H)$ ; separated Stewartson  $\beta = -0.19$  ( $H \approx 4.9$ ,  $Re_{\theta 0} \approx 26$ ).  $\max_y \hat{\Omega} \hat{I} = 0.162/0.078/0.037$  at  $\beta = -0.10/0/+0.10$ . Envelope  $k = 1$ ; model threshold  $k = 0.712$ .

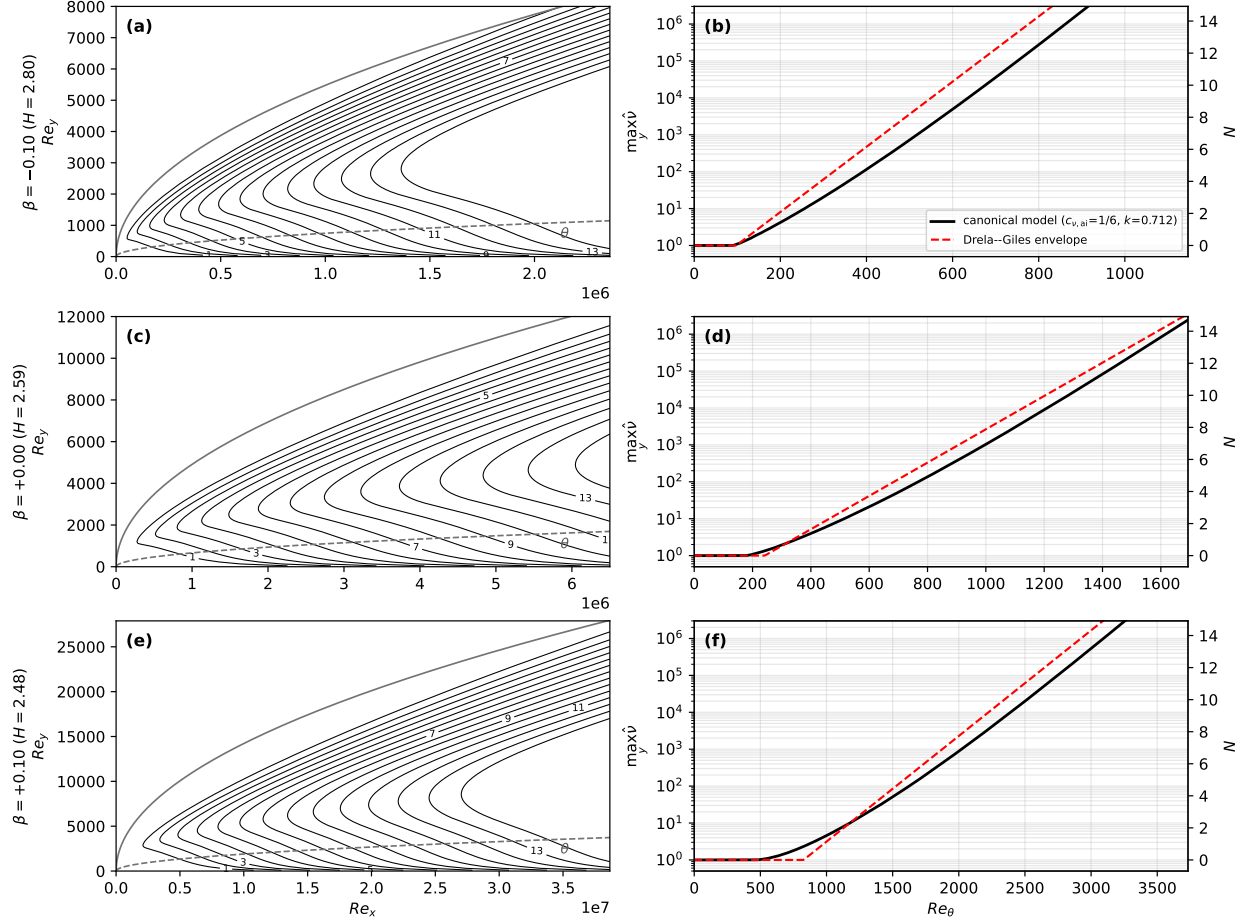


Figure 3: Disturbance transport, three wedges. Adverse  $\beta = -0.10$  ( $\max_y \hat{\Omega} \hat{I} = 0.162$ ), Blasius  $\beta = 0$  (0.078), favorable  $\beta = +0.10$  (0.037); seeded  $\hat{\nu}_\infty = 1$ ,  $a_{\max} = 0.19$ ,  $(c_{\nu, ai}, k) = (\frac{1}{6}, 0.712)$ . Anchor  $N = 9$  crossing 7% past Drela; favorable  $N = 1$  station  $\sim 11\%$  early.

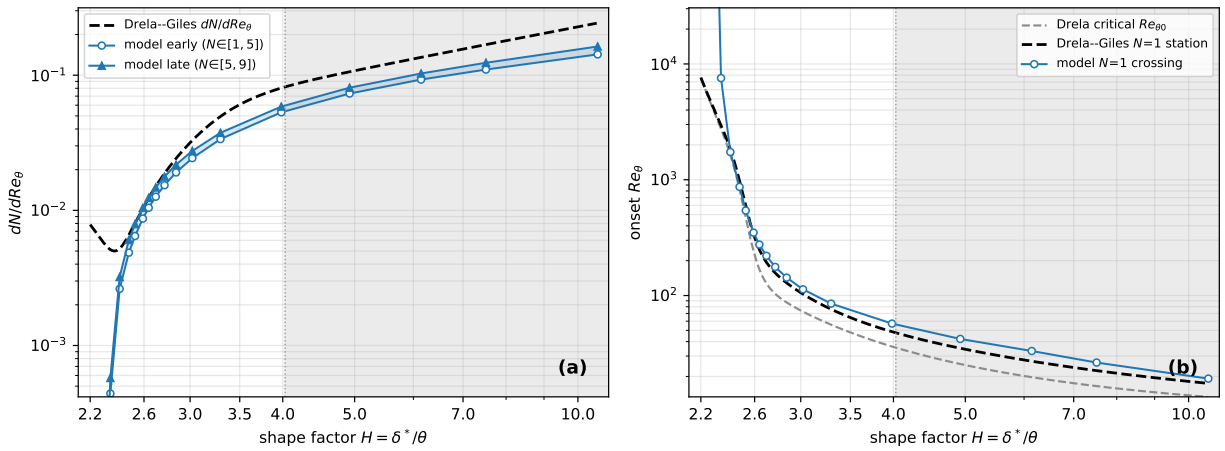


Figure 4: Calibrated model vs Drela-Giles. Stagnation  $H \approx 2.2$ , Blasius  $H \approx 2.59$ , separation limit  $H = 3.98$ , reversed-flow Stewartson  $H \approx 10.6$ ;  $a_{\max} = 0.19$ ,  $c_{\nu, ai} = 1/6$ . (b) onset tracks the Drela-Giles  $N = 1$  stations within  $\sim \pm 12\%$  from  $H = 2.44$  ( $\beta = +0.15$ ) to  $H = 3.98$ .



## 1.4 Reduced laminar diffusion

The molecular coefficient in the diffusion term of Eq. (1) only is scaled by  $c_{\nu,\text{ai}} < 1$ , leaving the mean-flow viscosity, the SA production and destruction, and  $\chi = \tilde{\nu}/\nu$  (formed with the true  $\nu$ ) untouched. The frozen-profile growth eigenvalue on Blasius at  $Re_\theta = 400$  delivers only 0.49 of the Drela–Giles rate, and the marched inception fails outright (the Blasius  $N = 1$  crossing sits at  $Re_\theta = 392$  even with the threshold removed entirely). Down the ladder  $c_{\nu,\text{ai}} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}$  that station’s eigenvalue ratio climbs 0.49, 0.75, 0.87, 1.02, 1.13: the value

$$c_{\nu,\text{ai}} = \frac{1}{6}$$

puts the just-past-critical stations on the correlation (the separation branch holds 0.82–0.91). The equilibrium disturbance-band thickness scales as  $\sqrt{c_{\nu,\text{ai}}}$ , so every halving thins the band the grid must resolve by  $\sqrt{2}$ .

## 1.5 Handover blend and destruction tie

The destruction is not gated off but tied:  $\sigma_D$  tracks  $\sigma_P$  with slope  $c_{b1}/(\kappa^2 c_{w1}) \approx 0.249$  from the floor  $(1 + c_{b2})/(\sigma c_{w1}) \approx 0.751$ , the two summing to one by the definition of  $c_{w1}$ . The tie is derived from the constant-stress wall layer, on which SA’s solution is the linear profile  $\tilde{\nu} = \kappa u_\tau y$  with height-independent terms  $P_{\text{SA}} = c_{b1} u_\tau^2$ ,  $D_{\text{SA}} = c_{w1} \kappa^2 u_\tau^2$ , and self-diffusion  $(1 + c_{b2}) \kappa^2 u_\tau^2 / \sigma$ ; requiring the gated balance to hold pointwise gives Eq. (4), under which the linear profile is an exact solution of the gated equation at every height and for any  $\tau$ —continuously and discretely (verified to below  $10^{-10}$  in profile and log-law intercept with two unrelated discretizations). In the laminar range the destruction floor is inert at the disturbance’s quadratic scale: its ratio to the amplification production stays below  $10^{-3}$  up to the  $\chi = 1$  crossing for every wedge and seed tested, and at  $\chi = c_{v1}$  is at most 1.4% of the  $\sigma_P$ -weighted SA production it is tied against; marching the calibration instrument with the floor included moves the anchor and prediction stations by less than 1%.

## 1.6 Eddy-viscosity assembly in lifted transitional layers

The near-wall damping  $f_{v1}$  enforces the attached-wall-layer identity  $\chi = \kappa y^+$ . Writing  $y^+ := \chi/\kappa$ , the reconstruction

$$q = \frac{|\mathbf{u}| d/\nu}{y^+ U^+(y^+)}, \quad U^+(y^+) = \frac{1}{\kappa} \ln(1 + \kappa y^+) + 7.8(1 - e^{-y^+/11} - \frac{y^+}{11} e^{-y^+/3}) \quad (\text{Reichardt}), \quad (11)$$

The assembly is blended,

$$\mu_t = [(1 - b) f_{v1}(\chi) + b] \rho \tilde{\nu}, \quad b = s(\chi) G(q), \quad s = \text{clip}(\chi - 1, 0, 1), \quad G = \text{clip}((q - 2)/2, 0, 1), \quad (12)$$

inert for  $\chi < 1$  (negligible  $\mu_t$  either way), for  $\chi \gtrsim 30$  ( $f_{v1} \rightarrow 1$ ), and in the attached inner layer ( $q = 1$ ,  $G$  shut). Controlled on/off batteries bound the net effect on attached-layer outer edges below 1.5 drag counts (flat plate and NLF at their cold-budget protocol).

## 1.7 Freestream seed

A case seeded with  $\chi_\infty$  reaches the handover where its envelope crosses  $N_{\text{crit}} = \ln(c_{v1}/\chi_\infty)$ .

$$N_{\text{crit}}(Tu) = -8.43 - 2.4 \ln(Tu_{\text{frac}}), \quad \chi_\infty = c_{v1} e^{-N_{\text{crit}}}, \quad (13)$$

One  $e$ -fold in  $\chi_\infty$  is one unit of  $N_{\text{crit}}$ , which on the Blasius envelope ( $dN/dRe_\theta \approx 10^{-2}$ ) shifts onset by  $\Delta Re_\theta \approx 100$ . Strongly bypass-dominated conditions ( $Tu \gtrsim 1\%$ ) are out of scope: at  $Tu = 1\%$  the map leaves only  $N_{\text{crit}} \approx 2.6$ , and by  $Tu \approx 3\%$  the seed exceeds  $c_{v1}$  outright.

## 1.8 Constants

Baseline SA, unchanged:

$$c_{b1} = 0.1355, \quad \sigma = \frac{2}{3}, \quad c_{b2} = 0.622, \quad \kappa = 0.41, \quad c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}, \quad c_{w2} = 0.3, \quad c_{w3} = 2, \quad c_{v1} = 7.1.$$

SA-AI:

| constant               | value          | determination                                                                                                   |
|------------------------|----------------|-----------------------------------------------------------------------------------------------------------------|
| $a_{\max}$             | 0.19           | tanh-layer Kelvin–Helmholtz temporal eigenvalue, $\omega_{i,\max}/\omega_{\text{peak}} = 0.1897$                |
| $(C, A, B)$            | (2600, 175, 2) | graze envelope of the attached Falkner–Skan family at its critical $Re_\theta$ (rms $\sim 6\%$ )                |
| $k$                    | 0.712          | Blasius march crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$ at $c_{\nu,\text{ai}} = \frac{1}{6}$ |
| $w$                    | 0.35           | onset ramp width; $N = 1$ stations move $< 1\%$ halved, $< 2.5\%$ doubled                                       |
| $c_{\nu,\text{ai}}$    | 1/6            | frozen-profile growth eigenvalue reaches the Drela–Giles rate just past critical (1.02 at $Re_\theta = 400$ )   |
| $\tau$                 | 4              | numerics bracket: handover 78% complete at $\chi = c_{v1}$ , 97% at $\chi \approx 14.8$                         |
| $\sigma_D$ tie         | Eq. (4)        | derived: constant-stress wall-layer balance; linear wall solution exact                                         |
| bypass $(s, G)$        | Eq. (12)       | parameter-free law-of-the-wall discriminator; $G$ ramps over $q \in (2, 4)$                                     |
| $(A, B)_{\text{Mack}}$ | (−8.43, 2.4)   | Mack 1977 $e^N$ correlation, wholesale                                                                          |

## 1.9 Running the model: branch selection and seed handling

On the frozen Hiemenz stagnation field with the freestream seed exactly zero, standard SA sustains the wedge above a critical nose Reynolds number bisected to  $Re_r = 4.66\text{--}4.74 \times 10^5$  (Fig. 5).

Table 1: Attachment-anchored branch: steady  $\max \chi$  versus nose Reynolds number  $Re_r = L^2$  on the frozen Hiemenz wedge, standard SA,  $\chi_\infty = 0$ . Below the bisected critical band the layer collapses to  $\chi = 0$ .

| $L = \sqrt{Re_r}$ | $Re_r$                          | steady $\max \chi$           |
|-------------------|---------------------------------|------------------------------|
| $\leq 683$        | $\leq 4.66 \times 10^5$         | $< 10^{-3}$ (collapses to 0) |
| 683–689           | $4.66\text{--}4.74 \times 10^5$ | critical band (bisected)     |
| 689               | $4.74 \times 10^5$              | 2.19                         |
| 700               | $4.90 \times 10^5$              | 6.15                         |
| 1000              | $1.00 \times 10^6$              | 24.1                         |
| 3000              | $9.00 \times 10^6$              | 75.7                         |

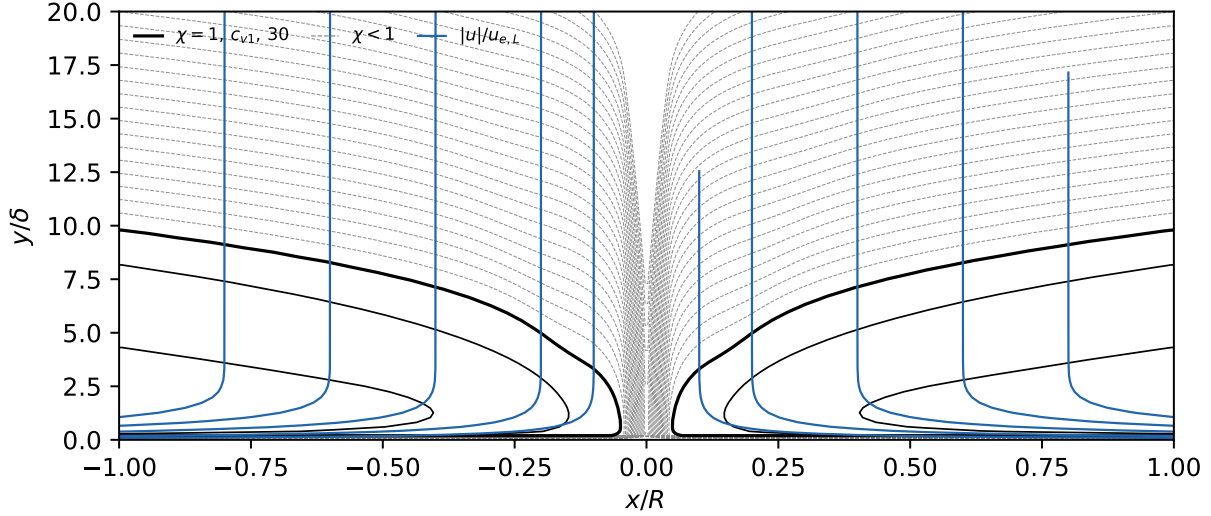


Figure 5: Attachment-anchored turbulent branch,  $L = 3000$  ( $Re_r = 9 \times 10^6$ ),  $\max \chi = 76$ . Standard SA, frozen Hiemenz,  $\chi_\infty = 0$ . Abscissa  $x/R$  (arc length in nose radii,  $R/\delta = L$ ); ordinate  $y/\delta$ .  $\chi$  contours dashed per decade for  $\chi < 1$  ( $10^{-24}$ – $10^{-1}$ ; the zero freestream leaves no floor, so  $\chi$  decays smoothly over decades), solid at 1,  $c_{v1}$ , 30; velocity magnitude  $|u|/u_{e,L}$  (blue, normalized by the  $x=L$  edge speed) at 0.1, 0.2, 0.4, 0.6, 0.8. Branch data in Table 1.

The protocol used for every result in this paper: the  $\tilde{\nu}$ -equation residual is scaled by  $f^{1-g}$ , where  $g$  rises smoothly from 0 (laminar) to 1 (turbulent) with  $\chi$  and  $f = 10^{-2}$  (the solver uses  $g = 1 - e^{-(\chi - c_{v1})/4}$  above  $\chi = c_{v1}$ , zero below). Because the freestream boundary condition flows through the same scaling, the freestream viscosity ratio supplied to the solver is  $\chi_\infty f$ ; the seeds quoted throughout are the physical  $\chi_\infty$ , recovered at the boundary-layer edge. (In the reference implementation the scaling is `AI_LAMINAR_SLOWDOWN` =  $f$ , and the case files carry the pre-multiplied boundary seed  $\chi_\infty f$ ; running a committed case at  $f = 1$  without rescaling the seed mis-seeds it by  $1/f$ .)

## 2 Results

Coupled RANS: Flow360, unstructured node-centered second-order finite volume, Roe flux,  $M = 0.1$ . Airfoil cases:  $\chi_\infty = c_{v1}e^{-9} \approx 8.76 \times 10^{-4}$  (the  $N_{\text{crit}} = 9$  anchor; via Mack’s map,  $Tu \approx 0.07\%$ ); two independently generated mesh families—all-triangular unstructured cavity meshes, constrained-Delaunay through the boundary layer, and all-quadrilateral structured O-grids—at three refinement levels,  $1.5 \times 10^4$  to  $2.6 \times 10^5$  nodes. Transition locations are the solver’s near-wall  $\chi = 1$  crossing (the  $\chi = c_{v1}$  crossing sits up to  $0.05c$  aft where the front grows slowly); bubble stations are signed- $C_f$  zero crossings. The cross-solver replication is a cell-centered, pressure-based, incompressible OpenFOAM implementation of the model on the same flat-plate grid specification and the same structured airfoil grids (the port omits the  $f_{v1}$  bypass); thirty airfoil cases plus the flat-plate sweep.

### 2.1 Flat plate, $Tu = 0.04$ – $0.60\%$

Structured  $320 \times 80$  grid,  $Re_x \leq 6 \times 10^6$ ,  $y^+ \lesssim 0.3$ ; five seeds from Mack’s map ( $\chi_\infty \approx 2.3 \times 10^{-4}$ ,  $1.2 \times 10^{-3}$ ,  $6.3 \times 10^{-3}$ ,  $2.9 \times 10^{-2}$ ,  $1.5 \times 10^{-1}$ ). Figure 6: at the  $\chi = c_{v1}$  crossing the onset brackets the Abu-Ghannam & Shaw correlation within 10% across the range. Figure 7:  $\chi = 1$  crossings within 7% in  $Re_\theta$ . Table 2: both solvers’ crossings.

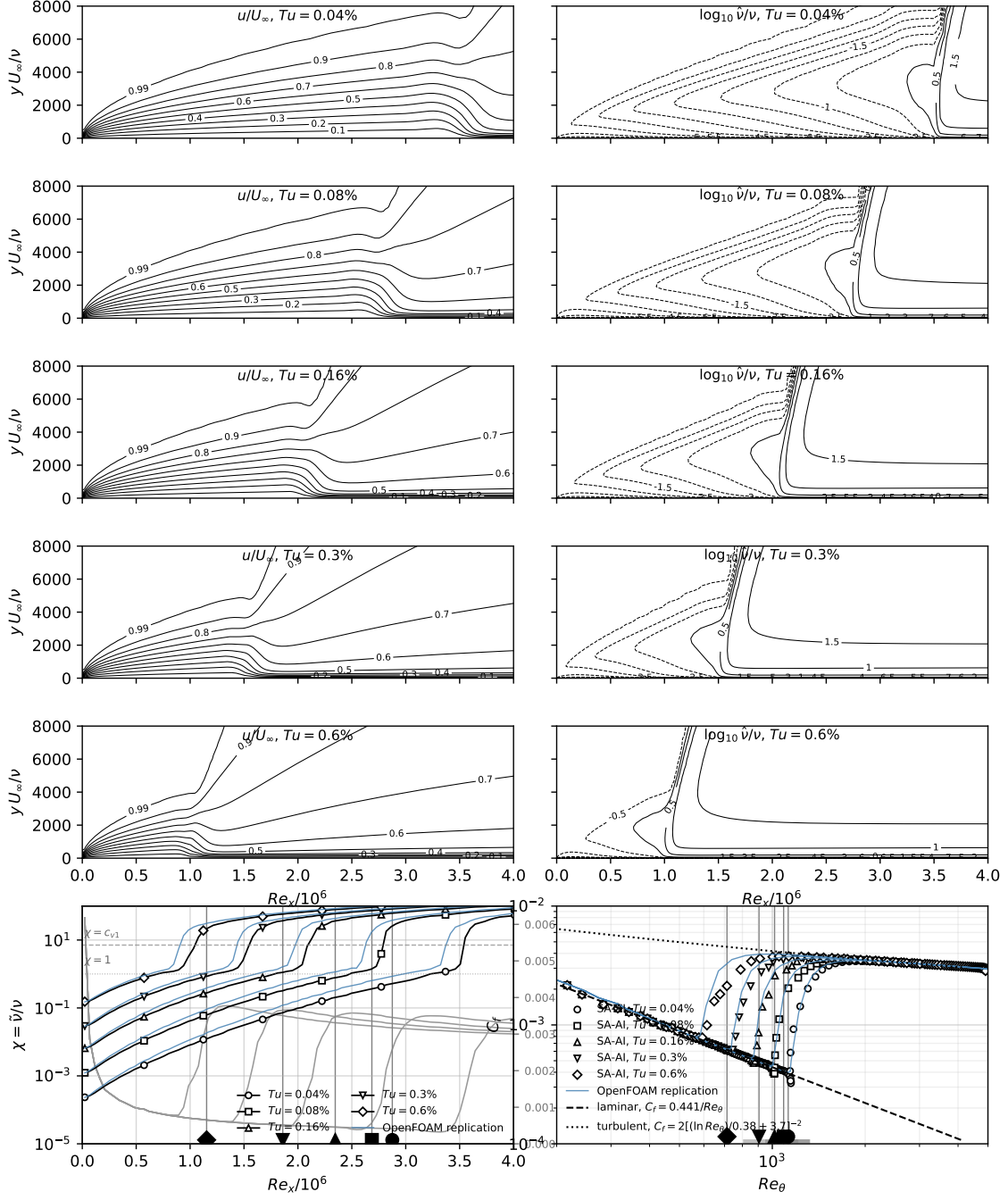


Figure 6: Flat plate, Flow360. Five levels  $Tu = 0.04\text{--}0.60\%$ ; references  $\chi = 1$  and  $\chi = c_{v1} = 7.1$ ; laminar  $C_f = 0.441/Re_\theta$ , turbulent  $C_f = 2[(\ln Re_\theta)/0.38 + 3.7]^{-2}$ ;  $\chi = c_{v1}$  onset brackets Abu-Ghannam & Shaw within 10%.

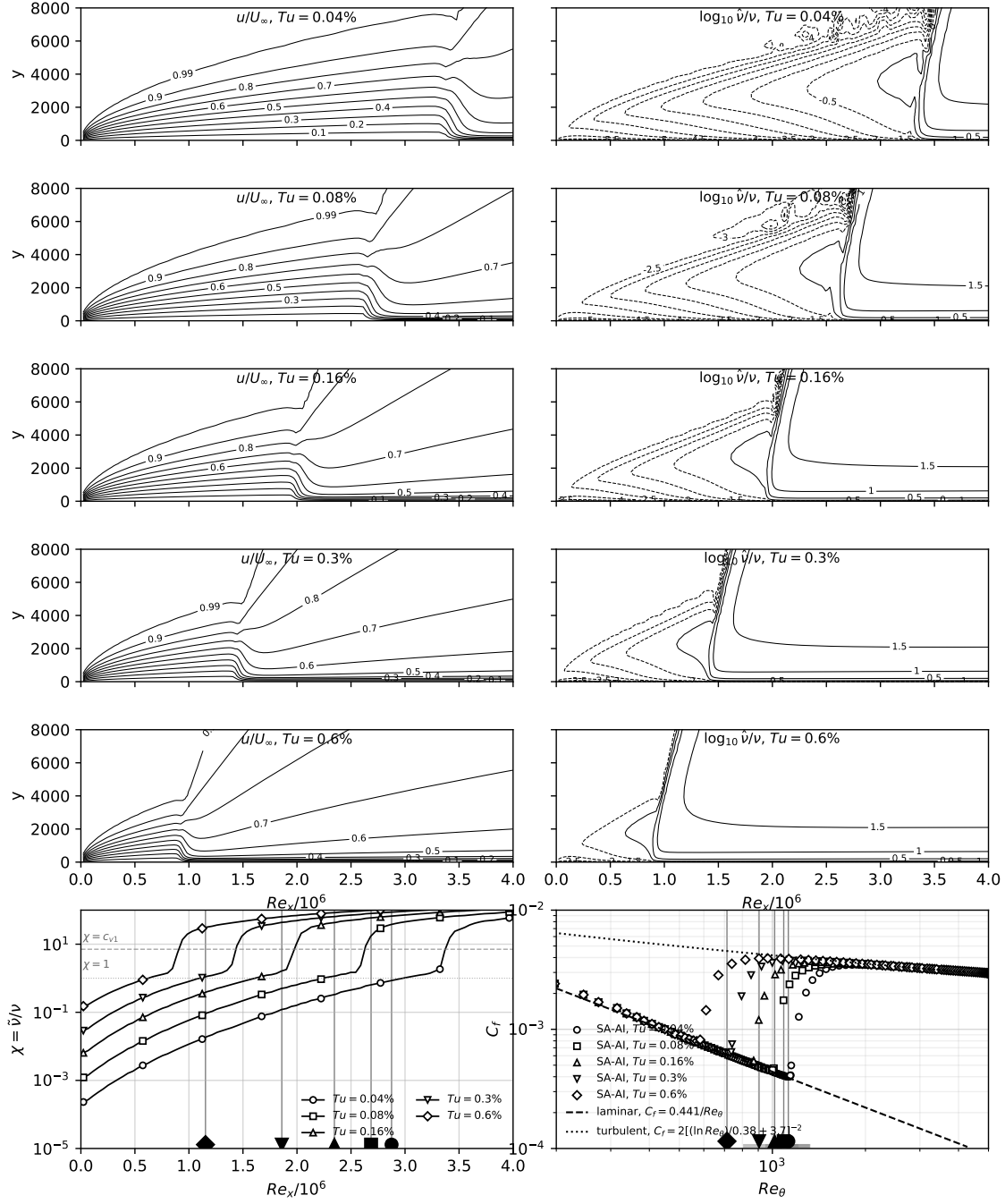


Figure 7: Flat plate, OpenFOAM. Same sweep as Fig. 6, five levels  $Tu = 0.04\text{--}0.60\%$ .

Table 2: Flat-plate transition-onset  $Re_\theta$  (integrated from the computed profiles) at the  $\chi = 1$  and  $\chi = c_{v1}$  crossings, Flow360 and the OpenFOAM replication (OF), against the Abu-Ghannam & Shaw correlation.

| $Tu$ [%] | AGS $Re_\theta$ | $\chi = 1$ | $\chi = c_{v1}$ | OF $\chi = 1$ | OF $\chi = c_{v1}$ |
|----------|-----------------|------------|-----------------|---------------|--------------------|
| 0.04     | 1126            | 1149       | 1203            | 1098          | 1154               |
| 0.08     | 1088            | 1002       | 1069            | 949           | 1004               |
| 0.16     | 1017            | 845        | 934             | 798           | 887                |
| 0.3      | 905             | 700        | 815             | 656           | 739                |
| 0.6      | 713             | 524        | 685             | 487           | 589                |

## 2.2 NLF(1)-0416 at $Re = 4 \times 10^6$

$\alpha \in \{0^\circ, 4^\circ, 9^\circ, 15^\circ\}$  on both families at L0–L2 plus  $\alpha = -8^\circ, -4^\circ$  on all six grids (thirty-six cases). Figure 8: transition fronts against the LTPT microphone measurements, Coder’s AFT (OVERFLOW) at  $N_{crit} = 10.07$  and recalibrated 7.18, the fourteen First AIAA Transition Modeling Workshop submittals, SA-LM2015, Langtry–Menter, and XFOIL. Figure 9: drag polar against Somers TP-1861, the  $e^9$  reference, and fully turbulent SA. Figures 10–12: surface distributions. Table 3: forces and fronts. Table 4: at L2 every front within  $0.016c$  and the lift within  $0.9\%$  of the SA-AI values.

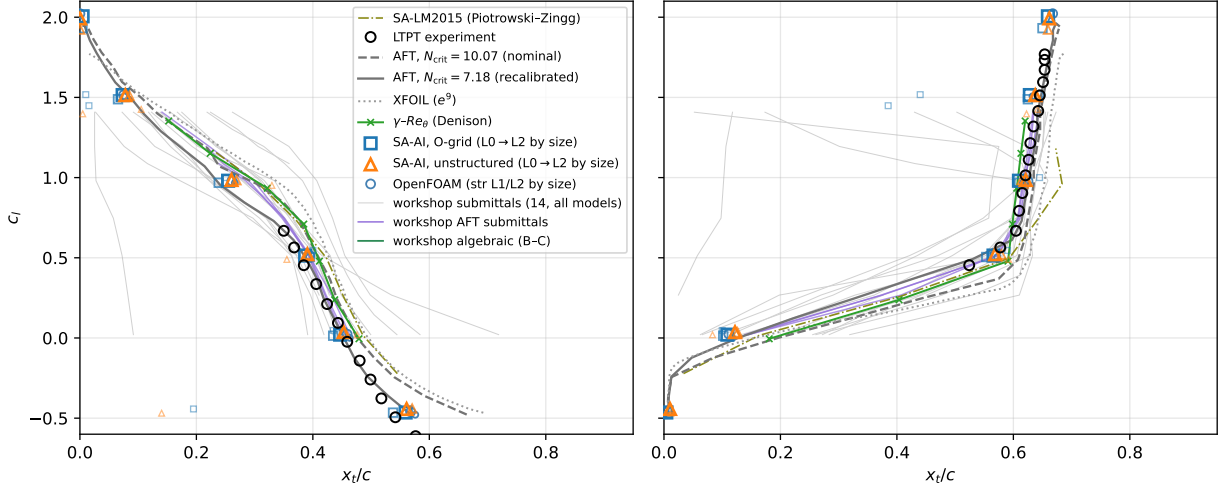


Figure 8: NLF(1)-0416 transition fronts,  $Re = 4 \times 10^6$ ; near-wall  $\chi = 1$  convention. SA-AI at  $\alpha = 0^\circ, 4^\circ, 9^\circ, 15^\circ, -8^\circ, -4^\circ$ ; workshop sweep  $\alpha = -4^\circ - 8^\circ$ .

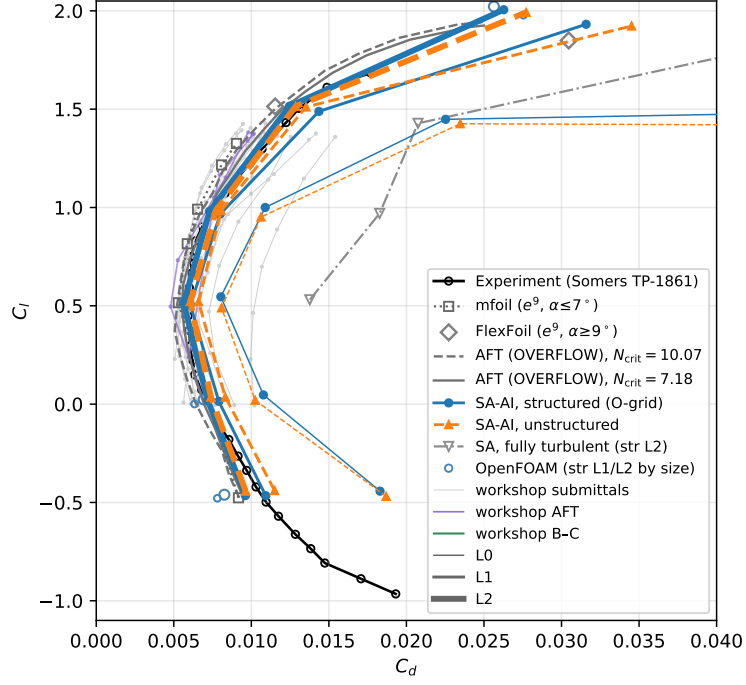


Figure 9: NLF(1)-0416 drag polar,  $Re = 4 \times 10^6$ . Fully turbulent SA ( $\chi_\infty = 3$ ): 1.6–2.5 $\times$  above the SA-AI drag, +144/+150% in the bucket at  $\alpha = 0^\circ/4^\circ$ , +67/+74% at  $9^\circ/15^\circ$ .

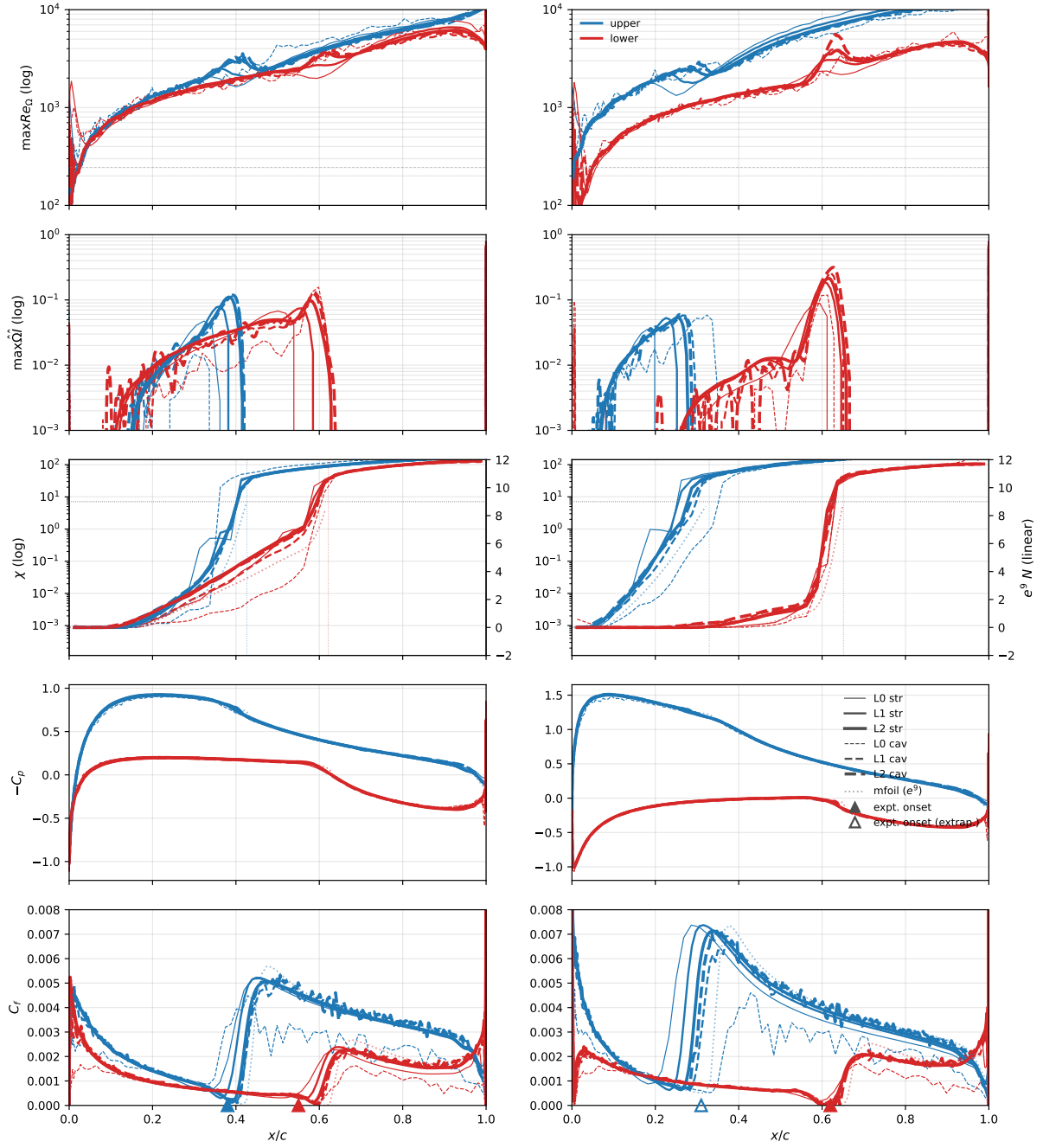


Figure 10: NLF(1)-0416,  $\alpha = 0^\circ, 4^\circ$ ,  $Re = 4 \times 10^6$ .



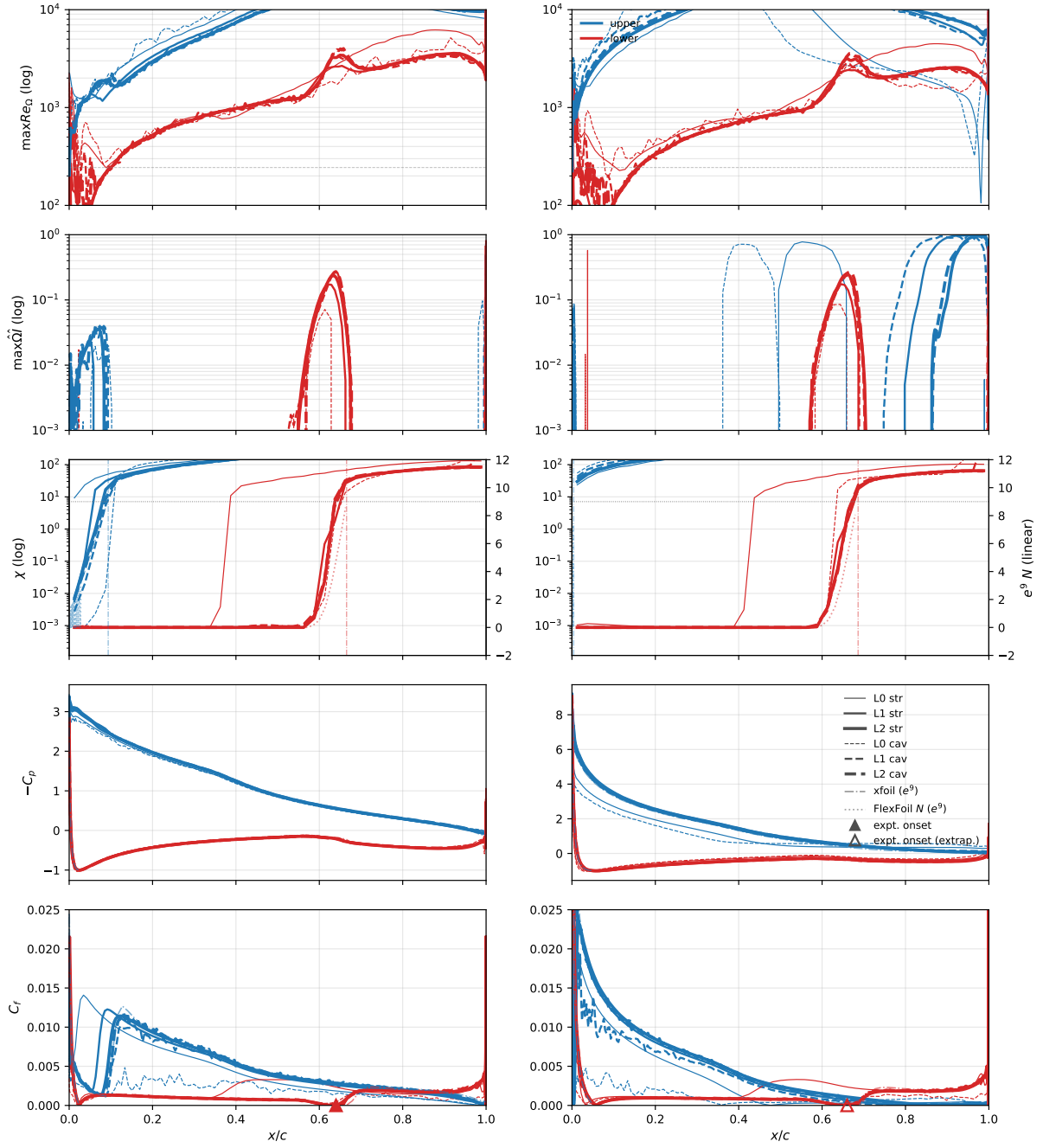


Figure 11: NLF(1)-0416,  $\alpha = 9^\circ, 15^\circ$ ,  $Re = 4 \times 10^6$ .

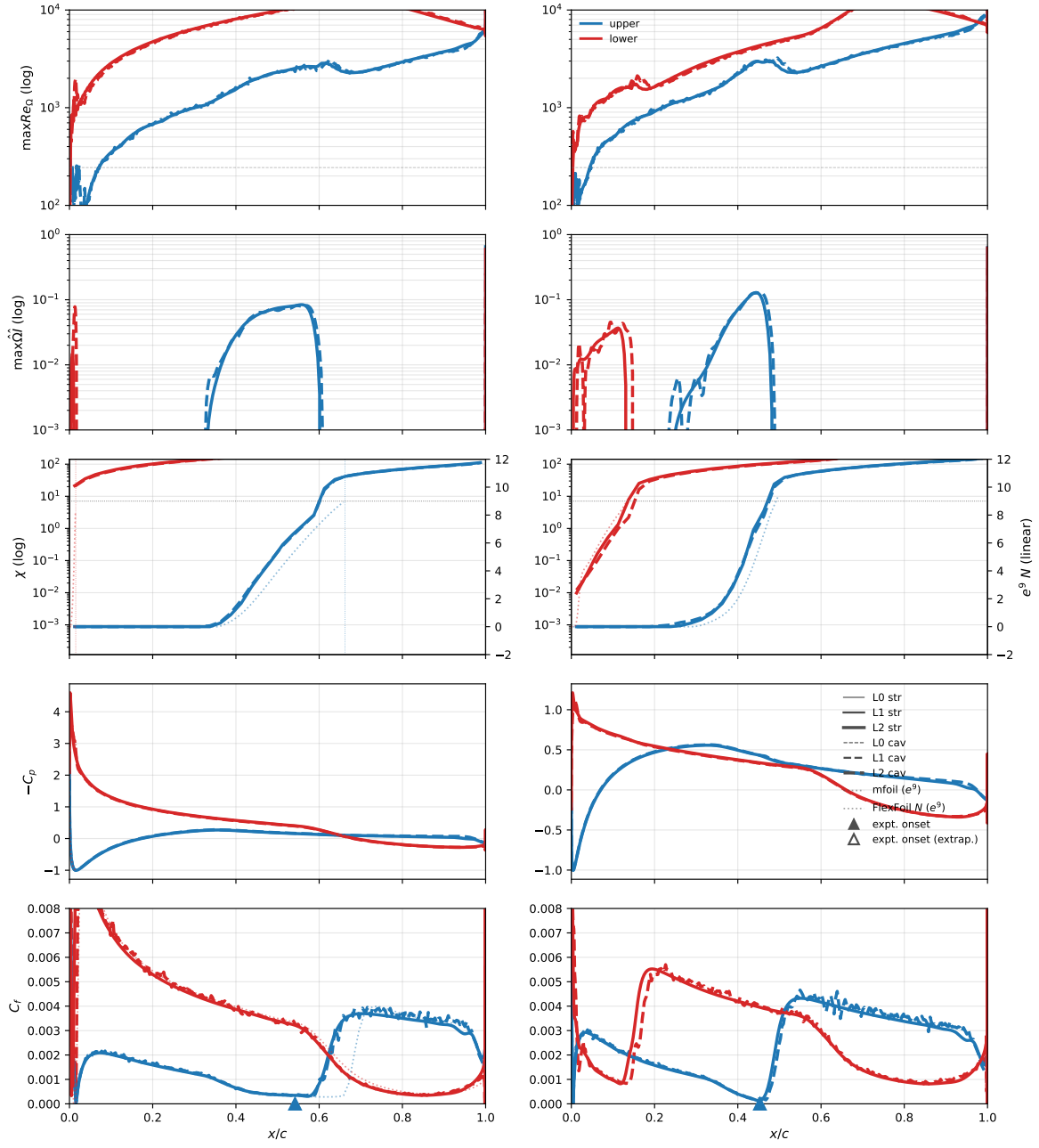


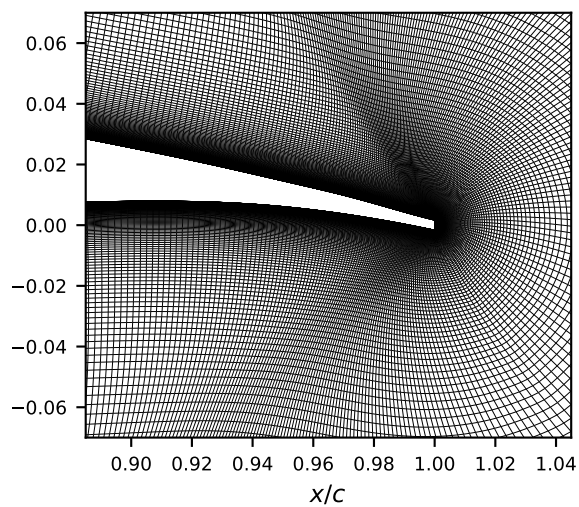
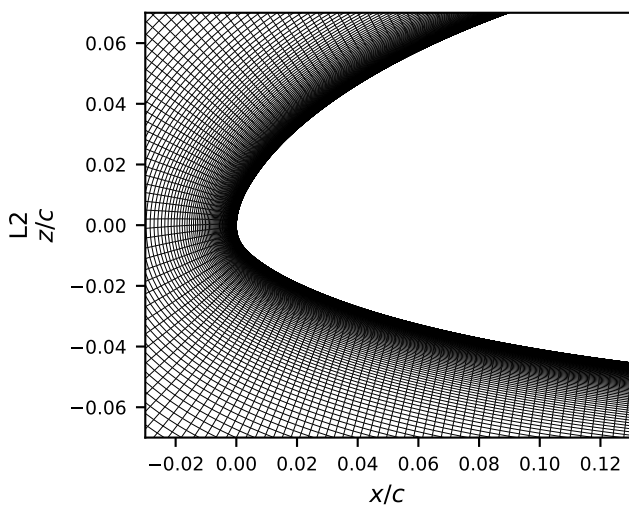
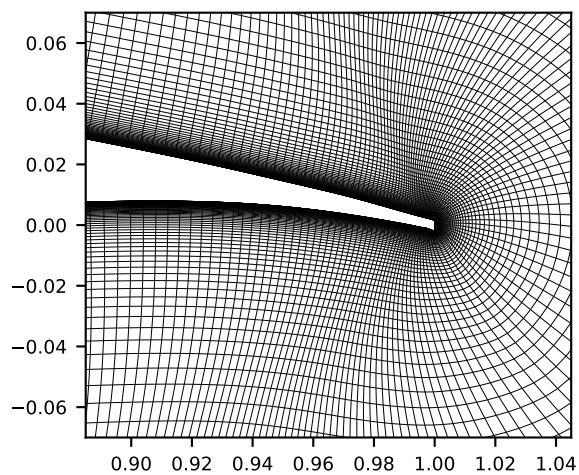
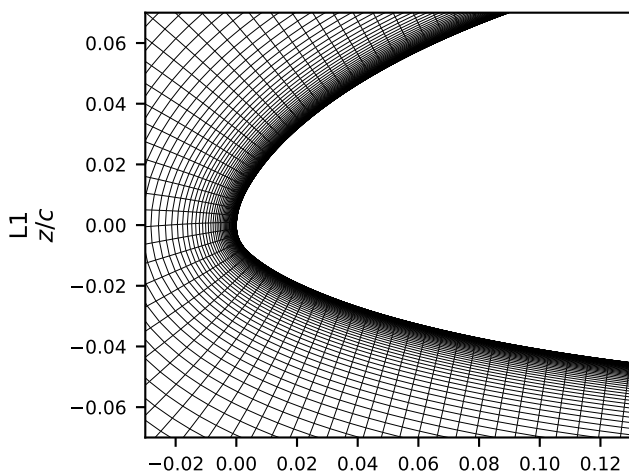
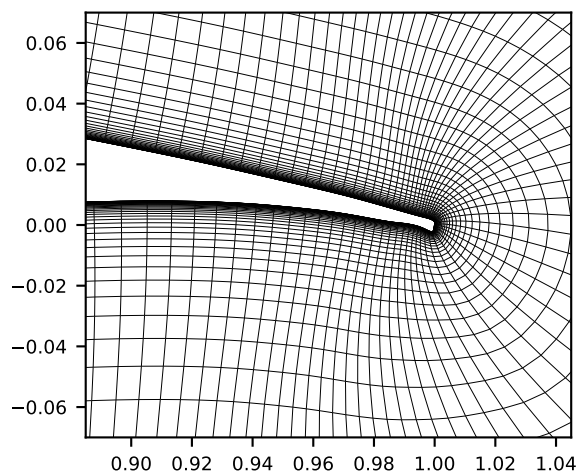
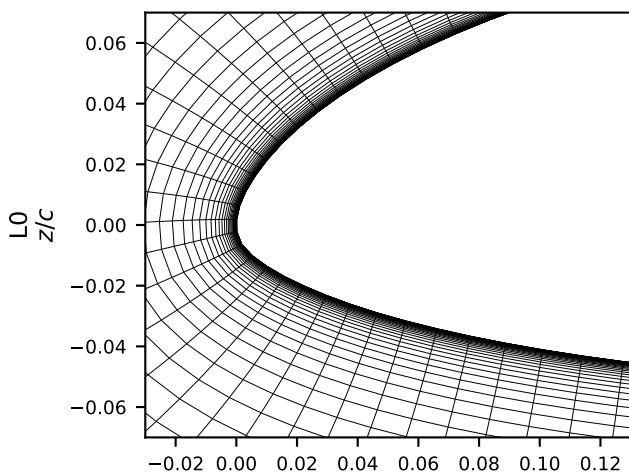
Figure 12: NLF(1)-0416,  $\alpha = -8^\circ, -4^\circ$ ,  $Re = 4 \times 10^6$ .

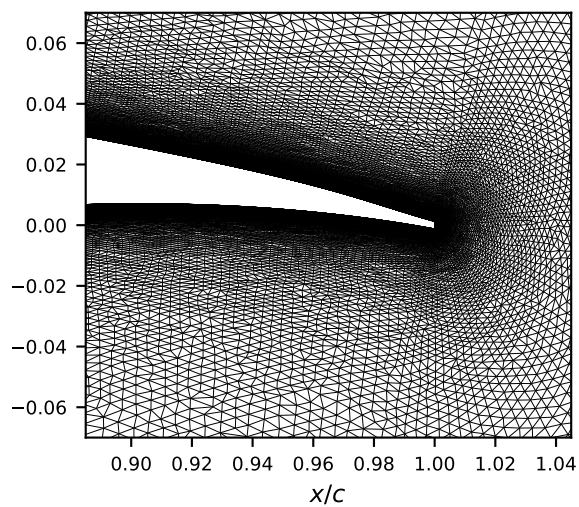
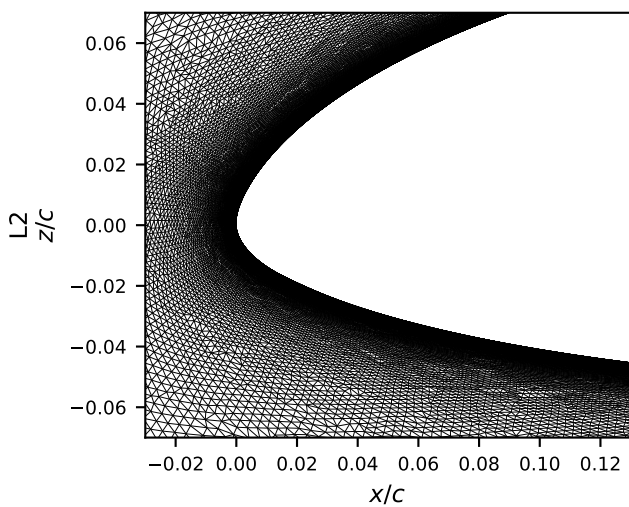
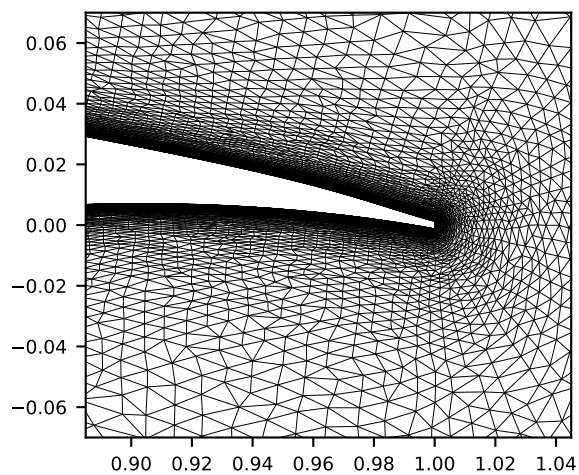
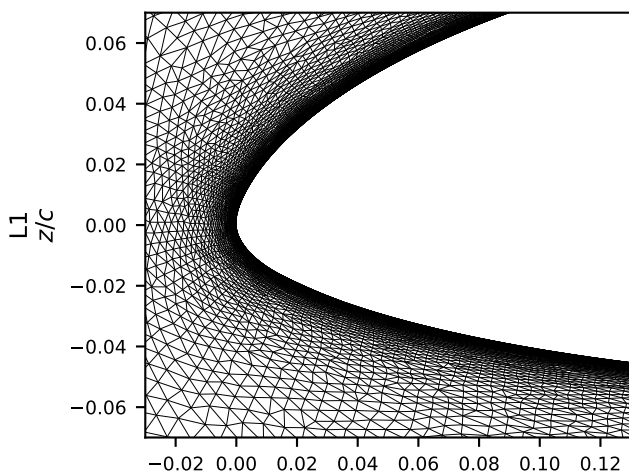
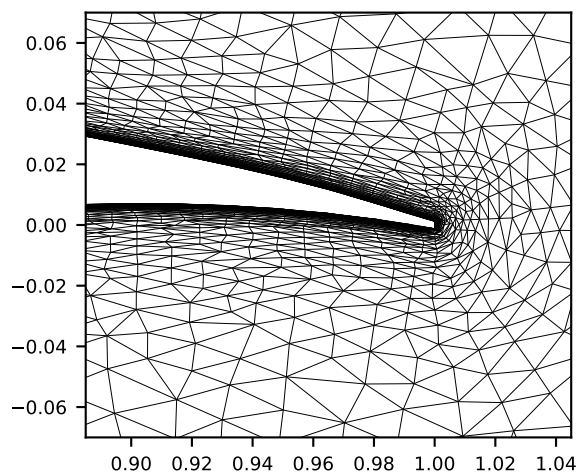
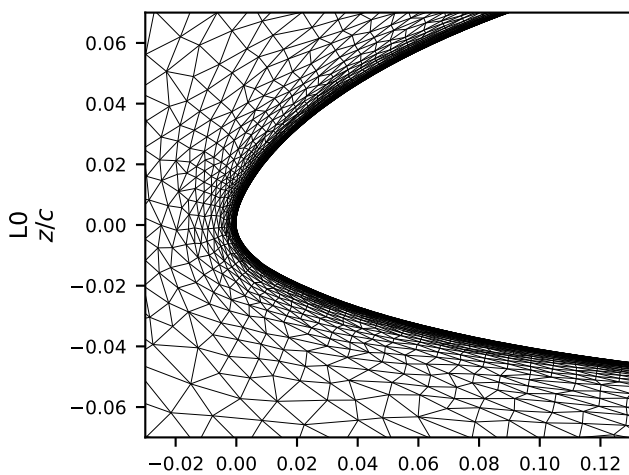
Table 3: NLF(1)-0416,  $Re=4\times 10^6$ : computed forces and transition locations (near-wall  $\chi=1$  front stations).

| $\alpha$ | grid | structured O-grid |         |               |               | unstructured cavity |         |               |               |
|----------|------|-------------------|---------|---------------|---------------|---------------------|---------|---------------|---------------|
|          |      | $C_L$             | $C_D$   | $x_{tr}^{up}$ | $x_{tr}^{lo}$ | $C_L$               | $C_D$   | $x_{tr}^{up}$ | $x_{tr}^{lo}$ |
| -8       | L0   | -0.4426           | 0.01830 | 0.195         | 0.011         | -0.4689             | 0.01869 | 0.140         | 0.012         |
| -8       | L1   | -0.4656           | 0.01093 | 0.538         | 0.006         | -0.4397             | 0.01150 | 0.571         | 0.011         |
| -8       | L2   | -0.4646           | 0.00962 | 0.559         | 0.004         | -0.4426             | 0.00964 | 0.561         | 0.010         |
| -4       | L0   | 0.0469            | 0.01078 | 0.435         | 0.100         | 0.0197              | 0.01025 | 0.476         | 0.083         |
| -4       | L1   | 0.0142            | 0.00790 | 0.435         | 0.102         | 0.0357              | 0.00832 | 0.452         | 0.126         |
| -4       | L2   | 0.0208            | 0.00718 | 0.447         | 0.109         | 0.0363              | 0.00739 | 0.453         | 0.122         |
| 0        | L0   | 0.5458            | 0.00803 | 0.400         | 0.590         | 0.4896              | 0.00808 | 0.355         | 0.592         |
| 0        | L1   | 0.5048            | 0.00607 | 0.395         | 0.553         | 0.5209              | 0.00660 | 0.392         | 0.579         |
| 0        | L2   | 0.5139            | 0.00564 | 0.387         | 0.566         | 0.5219              | 0.00606 | 0.391         | 0.569         |
| 4        | L0   | 1.0001            | 0.01090 | 0.250         | 0.645         | 0.9513              | 0.01062 | 0.330         | 0.621         |
| 4        | L1   | 0.9678            | 0.00800 | 0.238         | 0.610         | 0.9839              | 0.00805 | 0.269         | 0.614         |
| 4        | L2   | 0.9805            | 0.00730 | 0.255         | 0.610         | 0.9848              | 0.00772 | 0.260         | 0.621         |
| 9        | L0   | 1.4483            | 0.02252 | 0.015         | 0.385         | 1.4251              | 0.02345 | 0.105         | 0.641         |
| 9        | L1   | 1.4886            | 0.01436 | 0.065         | 0.625         | 1.5112              | 0.01353 | 0.085         | 0.635         |
| 9        | L2   | 1.5146            | 0.01245 | 0.073         | 0.628         | 1.5139              | 0.01286 | 0.078         | 0.638         |
| 15       | L0   | 1.5172            | 0.07109 | 0.010         | 0.440         | 1.3976              | 0.11602 | 0.005         | 0.623         |
| 15       | L1   | 1.9317            | 0.03159 | 0.004         | 0.650         | 1.9232              | 0.03454 | 0.001         | 0.659         |
| 15       | L2   | 2.0057            | 0.02628 | 0.005         | 0.659         | 1.9924              | 0.02772 | 0.000         | 0.662         |

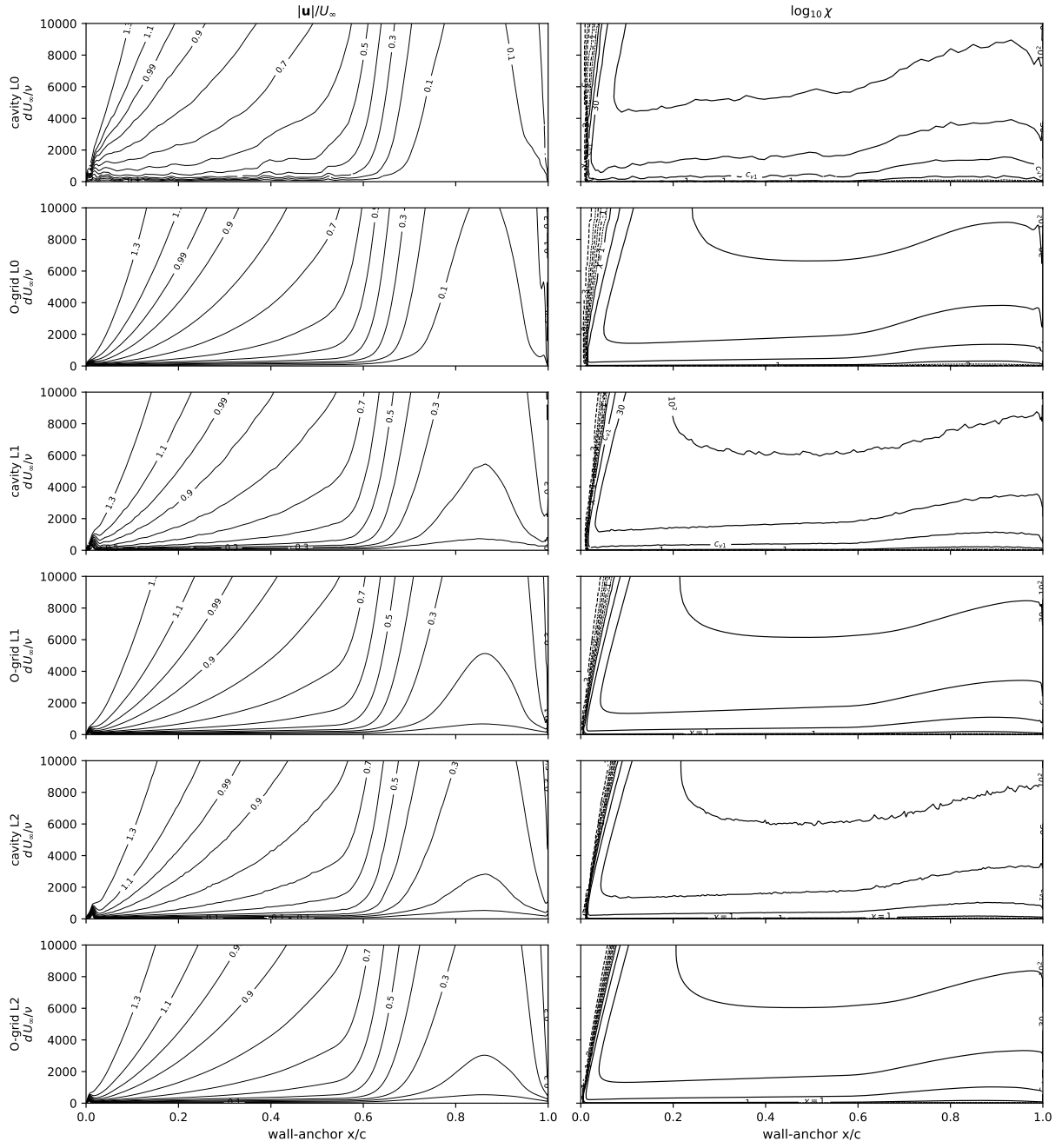
Table 4: NLF(1)-0416: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids, fronts at the same near-wall  $\chi=1$  convention (the port omits the  $f_{v1}$  bypass).

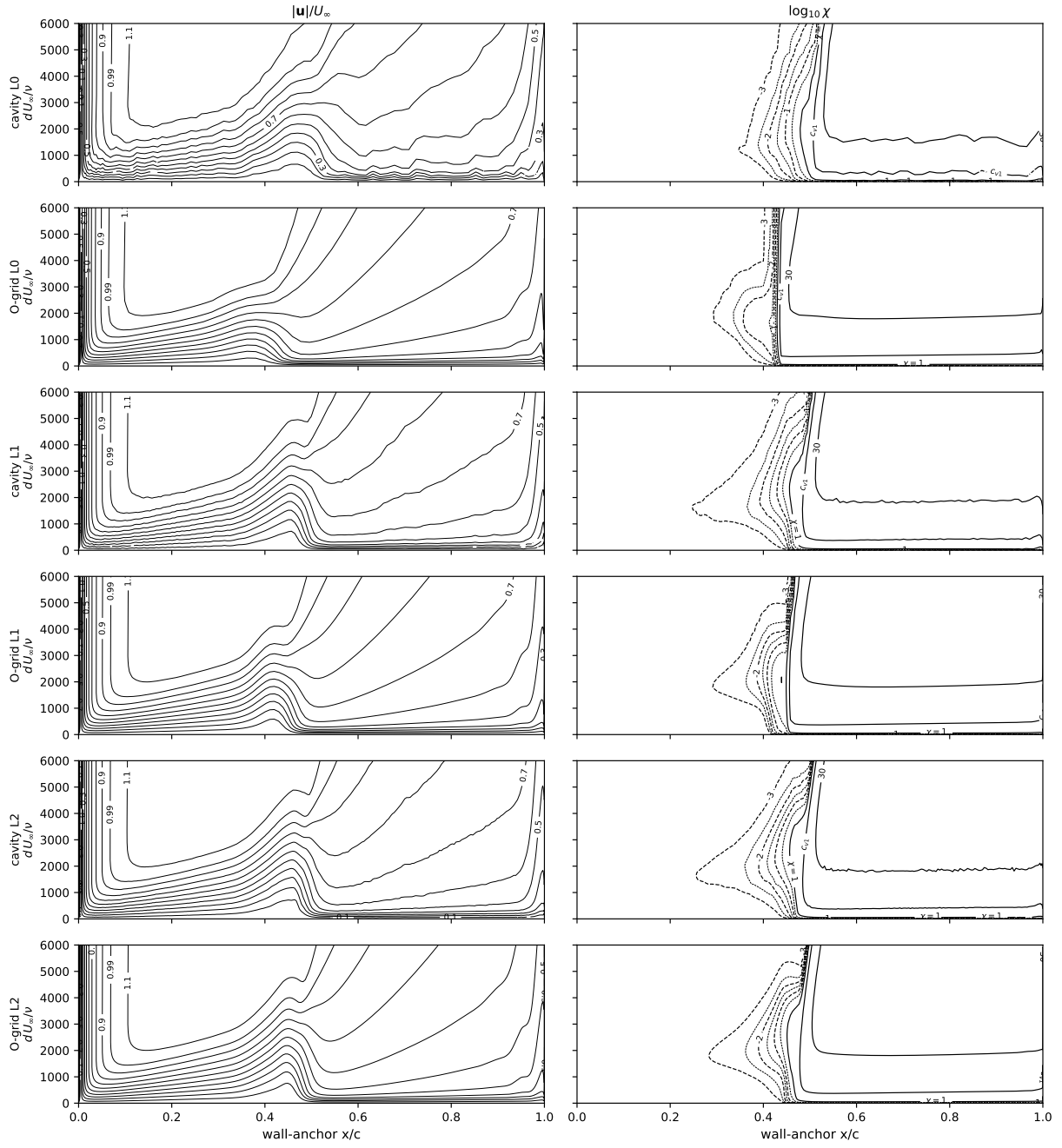
| $\alpha$ | grid | $C_L$   | $C_D$   | $x_{tr}^{up}$ | $x_{tr}^{lo}$ |
|----------|------|---------|---------|---------------|---------------|
| -8       | L0   | -0.4460 | 0.01241 | 0.556         | 0.007         |
| -8       | L1   | -0.4785 | 0.00781 | 0.576         | 0.000         |
| -8       | L2   | -0.4606 | 0.00826 | 0.568         | 0.000         |
| -4       | L0   | 0.0388  | 0.00748 | 0.424         | 0.018         |
| -4       | L1   | 0.0005  | 0.00634 | 0.464         | 0.117         |
| -4       | L2   | 0.0239  | 0.00693 | 0.459         | 0.111         |
| 0        | L0   | 0.5473  | 0.00669 | 0.387         | 0.564         |
| 0        | L1   | 0.5034  | 0.00603 | 0.393         | 0.572         |
| 0        | L2   | 0.5153  | 0.00573 | 0.395         | 0.568         |
| 4        | L0   | 1.0114  | 0.00955 | 0.271         | 0.618         |
| 4        | L1   | 0.9670  | 0.00778 | 0.266         | 0.616         |
| 4        | L2   | 0.9793  | 0.00749 | 0.261         | 0.625         |
| 9        | L0   | 1.5277  | 0.01496 | 0.063         | 0.618         |
| 9        | L1   | 1.5066  | 0.01273 | 0.076         | 0.633         |
| 9        | L2   | 1.5151  | 0.01270 | 0.075         | 0.644         |
| 15       | L0   | 1.7210  | 0.05409 | 0.000         | 0.634         |
| 15       | L1   | 1.9780  | 0.02754 | 0.000         | 0.665         |
| 15       | L2   | 2.0221  | 0.02564 | 0.000         | 0.667         |



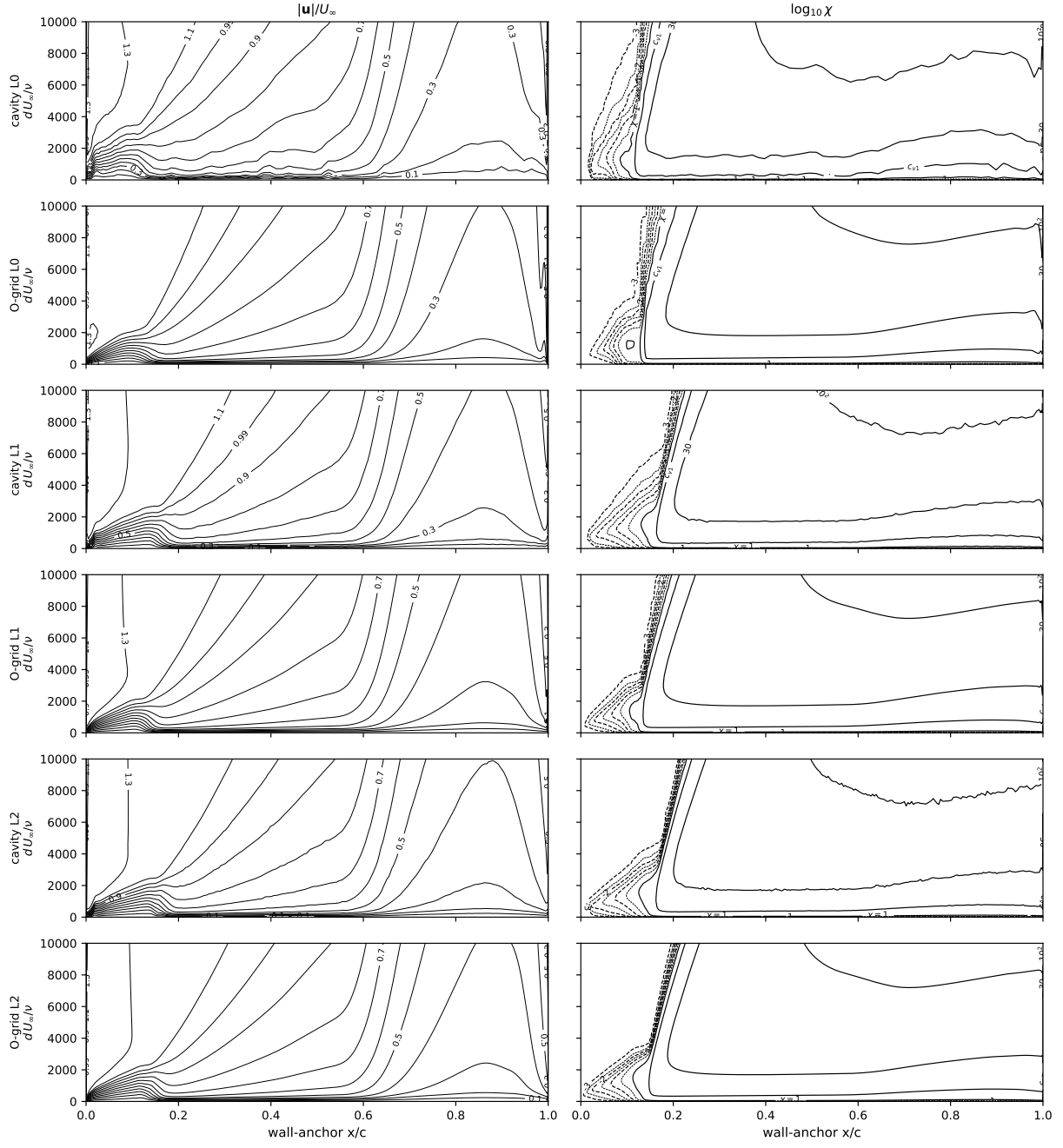


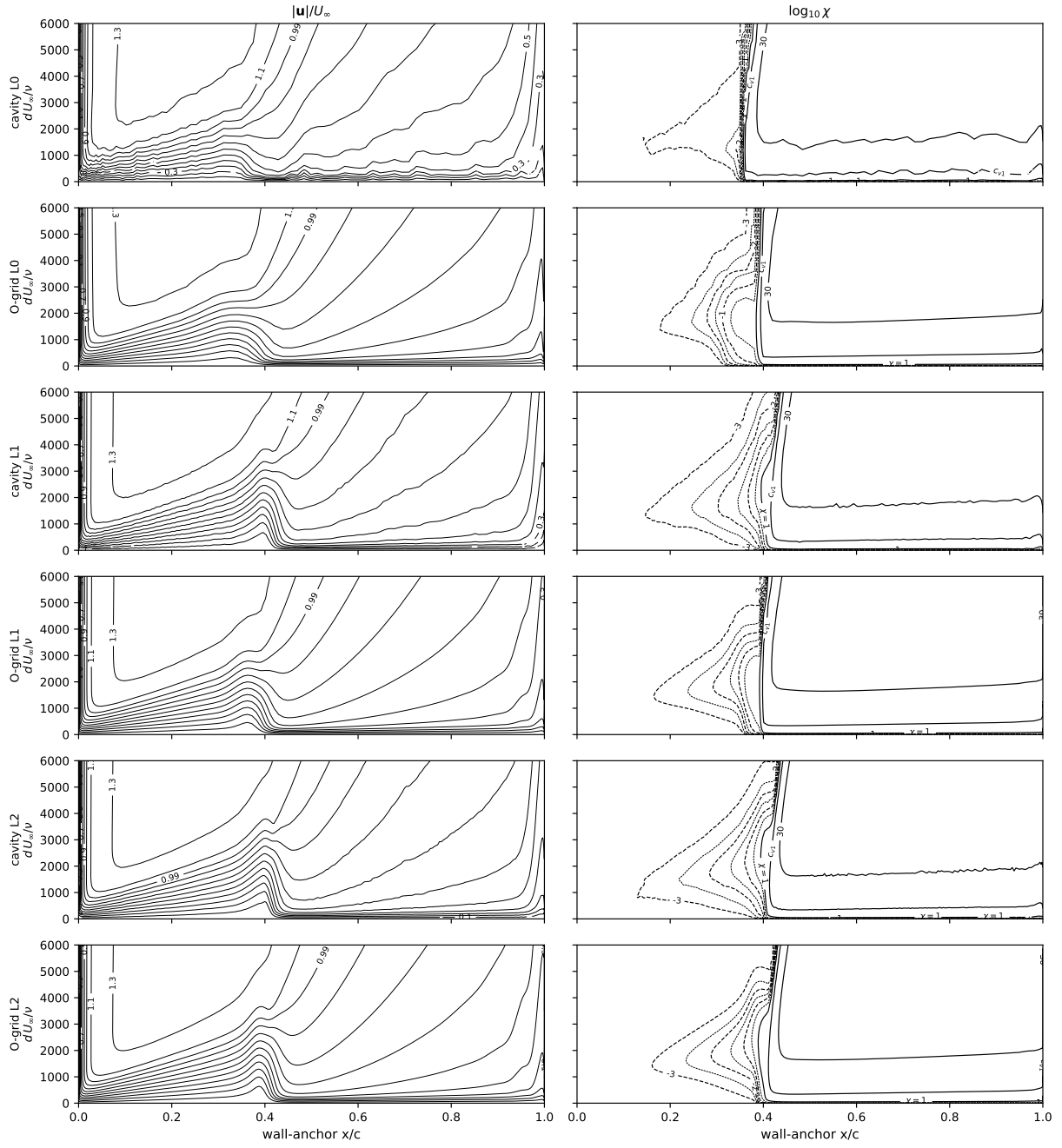


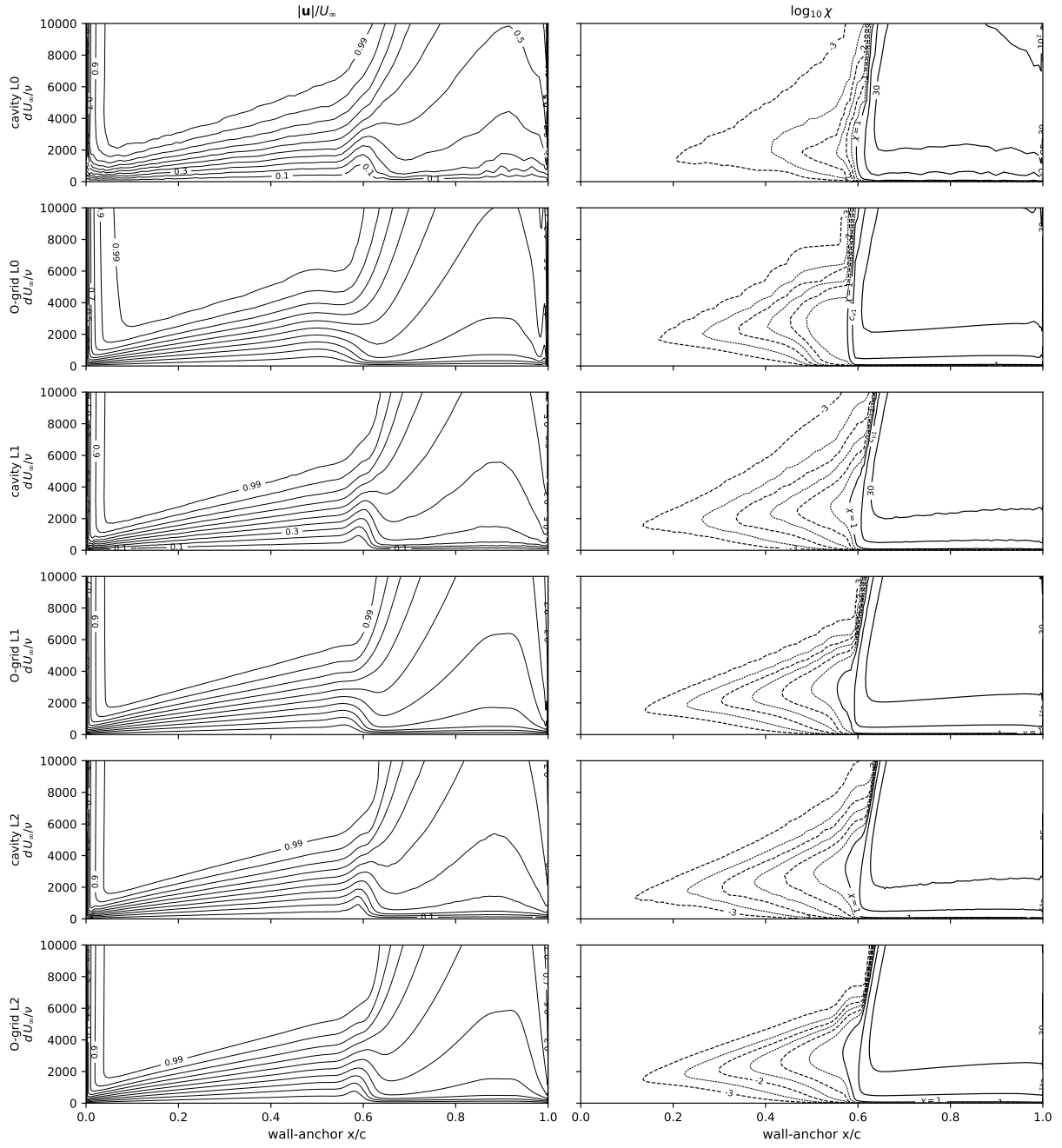


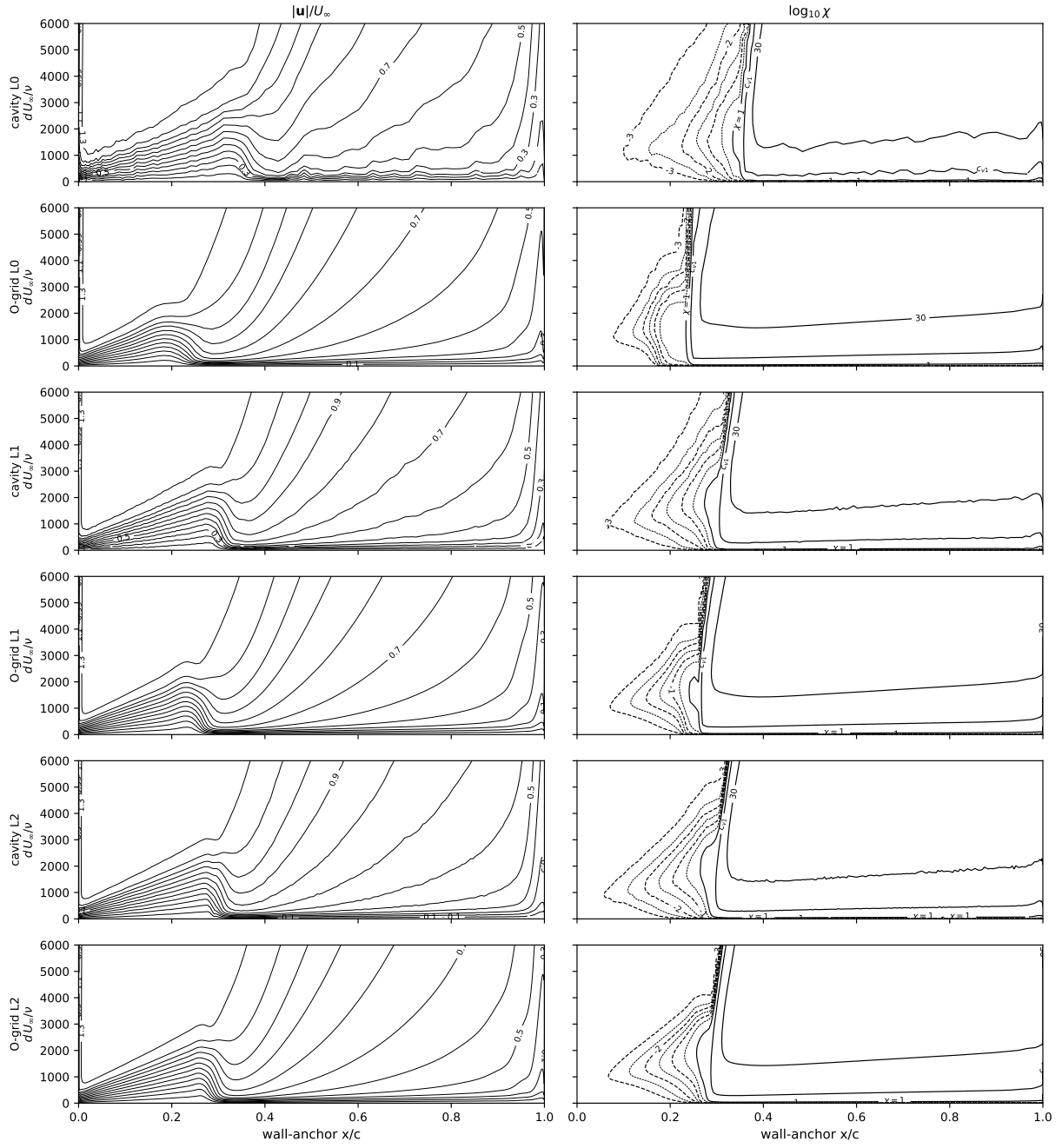


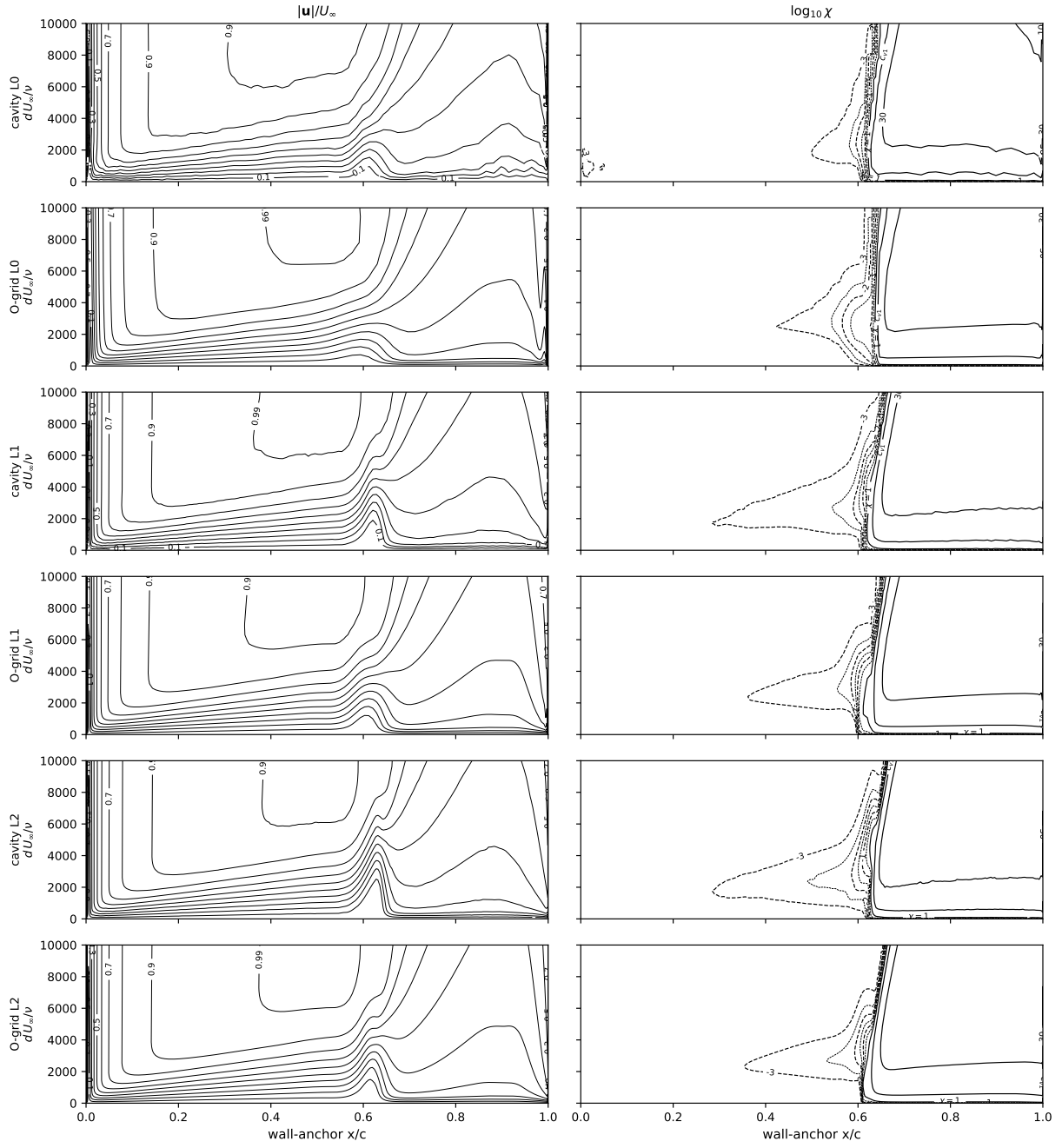


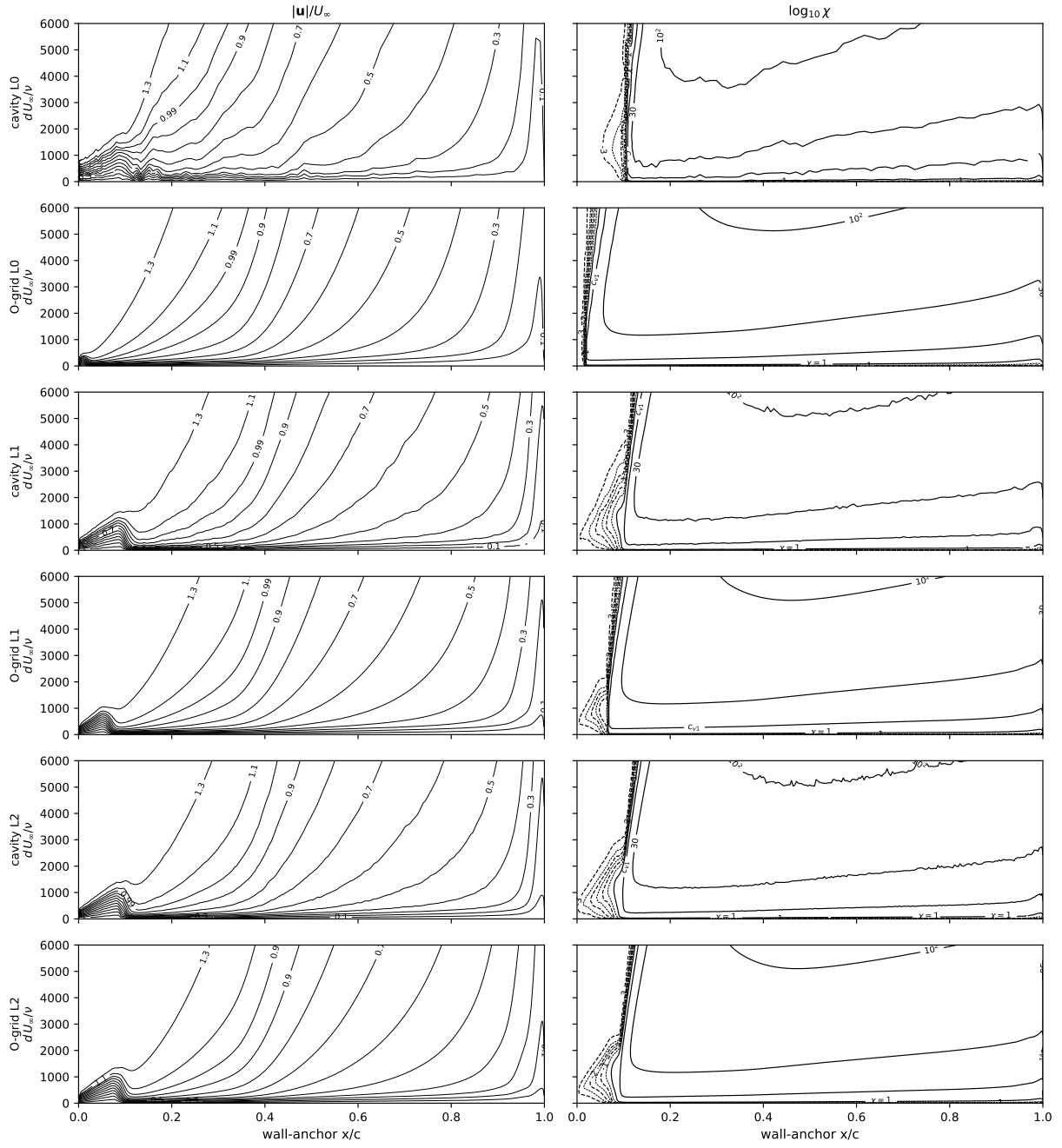


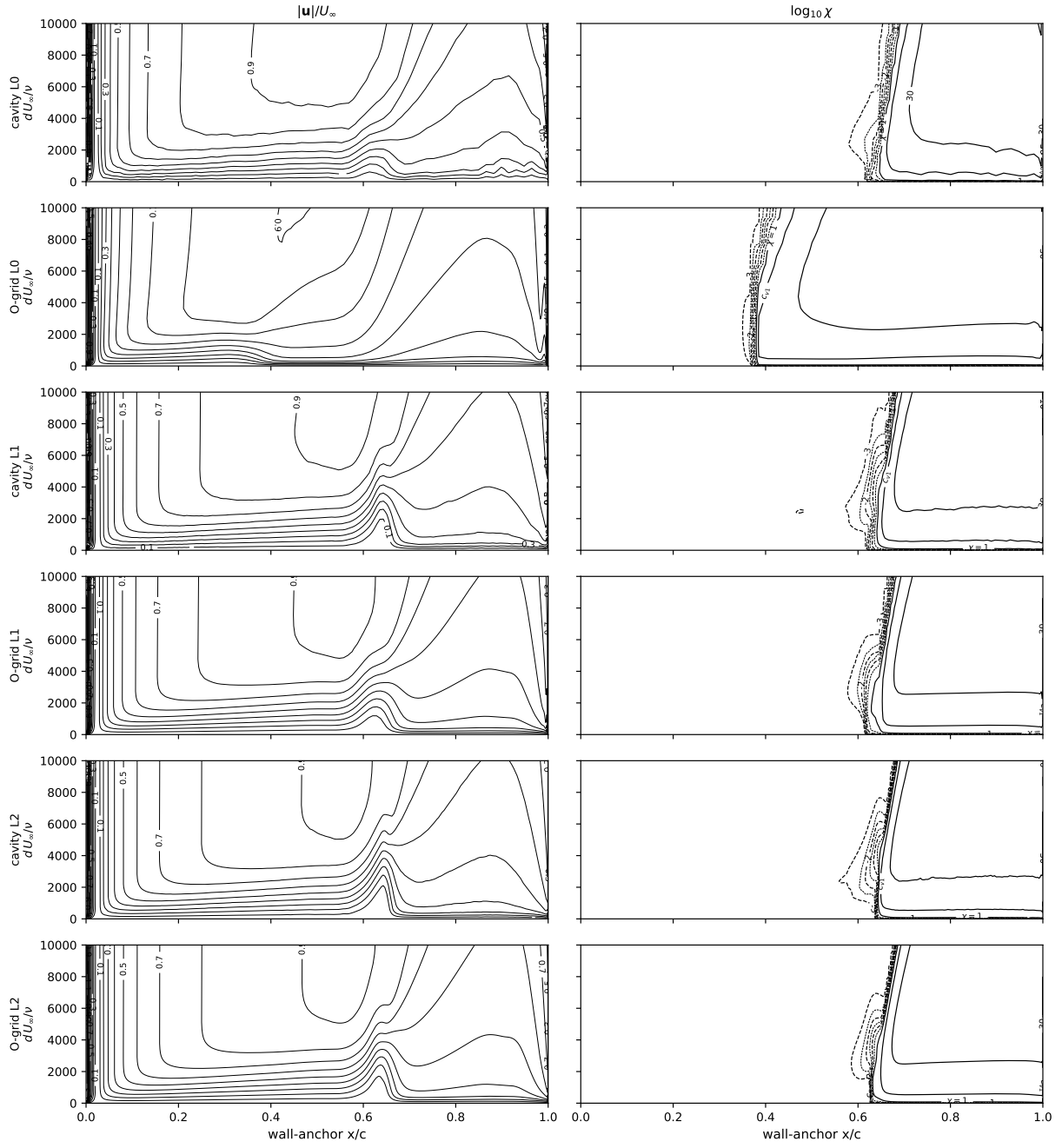






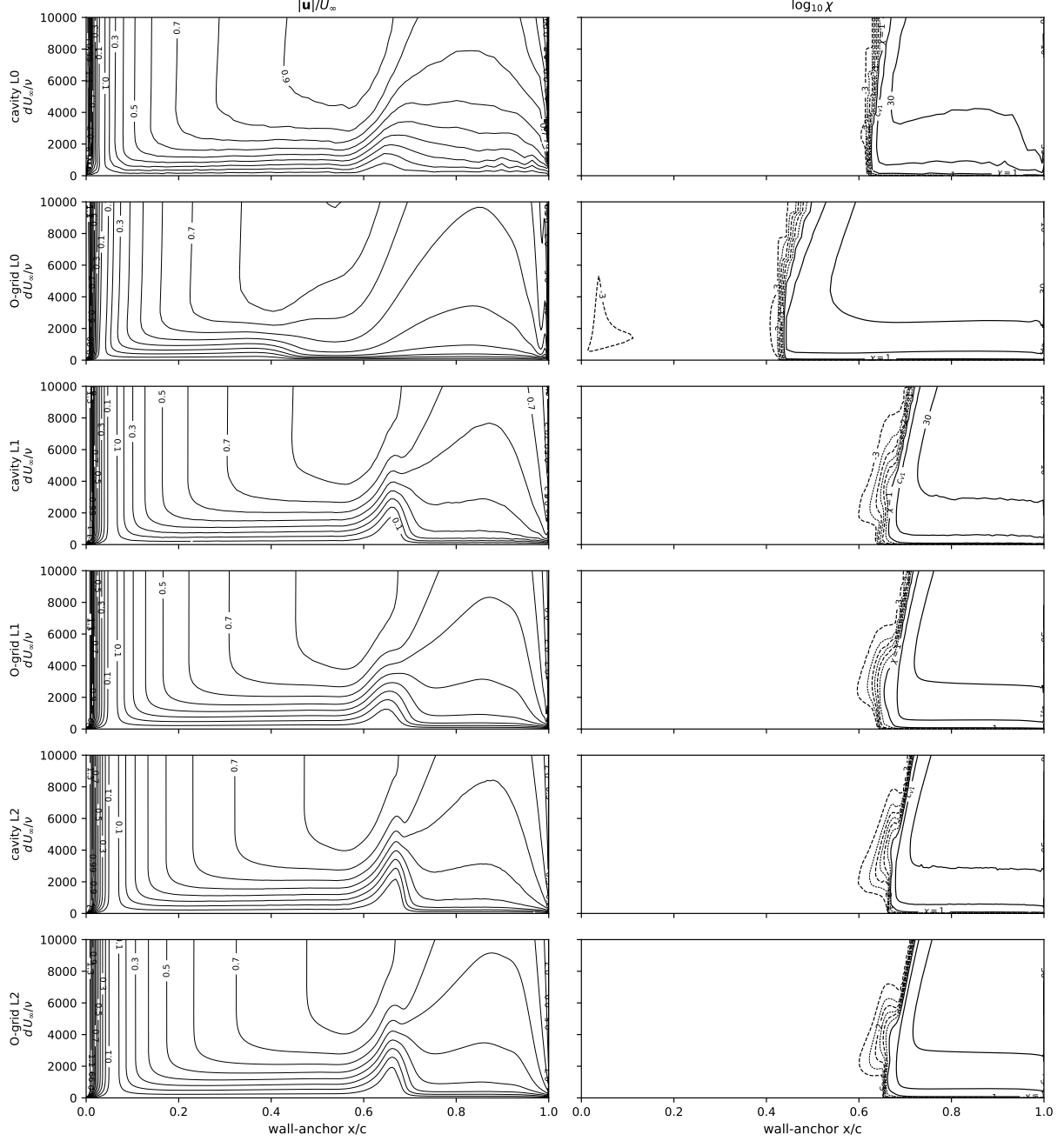












### 2.3 Eppler 387 at $Re = 2 \times 10^5$

$\alpha \in \{0^\circ, 2^\circ, 5^\circ, 7^\circ\}$  on both families at L0–L2 plus six extension incidences ( $-2^\circ, 1^\circ, 3^\circ, 4^\circ, 6^\circ, 8.5^\circ$ ) on the finest pair; same seed. Figure 13: drag polar against the LTPT measurement (McGhee et al., TM-4062), the  $e^9$  reference, fully turbulent SA, and the digitized  $\gamma-Re_\theta$  and SA-BC curves ( $-6.4\%$  to  $+3.7\%$  over  $\alpha = 0^\circ$ – $7^\circ$ ). Figure 14: bubble stations against the oil flow, Selig et al., Cole–Mueller, and the  $e^9$  references—reattachment  $0.02$ – $0.04c$  late at  $\alpha \leq 5^\circ$ , at the edge of bubble collapse at  $7^\circ$ . Figures 15–16: surface suites. Table 5: forces. Table 6: bubble stations. Table 7: at L2 lift within  $0.5\%$ , separation  $0.030$ – $0.037c$  ahead of and reattachment  $0.033$ – $0.045c$  behind the SA-AI stations.

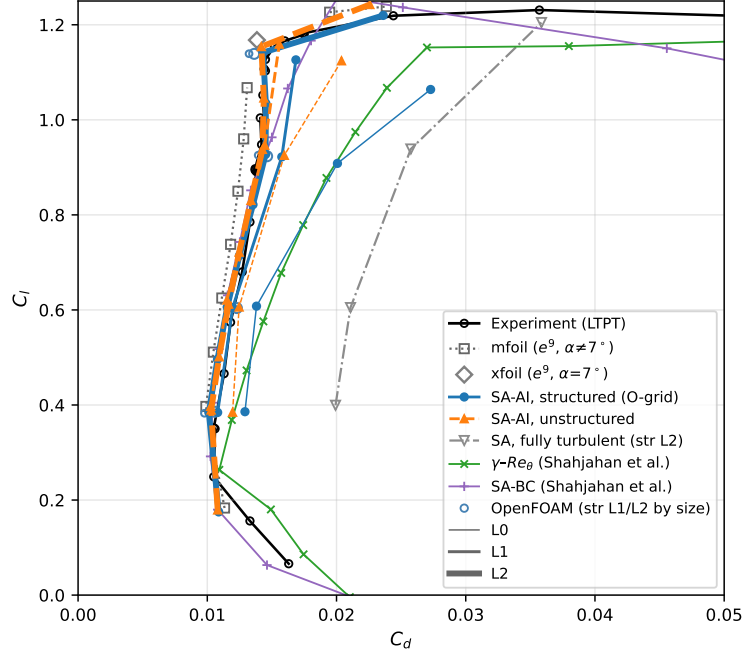


Figure 13: Eppler 387 drag polar,  $Re = 2 \times 10^5$ . Fully turbulent SA ( $\chi_\infty = 3$ ) 77–97% above the SA-AI drag in the bucket, +151% at  $\alpha = 7^\circ$ ; mesh families agree to  $\Delta C_d \leq 2.4 \times 10^{-4}$  over  $\alpha = -2^\circ$ – $7^\circ$  and  $1.0 \times 10^{-3}$  at  $8.5^\circ$ ; OpenFOAM port omits the bypass, worth  $-6$  to  $-10$  counts at  $\alpha = 0$ – $5^\circ$ .

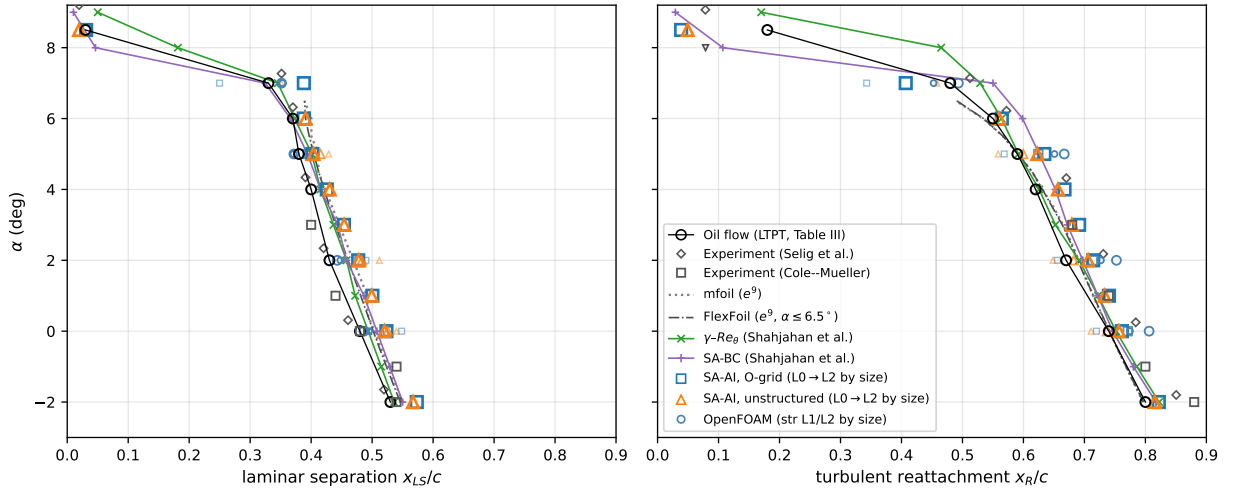


Figure 14: Eppler 387 bubble stations,  $Re = 2 \times 10^5$ ; signed- $C_f$  zero crossings against oil flow (TM-4062 Table III), Selig, Cole–Mueller,  $e^9$ . At  $\alpha = 7^\circ$  XFOIL  $C_l = 1.17$ , mfoil  $C_l = 0.928$ .

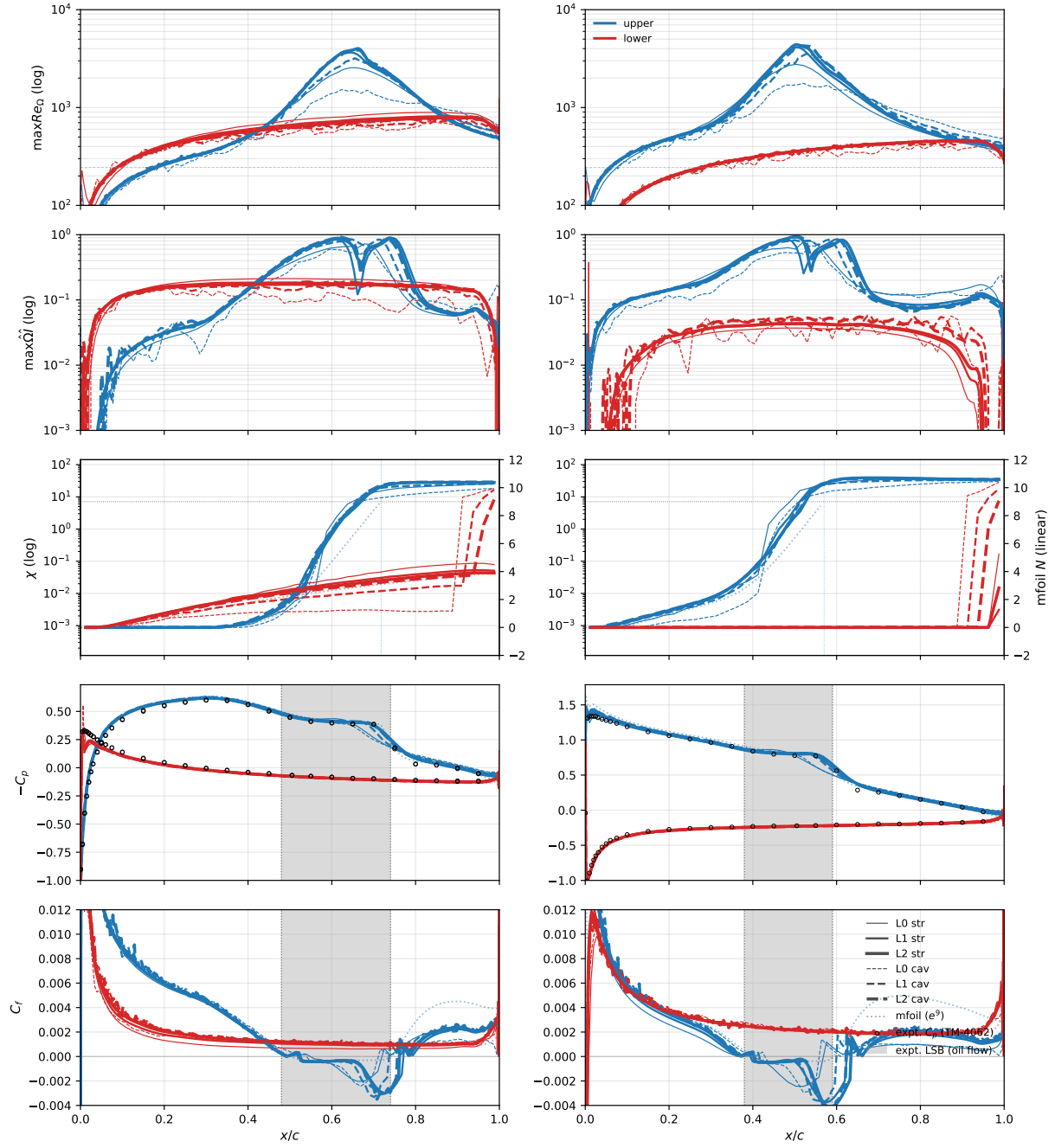


Figure 15: Eppler 387,  $\alpha = 0^\circ, 5^\circ$ ,  $Re = 2 \times 10^5$ .

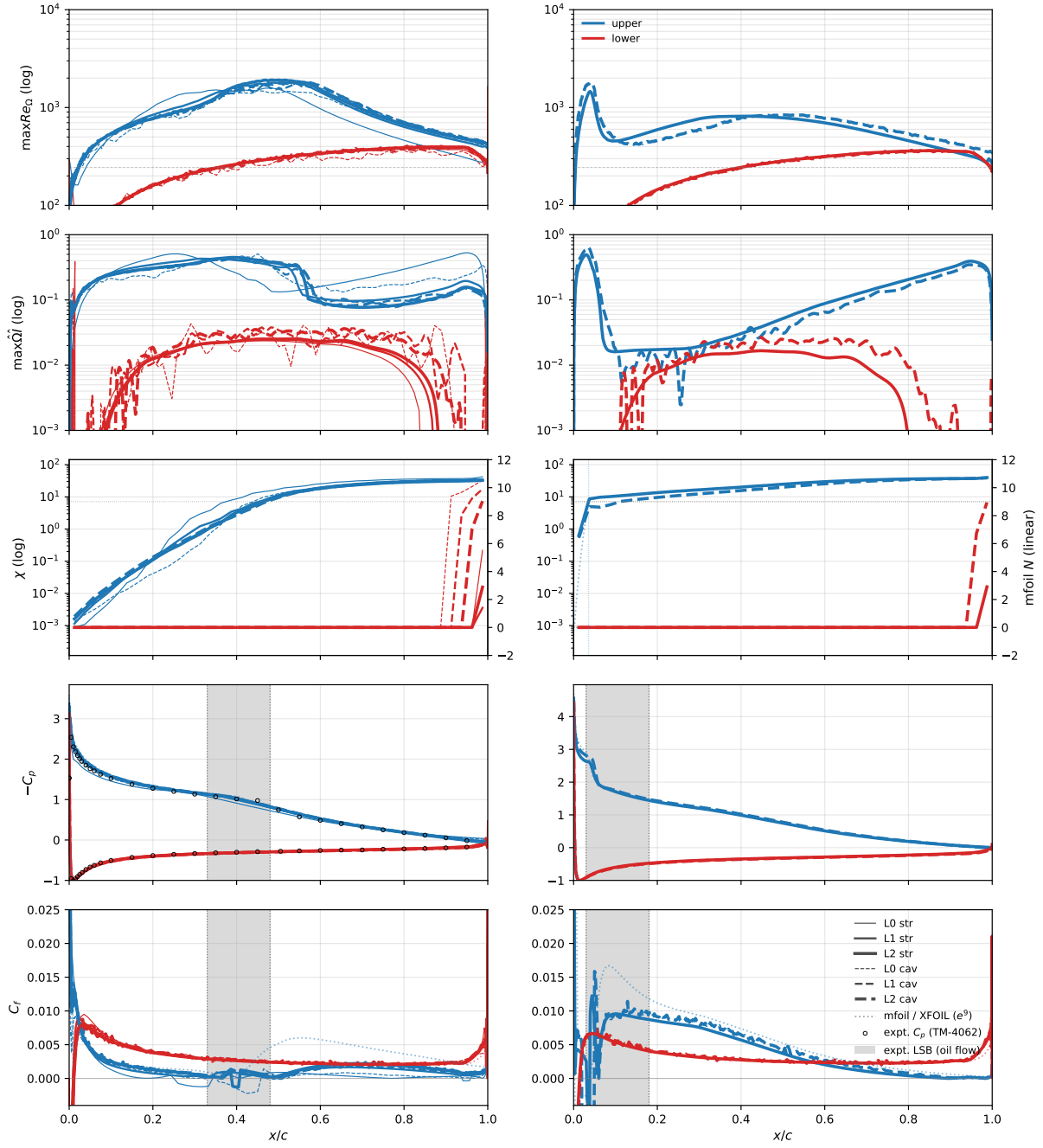


Figure 16: Eppler 387,  $\alpha = 7^\circ, 8.5^\circ$ ,  $Re = 2 \times 10^5$ .

Table 5: Eppler 387,  $Re = 2 \times 10^5$ : computed forces (the  $-2^\circ$ – $8.5^\circ$  extension incidences exist on the finest pair only).

| $\alpha$ | grid | structured<br>$C_L$ | O-grid<br>$C_D$ | unstructured<br>$C_L$ | cavity<br>$C_D$ |
|----------|------|---------------------|-----------------|-----------------------|-----------------|
| −2       | L2   | 0.1752              | 0.01089         | 0.1791                | 0.01080         |
| 0        | L0   | 0.3859              | 0.01292         | 0.3847                | 0.01196         |
| 0        | L1   | 0.3843              | 0.01080         | 0.3969                | 0.01026         |
| 0        | L2   | 0.3844              | 0.01010         | 0.3894                | 0.01030         |
| 1        | L2   | 0.4957              | 0.01078         | 0.5012                | 0.01092         |
| 2        | L0   | 0.6079              | 0.01381         | 0.6061                | 0.01246         |
| 2        | L1   | 0.6048              | 0.01196         | 0.6207                | 0.01152         |
| 2        | L2   | 0.6059              | 0.01154         | 0.6124                | 0.01167         |
| 3        | L2   | 0.7146              | 0.01247         | 0.7218                | 0.01245         |
| 4        | L2   | 0.8218              | 0.01354         | 0.8304                | 0.01341         |
| 5        | L0   | 0.9083              | 0.02007         | 0.9253                | 0.01596         |
| 5        | L1   | 0.9218              | 0.01575         | 0.9465                | 0.01444         |
| 5        | L2   | 0.9274              | 0.01451         | 0.9375                | 0.01428         |
| 6        | L2   | 1.0346              | 0.01453         | 1.0456                | 0.01439         |
| 7        | L0   | 1.0639              | 0.02728         | 1.1241                | 0.02038         |
| 7        | L1   | 1.1262              | 0.01686         | 1.1593                | 0.01557         |
| 7        | L2   | 1.1415              | 0.01431         | 1.1535                | 0.01419         |
| 8.5      | L2   | 1.2190              | 0.02310         | 1.2611                | 0.02150         |

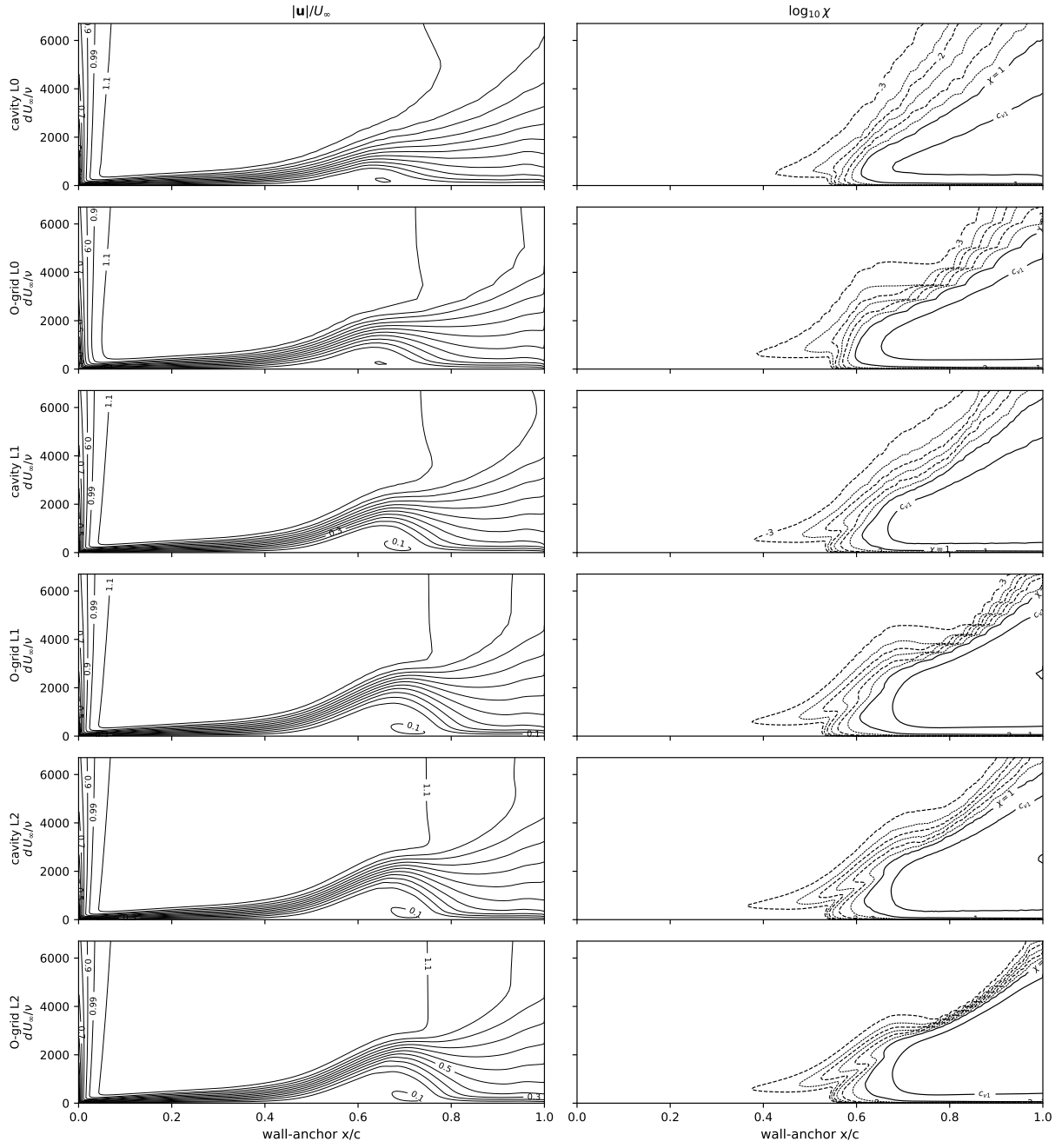
Table 6: Eppler 387,  $Re = 2 \times 10^5$ : computed upper-surface laminar-separation and turbulent-reattachment stations (signed- $C_f$  zero crossings; missing entries transition ahead of separation and have no bubble). The measured oil-flow reattachments (TM-4062 Table III) at  $\alpha = 0^\circ/2^\circ/5^\circ$  are 0.74/0.67/0.59  $c$ .

| $\alpha$ | grid | structured<br>$x_{LS}/c$ | O-grid<br>$x_R/c$ | unstructured<br>$x_{LS}/c$ | cavity<br>$x_R/c$ |
|----------|------|--------------------------|-------------------|----------------------------|-------------------|
| −2       | L2   | 0.573                    | 0.822             | 0.567                      | 0.815             |
| 0        | L0   | 0.549                    | 0.720             | 0.540                      | 0.711             |
| 0        | L1   | 0.526                    | 0.771             | 0.526                      | 0.737             |
| 0        | L2   | 0.523                    | 0.761             | 0.520                      | 0.757             |
| 1        | L2   | 0.500                    | 0.740             | 0.499                      | 0.733             |
| 2        | L0   | 0.490                    | 0.656             | 0.512                      | 0.649             |
| 2        | L1   | 0.479                    | 0.713             | 0.483                      | 0.682             |
| 2        | L2   | 0.477                    | 0.714             | 0.478                      | 0.706             |
| 3        | L2   | 0.454                    | 0.691             | 0.454                      | 0.679             |
| 4        | L2   | 0.426                    | 0.667             | 0.430                      | 0.657             |
| 5        | L0   | 0.393                    | 0.568             | 0.429                      | 0.558             |
| 5        | L1   | 0.394                    | 0.624             | 0.415                      | 0.600             |
| 5        | L2   | 0.402                    | 0.634             | 0.404                      | 0.623             |
| 6        | L2   | 0.388                    | 0.565             | 0.391                      | 0.560             |
| 7        | L0   | 0.250                    | 0.343             | 0.351                      | 0.458             |
| 7        | L1   | —                        | —                 | —                          | —                 |
| 7        | L2   | 0.388                    | 0.407             | —                          | —                 |
| 8.5      | L2   | 0.031                    | 0.039             | 0.019                      | 0.049             |

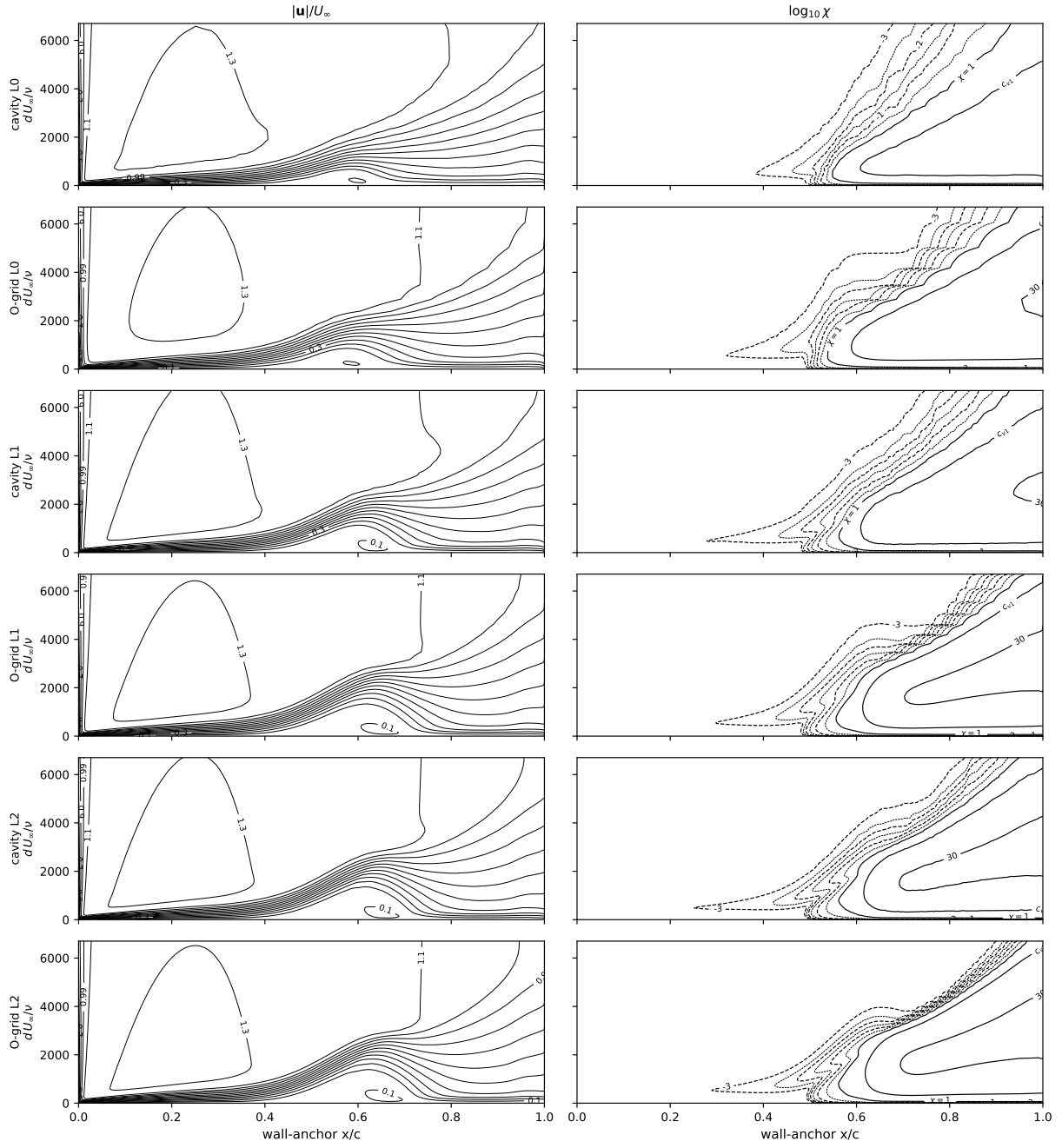
Table 7: Eppler 387: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids (fronts at the near-wall  $\chi = 1$  convention, stations at the signed- $C_f$  zero crossings; the port omits the  $f_{v1}$  bypass).

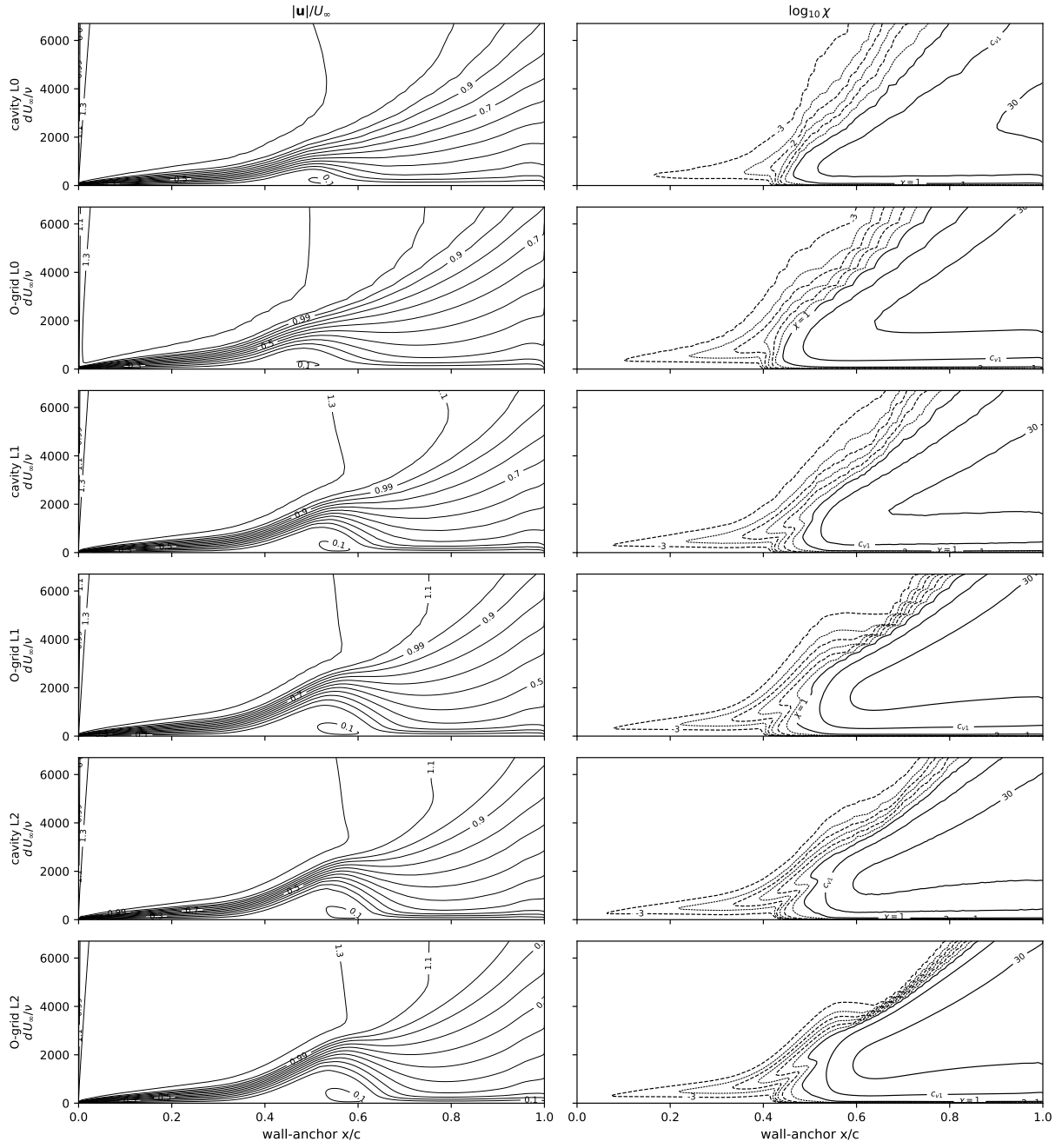
| $\alpha$ | grid | $C_L$  | $C_D$   | $x_{tr}^{up}$ | $x_{LS}$ | $x_R$ |
|----------|------|--------|---------|---------------|----------|-------|
| 0        | L0   | 0.3887 | 0.01167 | 0.638         | 0.493    | 0.784 |
| 0        | L1   | 0.3826 | 0.00978 | 0.605         | 0.495    | 0.775 |
| 0        | L2   | 0.3857 | 0.01058 | 0.621         | 0.486    | 0.806 |
| 2        | L0   | 0.6133 | 0.01406 | 0.568         | 0.447    | 0.739 |
| 2        | L1   | 0.6037 | 0.01174 | 0.543         | 0.448    | 0.727 |
| 2        | L2   | 0.6039 | 0.01229 | 0.564         | 0.443    | 0.752 |
| 5        | L0   | 0.9292 | 0.01665 | 0.457         | 0.344    | 0.654 |
| 5        | L1   | 0.9256 | 0.01394 | 0.468         | 0.371    | 0.651 |
| 5        | L2   | 0.9234 | 0.01463 | 0.487         | 0.372    | 0.667 |
| 7        | L0   | 1.1173 | 0.01817 | 0.139         | —        | 0.201 |
| 7        | L1   | 1.1391 | 0.01323 | 0.329         | 0.353    | 0.453 |
| 7        | L2   | 1.1388 | 0.01363 | 0.379         | 0.351    | 0.493 |

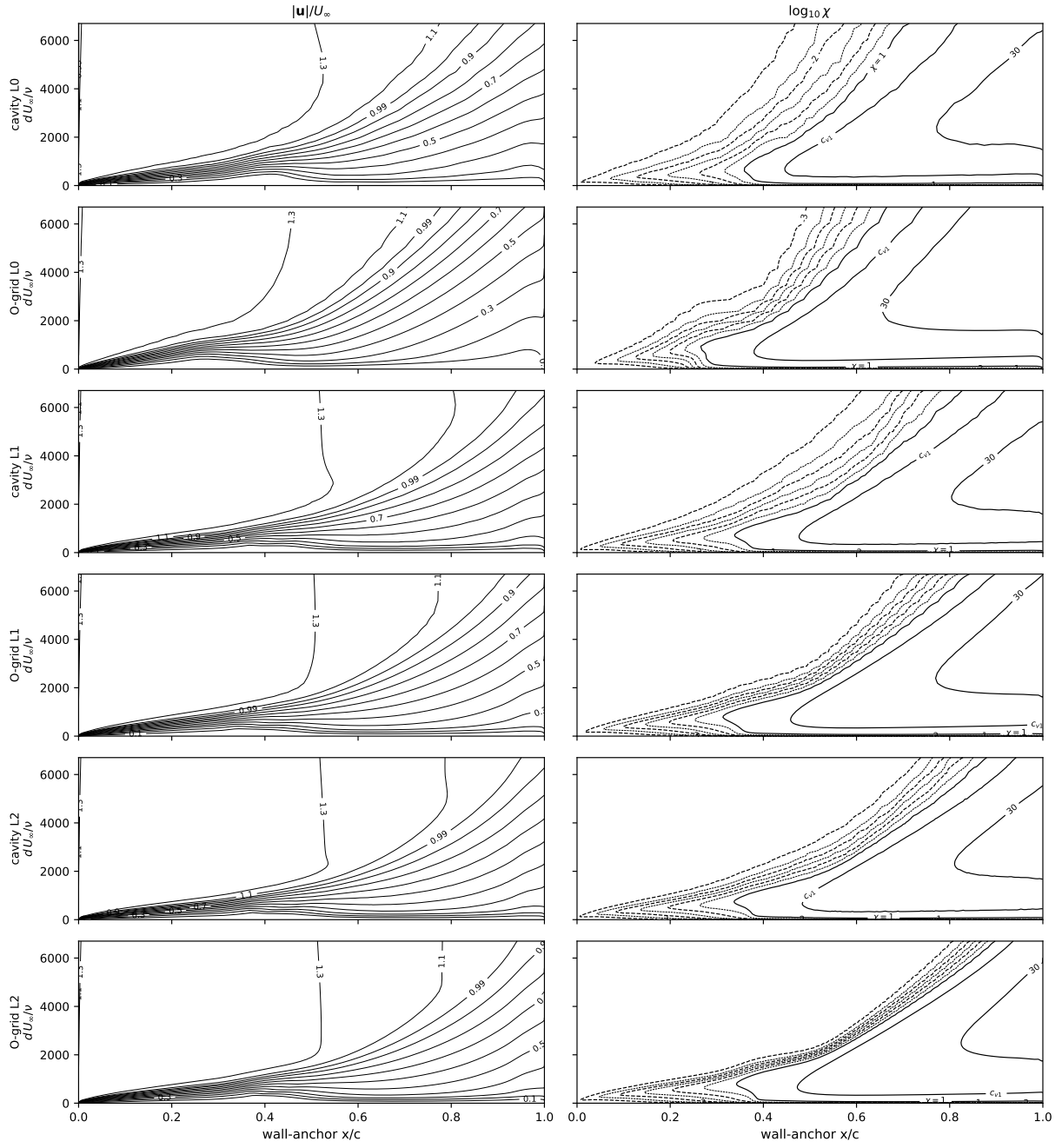


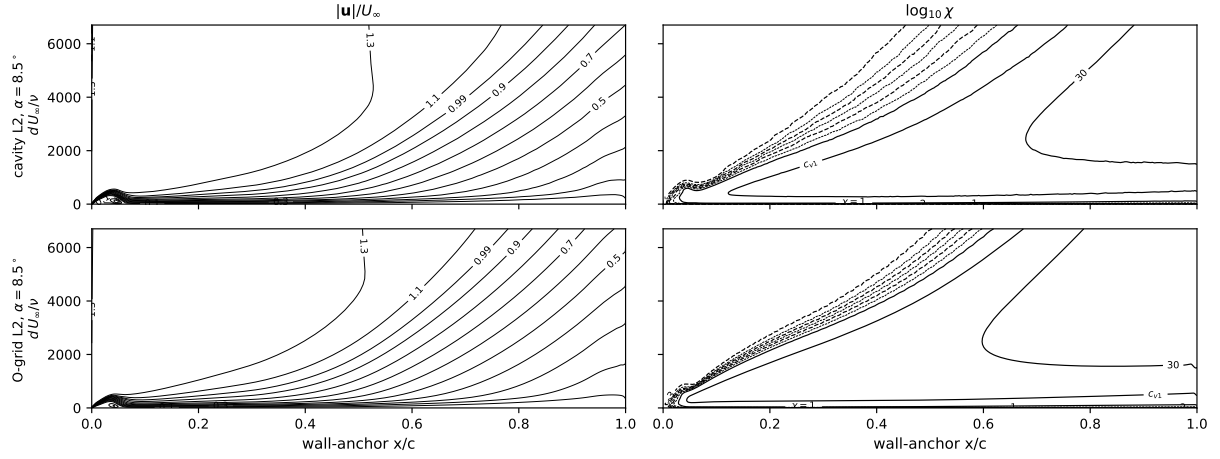












## 2.4 Eppler 387 Reynolds sweep at $\alpha = 5^\circ$

$Re \in \{6 \times 10^4, 10^5, 3 \times 10^5, 4.6 \times 10^5\}$  on the same grids (24 further solutions). Figure 17: the refinement fan closes onto the measurement at  $2\text{--}4.6 \times 10^5$  and opens at  $10^5$ , the model's bursting boundary, one Reynolds step above the measurement's ( $6 \times 10^4$ ). Figures 18–19: surface suites. Table 8: L2 coefficients.

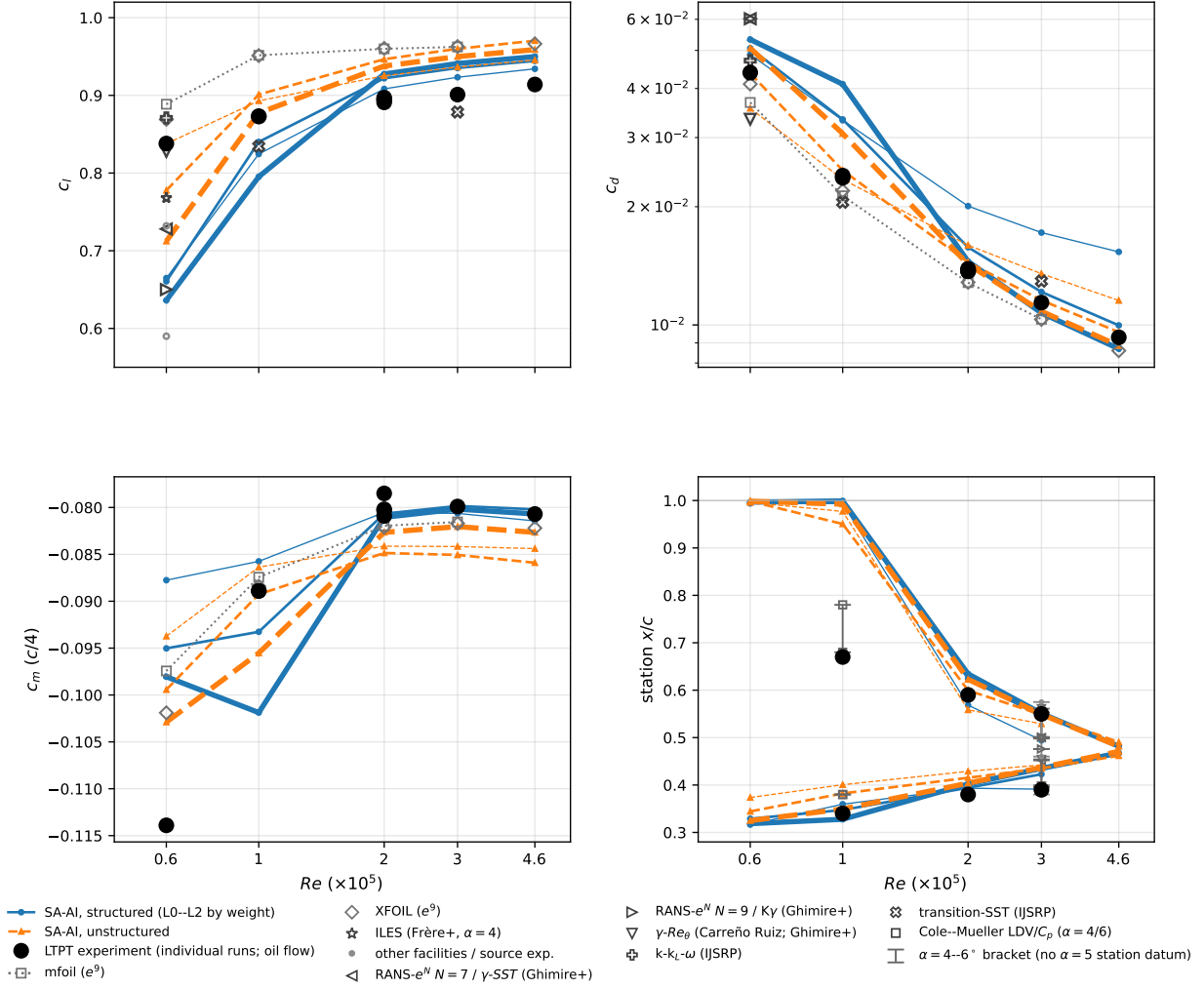


Figure 17: Reynolds-sweep forces and stations,  $\alpha = 5^\circ$ ,  $Re = 6 \times 10^4 - 4.6 \times 10^5$ . LTPT runs 1/2/4/1/1 at  $0.6/1/2/3/4.6 \times 10^5$ .

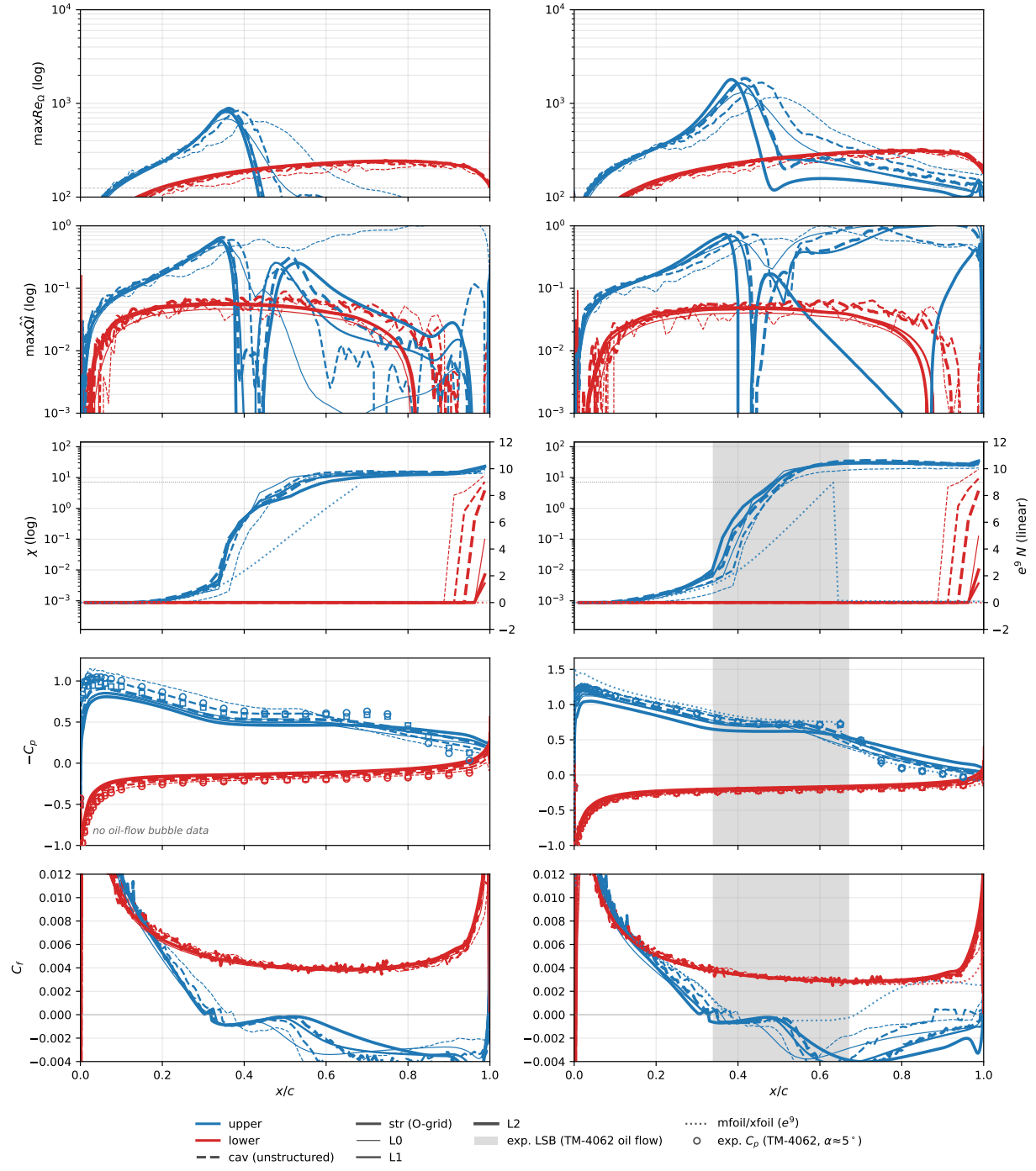


Figure 18: Reynolds sweep,  $Re = 6 \times 10^4, 10^5$ .

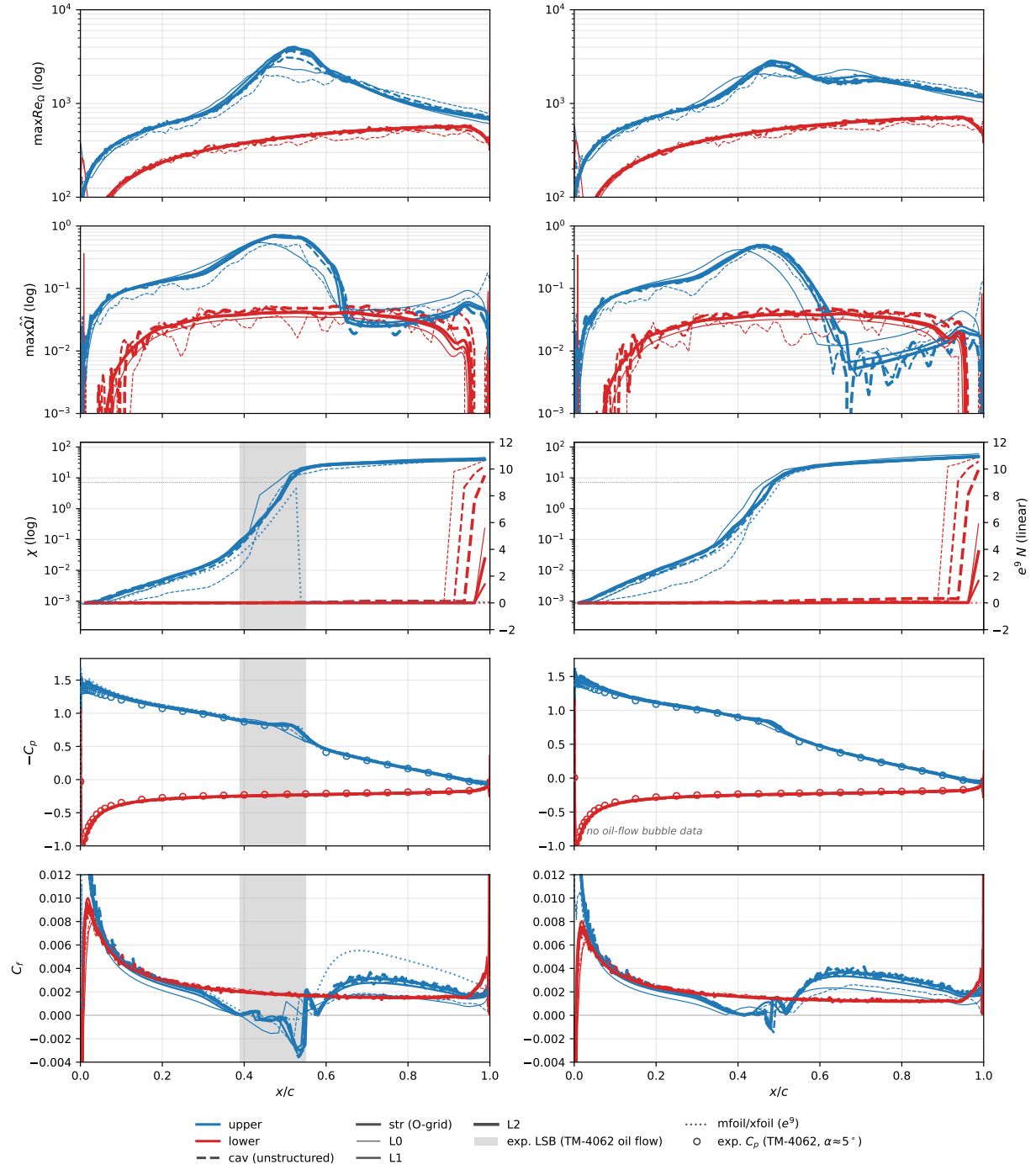


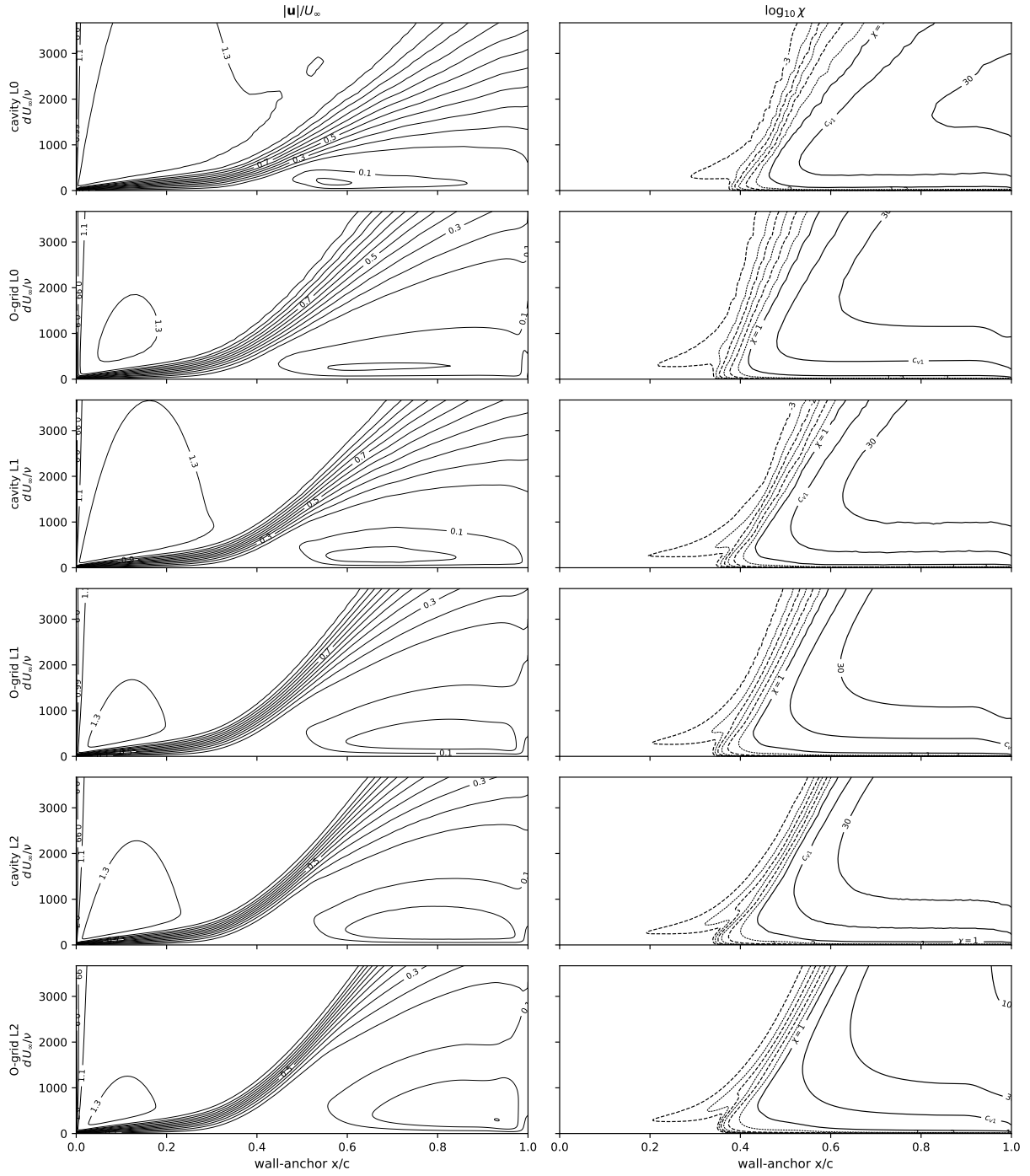
Figure 19: Reynolds sweep,  $Re = 3, 4.6 \times 10^5$ .

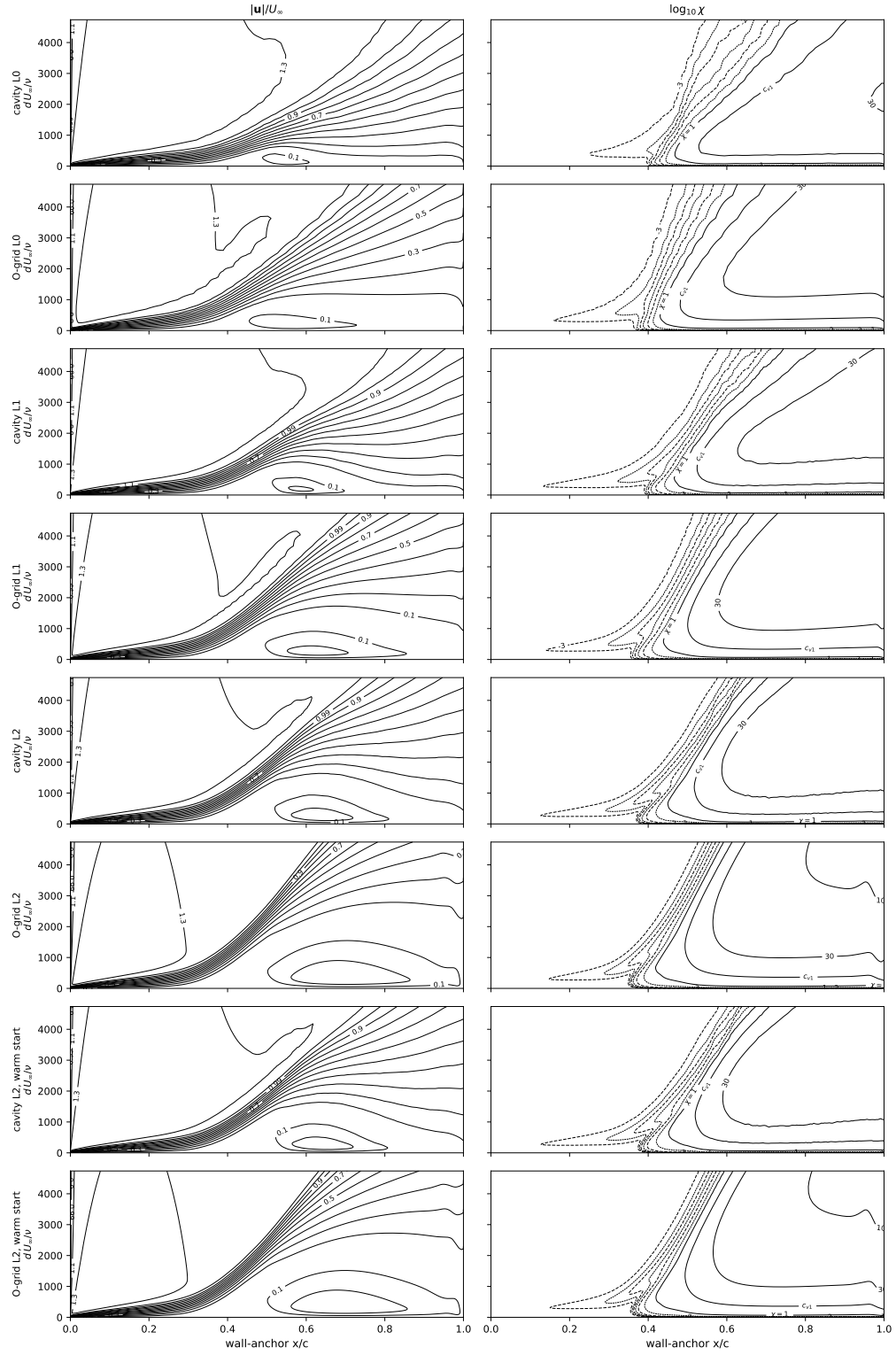
Table 8: Eppler 387 at  $\alpha = 5^\circ$  across the Reynolds sweep: L2 section coefficients against the  $e^9$  panel reference and the LTPT measurement (read at the tabulated angle nearest  $5^\circ$ ). Computed moments transferred to the quarter chord.

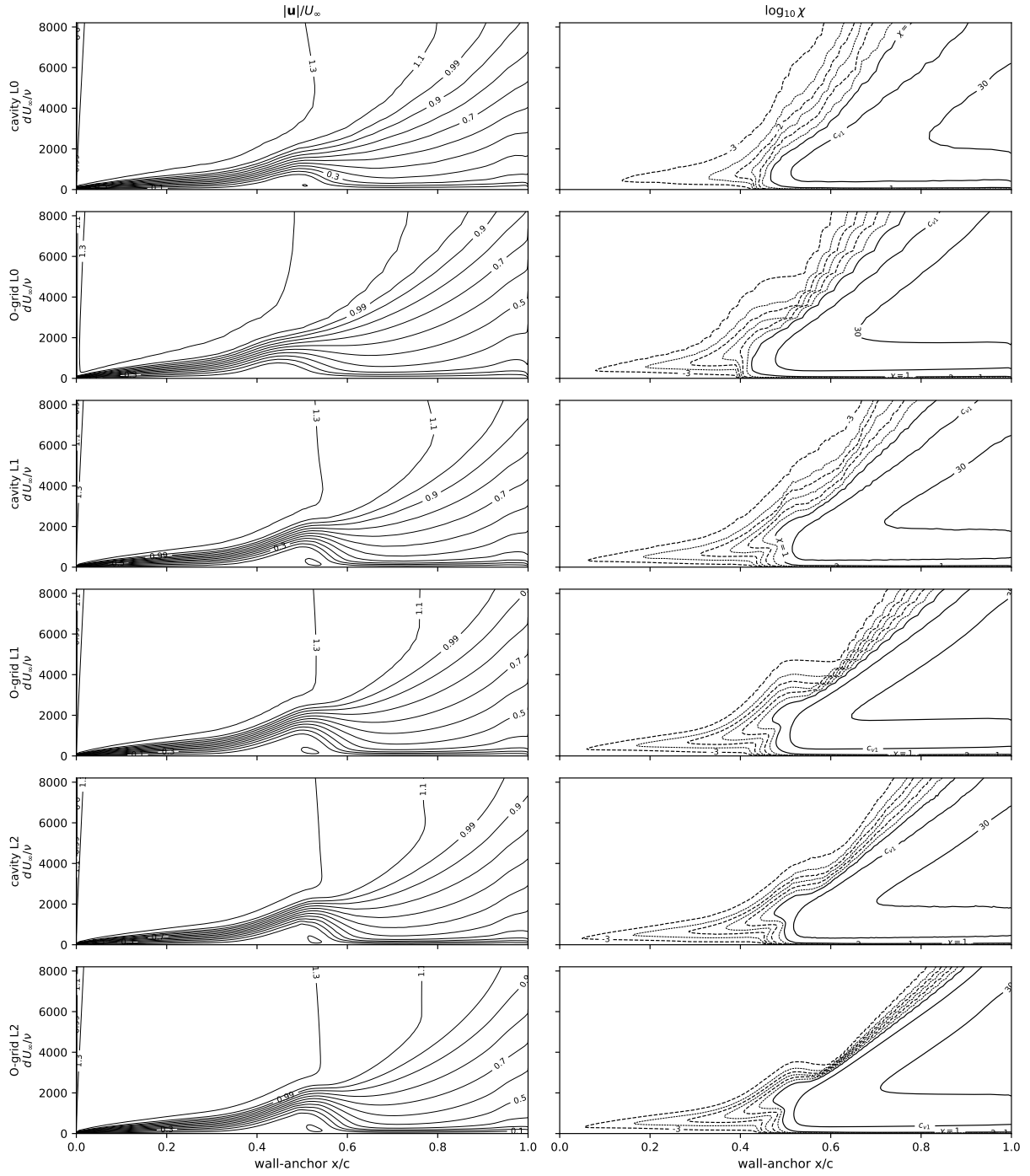
| $Re$              | SA-AI, structured L2 |        |        | SA-AI, unstructured L2 |        |        | $e^9$ panel        |                     |                     | Exp (LTPT) |        |        |
|-------------------|----------------------|--------|--------|------------------------|--------|--------|--------------------|---------------------|---------------------|------------|--------|--------|
|                   | $c_l$                | $c_d$  | $c_m$  | $c_l$                  | $c_d$  | $c_m$  | $c_l$              | $c_d$               | $c_m$               | $c_l$      | $c_d$  | $c_m$  |
| $6 \times 10^4$   | 0.636                | 0.0533 | -0.098 | 0.712                  | 0.0506 | -0.103 | 0.869 <sup>‡</sup> | 0.0411 <sup>‡</sup> | -0.102 <sup>‡</sup> | 0.838      | 0.0439 | -0.114 |
| $1 \times 10^5$   | 0.795                | 0.0410 | -0.102 | 0.877                  | 0.0307 | -0.096 | 0.952              | 0.0213              | -0.087              | 0.873      | 0.0237 | -0.089 |
| $2 \times 10^5$   | 0.927                | 0.0145 | -0.081 | 0.938                  | 0.0143 | -0.083 | 0.960              | 0.0128              | -0.082              | 0.891      | 0.0138 | -0.081 |
| $3 \times 10^5$   | 0.941                | 0.0107 | -0.080 | 0.950                  | 0.0108 | -0.082 | 0.962              | 0.0103              | -0.082              | 0.901      | 0.0114 | -0.080 |
| $4.6 \times 10^5$ | 0.950                | 0.0087 | -0.081 | 0.959                  | 0.0089 | -0.083 | 0.966              | 0.0086              | -0.082              | 0.914      | 0.0093 | -0.081 |

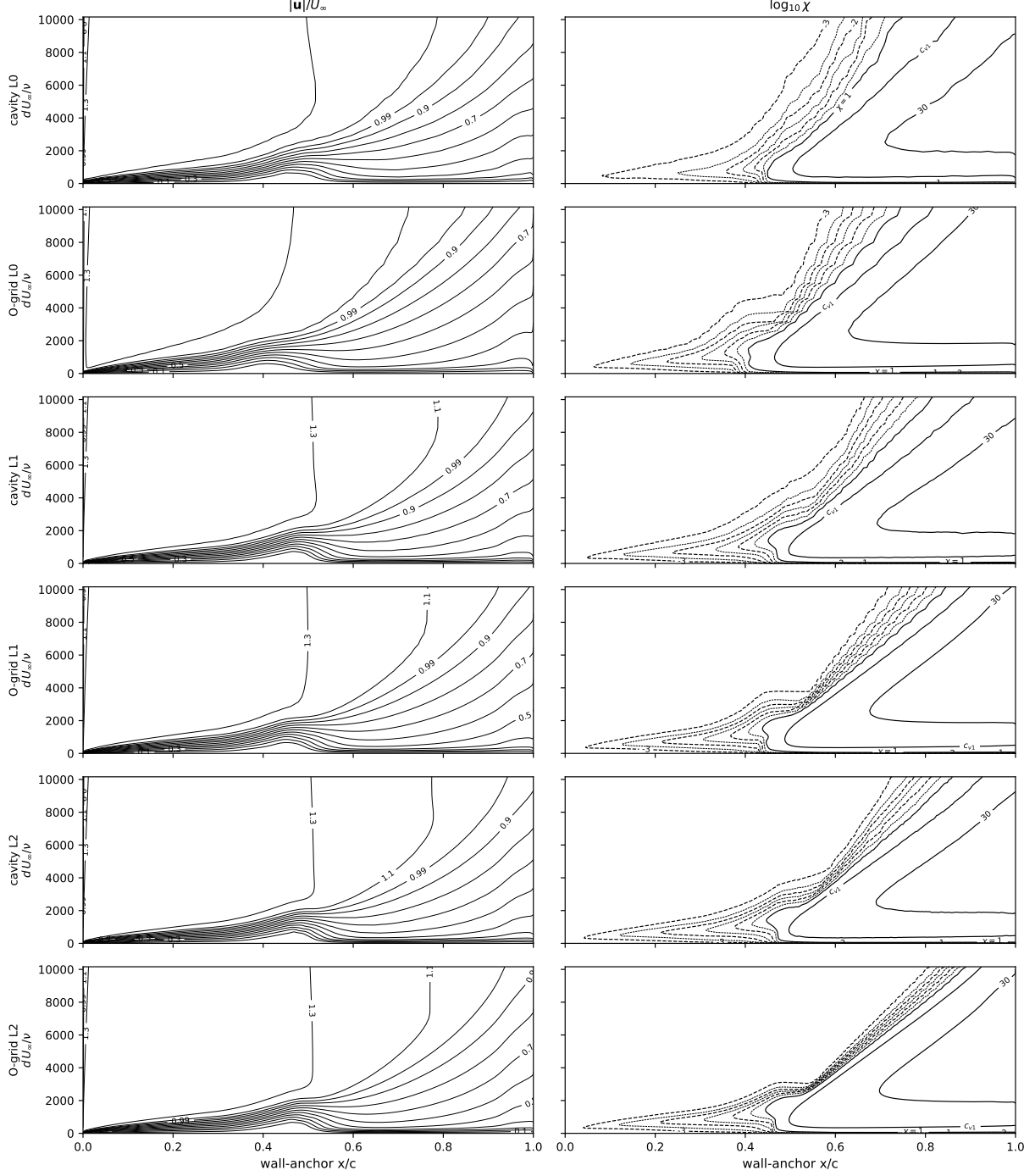
<sup>‡</sup>XFOIL, whose forces sit closest to the measurement, is the reference at this bistable condition; mfoil gives  $c_l = 0.889$ ,  $c_d = 0.0368$ ,  $c_m = -0.097$ , and FlexFoil sides with XFOIL ( $c_l = 0.868$ ,  $c_d = 0.0406$ ,  $c_m = -0.102$ ).











## 2.5 Cylinder drag crisis, $Re_D = 1\text{--}10^{10}$

A steady-branch traverse over ten decades at three seeds ( $\chi_\infty = 3.9 \times 10^{-4}$ ,  $1.1 \times 10^{-2}$ ,  $0.22$ ; Mack's map at  $Tu = 0.05, 0.2, 0.7\%$ ), each seed swept as upward and downward continuation ladders on matched quasi-two-dimensional O-grids ( $y^+ \lesssim 1$ ; a large-domain laminar grid for  $Re_D \lesssim 10^3$ ,  $1200 \times 148$  for the matrix, and  $1600 \times 170$  fine grids for the supercritical and ultra range). Every case lands on the symmetric branch,  $|C_L| < 2 \times 10^{-7}$ .

Figure 20: the crisis moves from  $Re_D \approx 5-7 \times 10^5$  at the two quiet seeds to  $\approx 3-4 \times 10^5$  at  $Tu = 0.7\%$ ; the supercritical floor ( $C_d = 0.217-0.233$  at  $2 \times 10^6$ ) rises with the seed. Figure 21: transition and separation angles. Figure 22: fields across the traverse. Table 9: the matrix  $C_d$ . Below shedding onset ( $Re_D \approx 46$ ) the steady solution is physical:  $C_d = 10.67$  at  $Re_D = 1$ , 2.797 at 10 ( $-1.7\%$  vs Dennis & Chang).

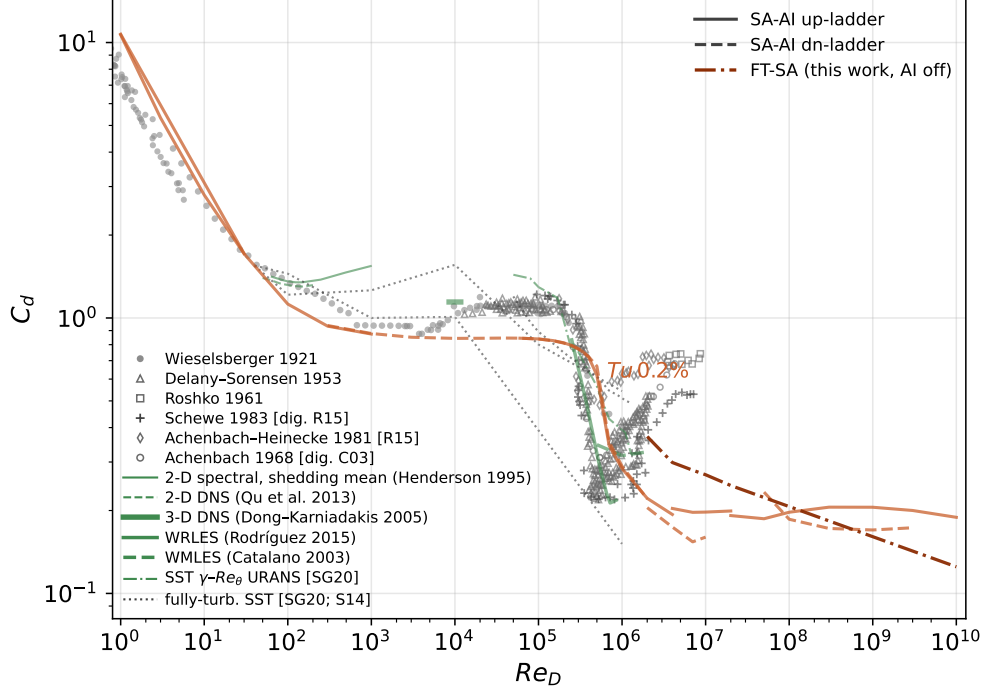


Figure 20: Cylinder drag crisis,  $Re_D = 1-10^{10}$ ; middle seed  $Tu = 0.2\%$ , up (solid)/down (dashed) ladders.

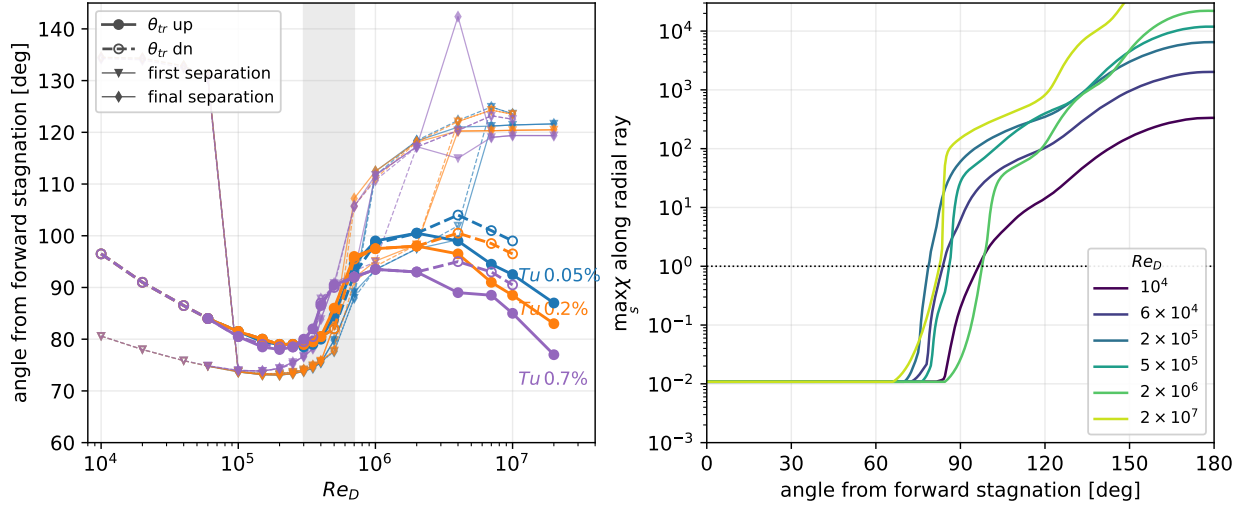


Figure 21: Transition and separation angles; three seeds  $Tu = 0.05, 0.2, 0.7\%$ .

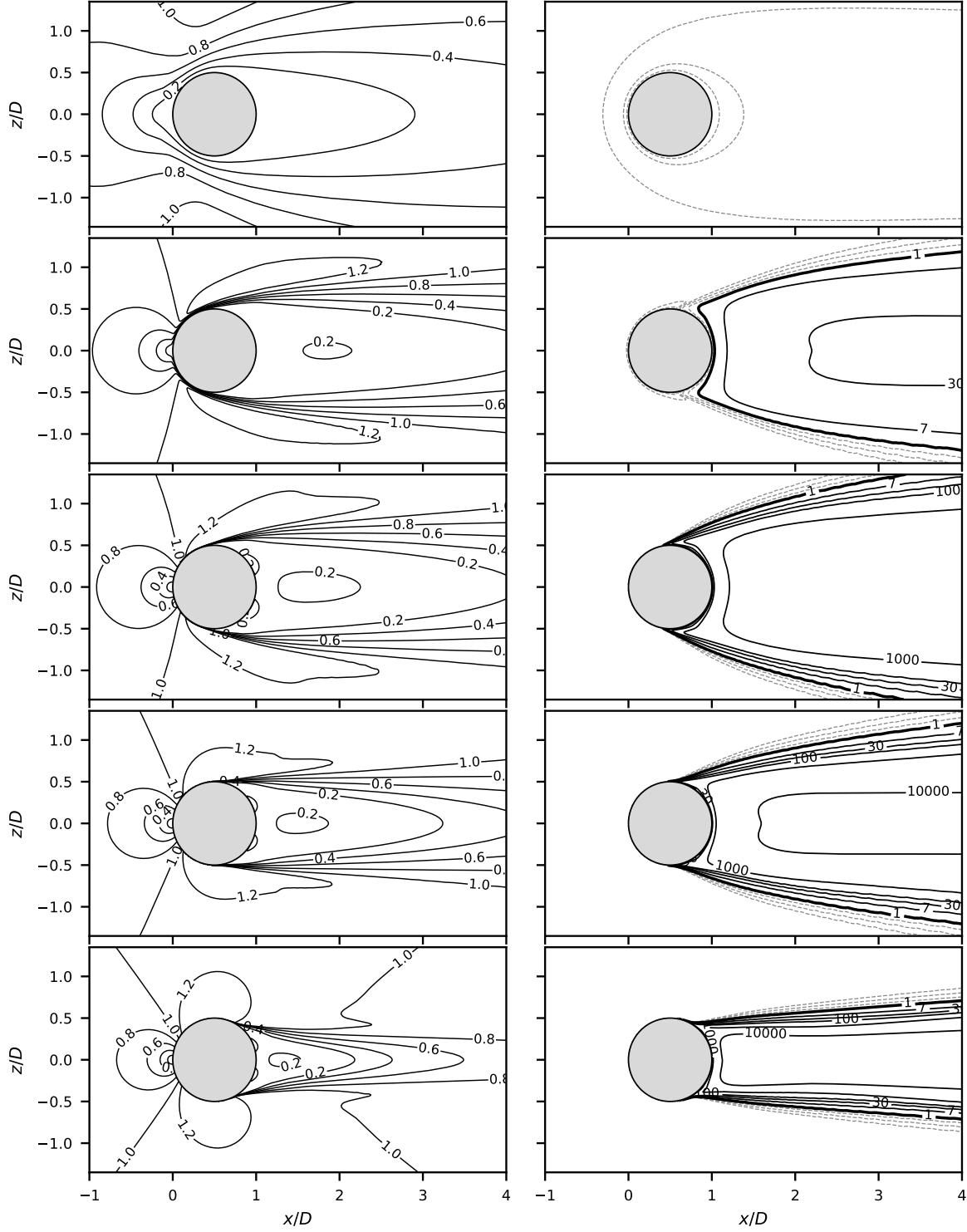


Figure 22: Drag-crisis fields,  $Re_D = 10, 10^3, 10^5, 5 \times 10^5, 2 \times 10^6$  (rows, up-ladder  $Tu = 0.2\%$ ); columns  $|\mathbf{u}|/U_\infty$  and  $\chi$  (dashed  $\chi < 1$ , solid  $\chi = 1$ ,  $c_{v1}$ , 30,  $10^2$ – $10^4$ ).  $C_d = 2.80, 0.87, 0.84, 0.62, 0.22$ .

Table 9: Cylinder steady drag-crisis matrix:  $C_d$  (median over the final-stage tail window) for the 84 matrix cases. Bracketed values: tail-window  $C_d$  peak-to-peak for cases flagged as steady limit cycles;  $|C_L| < 2 \times 10^{-7}$  in all 84 cases.

| $Re_D$<br>( $\times 10^5$ ) | $Tu$ 0.05%    |               | $Tu$ 0.2%     |               |               | $Tu$ 0.7%     |               |
|-----------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                             | up            | dn            | up            | dn            | cold          | up            | dn            |
| 0.6                         | 0.845         | 0.845         | 0.845         | 0.845         | 0.845         | 0.842         | 0.842         |
| 1                           | 0.840         | 0.839         | 0.839         | 0.839         | 0.839         | 0.831         | 0.832         |
| 1.5                         | 0.828         | 0.827         | 0.828         | 0.827         | 0.827         | 0.809         | 0.812         |
| 2                           | 0.814         | 0.812         | 0.813         | 0.811         | 0.811         | 0.781         | 0.786         |
| 2.5                         | 0.796         | 0.794         | 0.795         | 0.793         | 0.793         | 0.745         | 0.755         |
| 3                           | 0.776         | 0.774         | 0.774         | 0.771         | 0.772         | 0.677 [0.075] | 0.709         |
| 3.5                         | 0.753         | 0.750         | 0.749         | 0.748         | 0.749         | 0.616 [0.199] | 0.671 [0.030] |
| 4                           | 0.725         | 0.726         | 0.719         | 0.722         | 0.724         | 0.495 [0.122] | 0.547 [0.086] |
| 5                           | 0.644 [0.076] | 0.670 [0.036] | 0.620 [0.129] | 0.668 [0.047] | 0.650 [0.136] | 0.430 [0.008] | 0.439         |
| 7                           | 0.409 [0.051] | 0.407 [0.024] | 0.344 [0.002] | 0.355         | 0.347 [0.003] | 0.365 [0.006] | 0.356         |
| 10                          | 0.286 [0.006] | 0.285         | 0.284 [0.005] | 0.290         | 0.282 [0.004] | 0.294 [0.007] | 0.303         |
| 20                          | 0.219 [0.013] | 0.217 [0.008] | 0.221 [0.011] | 0.220 [0.008] | 0.220 [0.008] | 0.233 [0.010] | 0.232 [0.005] |

## 2.6 Daedalus wing (three-dimensional deployment)

The Daedalus human-powered-aircraft wing: half-span 17.07 m, root chord 0.917 m, aspect ratio 37.8, DAE-11/21/31 sections;  $Re_{root} = 5 \times 10^5$ ,  $\alpha \in \{4^\circ, 5^\circ, 6^\circ\}$  ( $C_L \approx 1.02$ – $1.23$ ), flight-quiet seed  $\chi_\infty = 8.76 \times 10^{-6}$  ( $N_{crit} \approx 13.6$ ); two mesh families (spanwise-stacked structured O-grid, 0.72–36.7M nodes; unstructured prism/tet/hex, 1.70–54.9M nodes) at three levels, eighteen solutions; the reference is AVL with strip-wise XFOIL section polars at matched local  $c_l$  and  $N_{crit}$ , trimmed to the RANS lift. Figure 23: the families agree at L2 to 0.35–0.63% in  $C_L$  and 0.7–1.7 counts in  $C_D$ , and the finest-grid drag runs 15.6/10.3/5.6 counts (7.3/4.4/2.2%) below the matched-lift reference at  $\alpha = 4^\circ/5^\circ/6^\circ$ . Figure 24: surface maps at  $\alpha = 4^\circ$  against the  $e^N$  strips—at  $\eta = 0.31$ , separation within  $0.017c$  and reattachment within  $0.009c$  at every incidence, the  $\chi = 1$  front sits  $\sim 0.09c$  ahead of the strips’  $N_{crit}$  contour, and the tip vortex trips transition to the leading edge for  $\eta \gtrsim 0.992$ . Table 10: totals by level ( $\alpha = 5^\circ, 6^\circ$  surface maps and section sheets below).

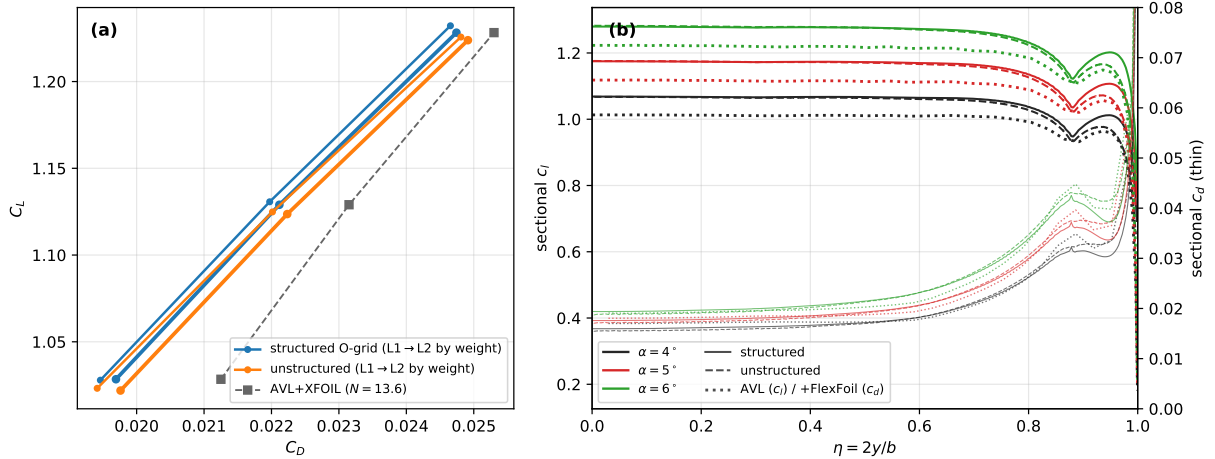


Figure 23: Daedalus polar and sections. Families agree at L1 to 0.46–0.53% in  $C_L$  and 0.4–1.6 counts in  $C_D$ , at L0 to 0.74–0.97% and 18–21 counts; taper break at  $\eta = 0.88$ .

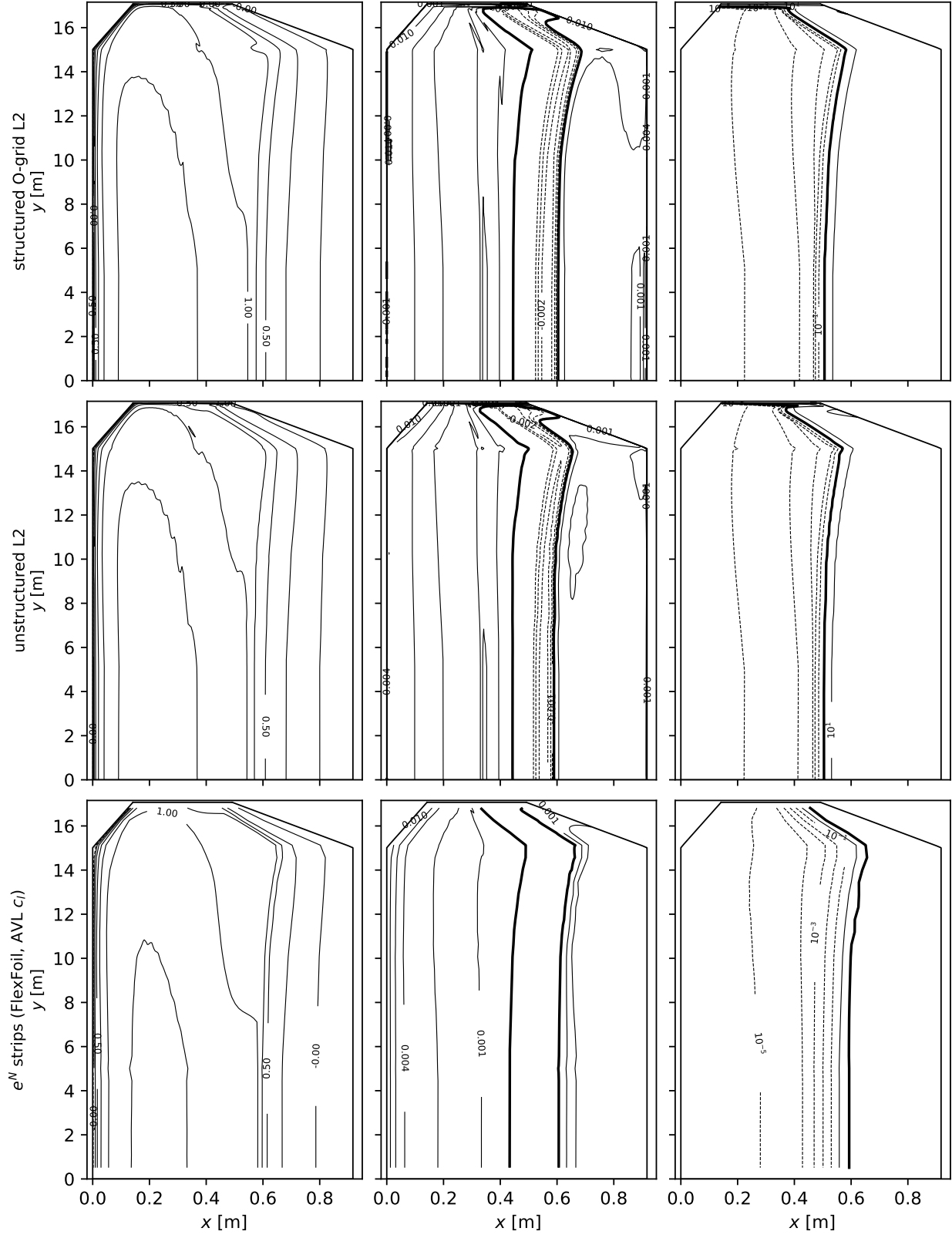
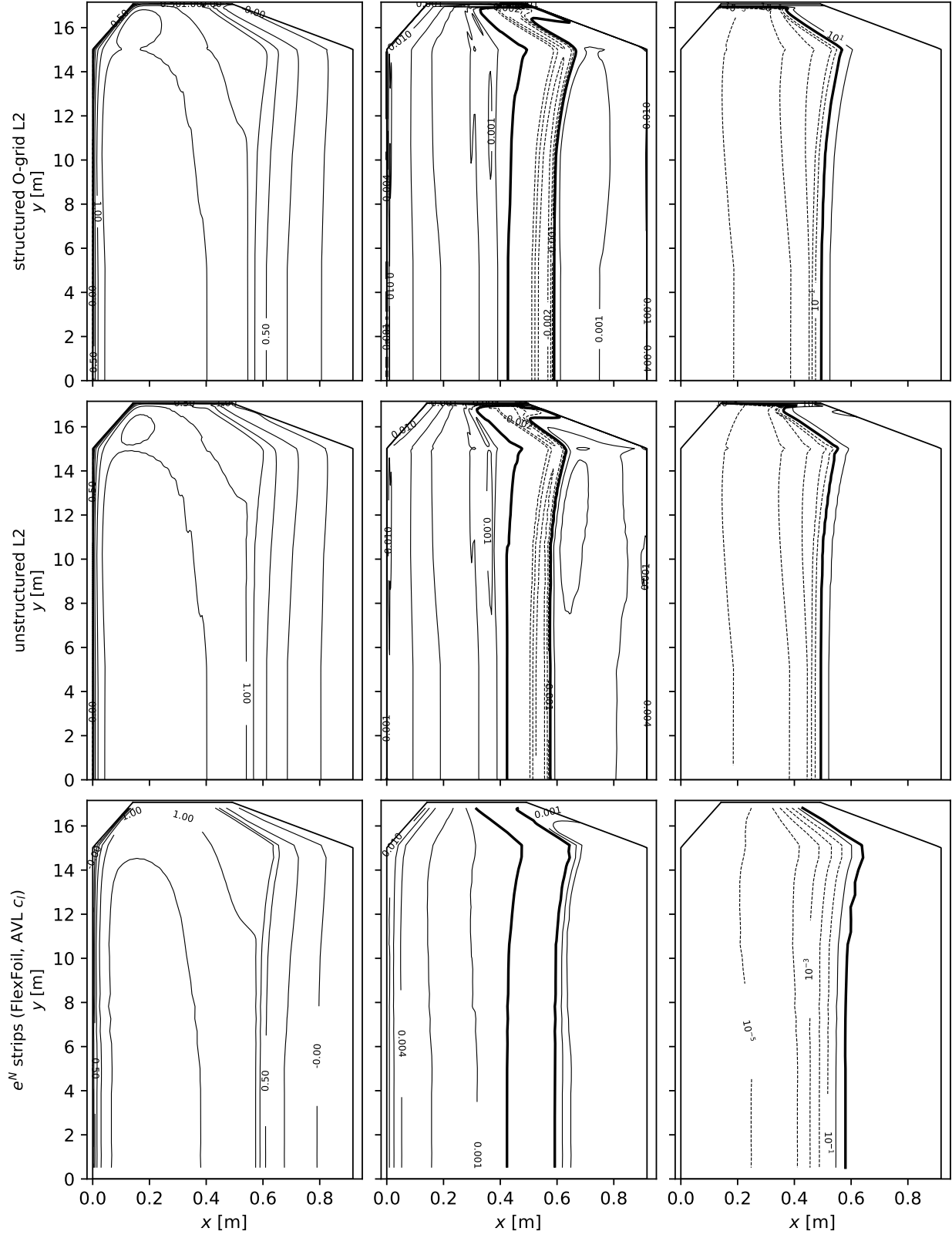


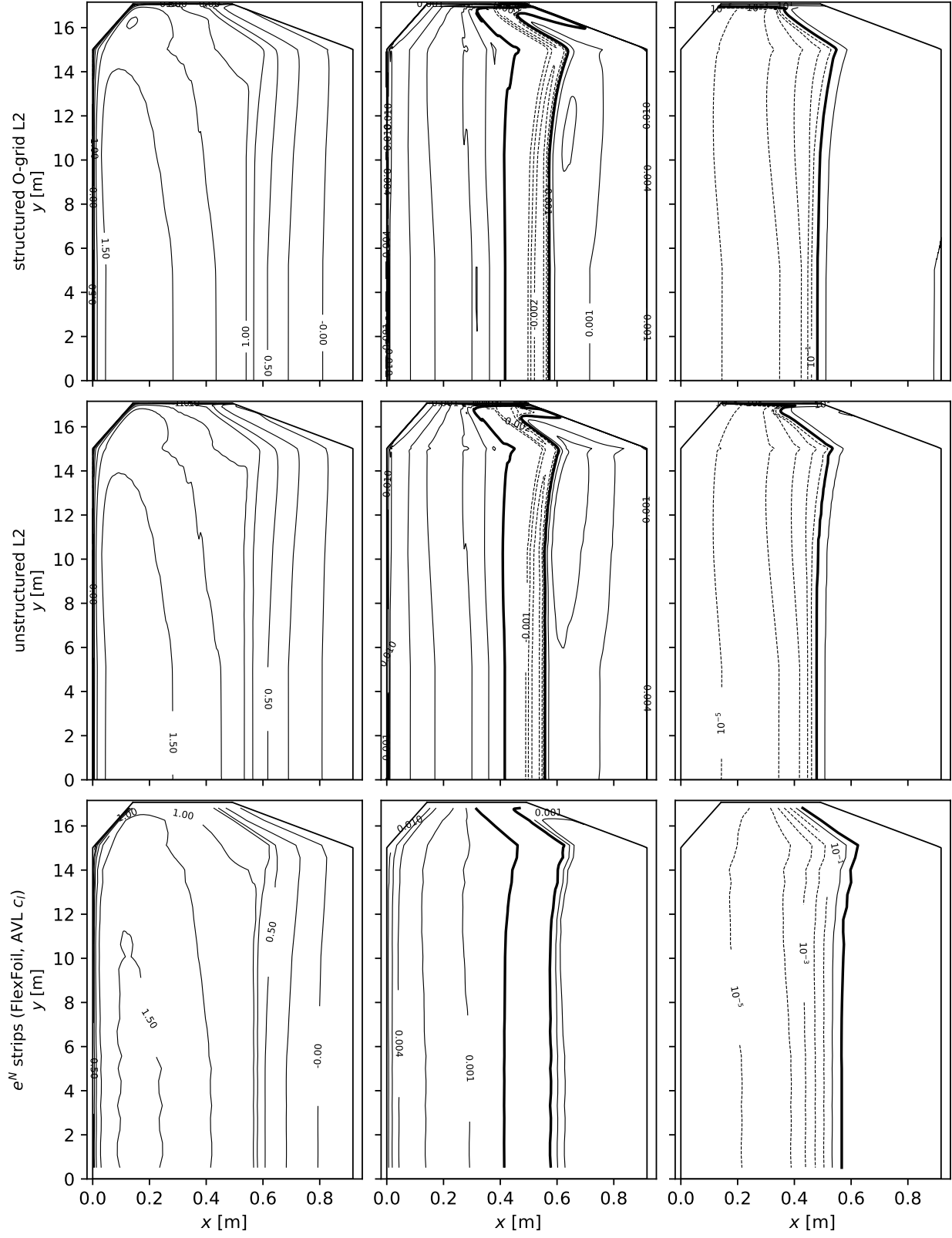
Figure 24: Daedalus surface maps,  $\alpha = 4^\circ$ ,  $Re_{\text{root}} = 5 \times 10^5$ .

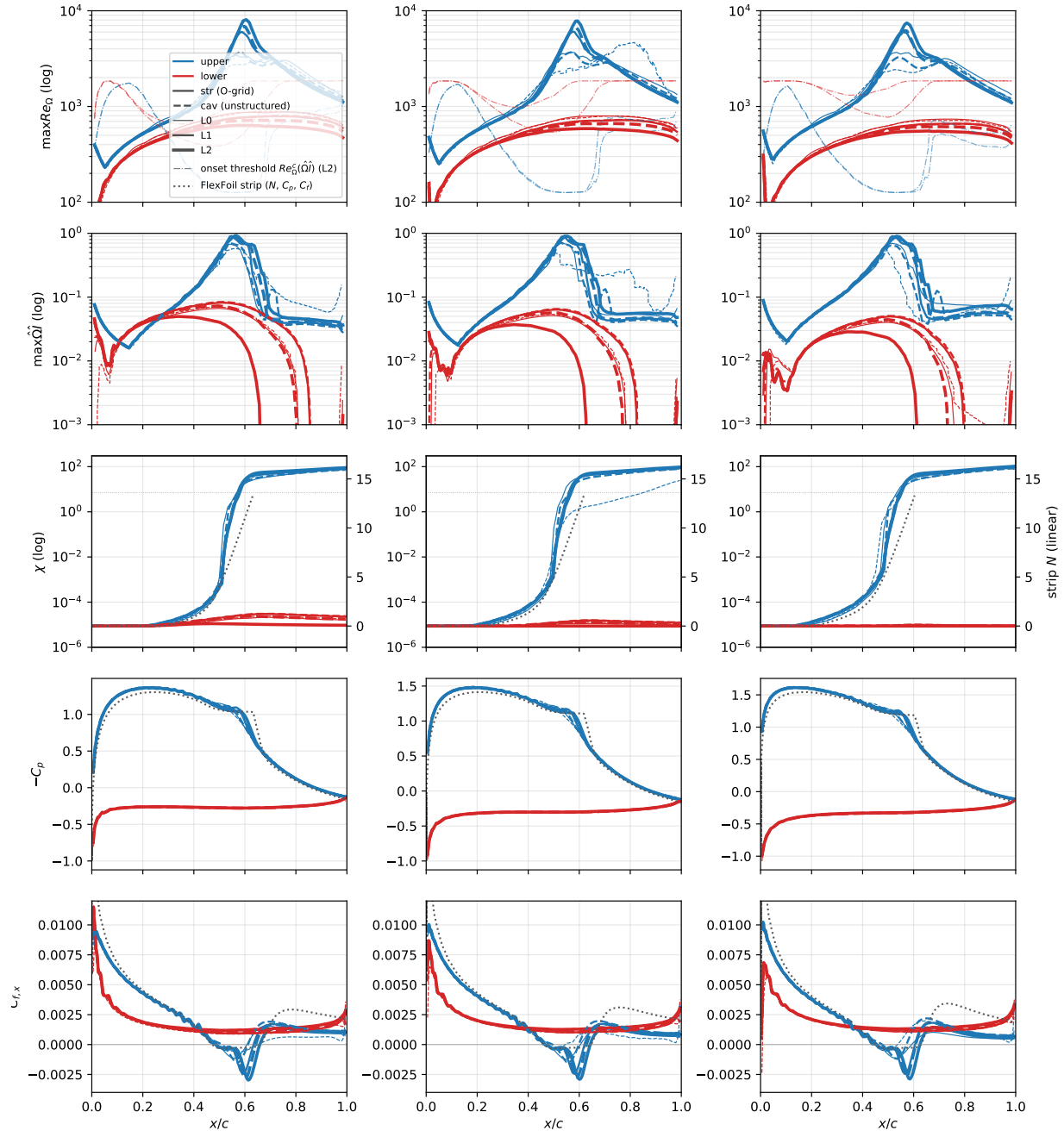


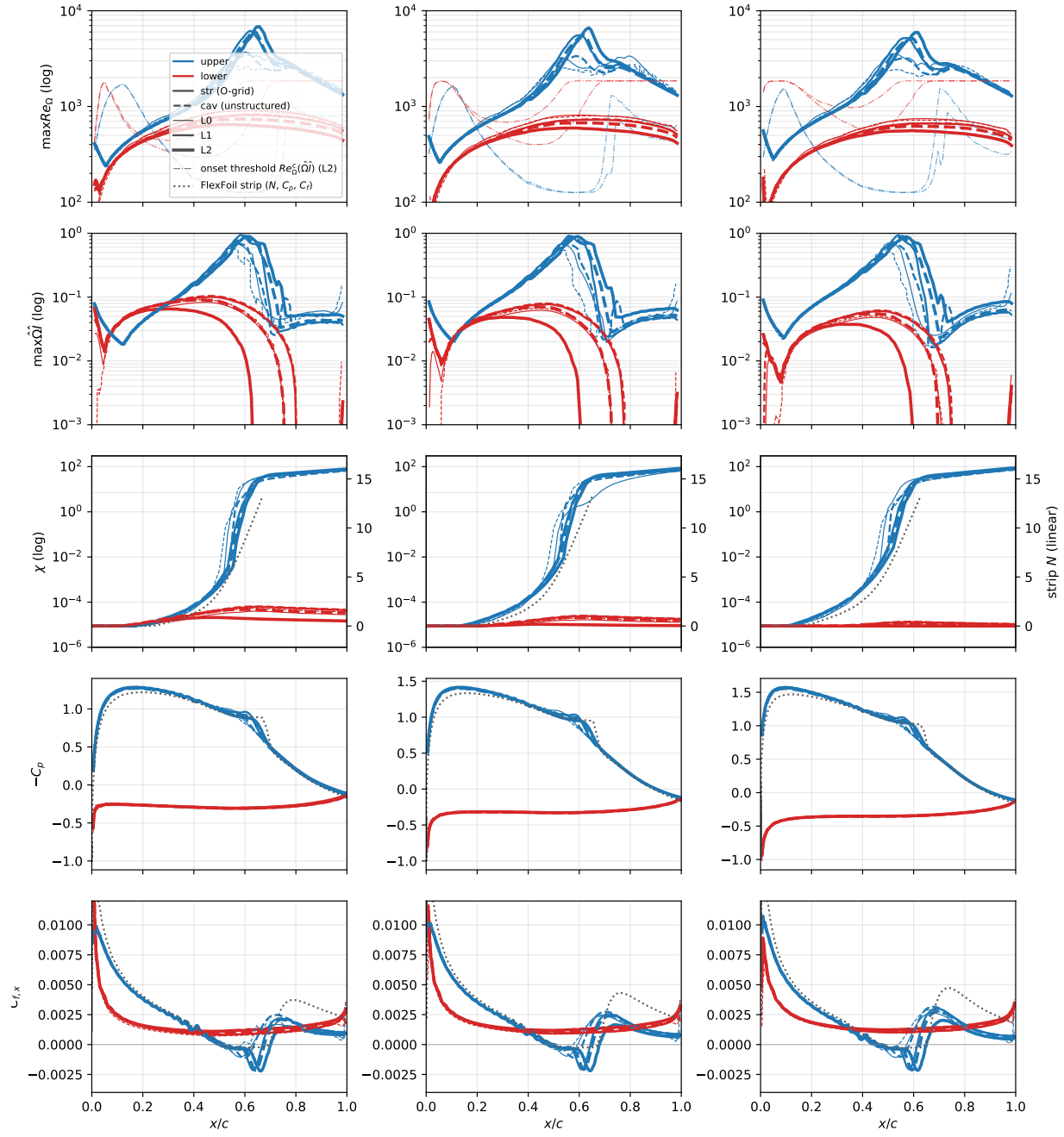
Table 10: Daedalus wing totals by grid level: SA-AI on both mesh families against the AVL+XFOIL reference trimmed to the RANS lift at each incidence.

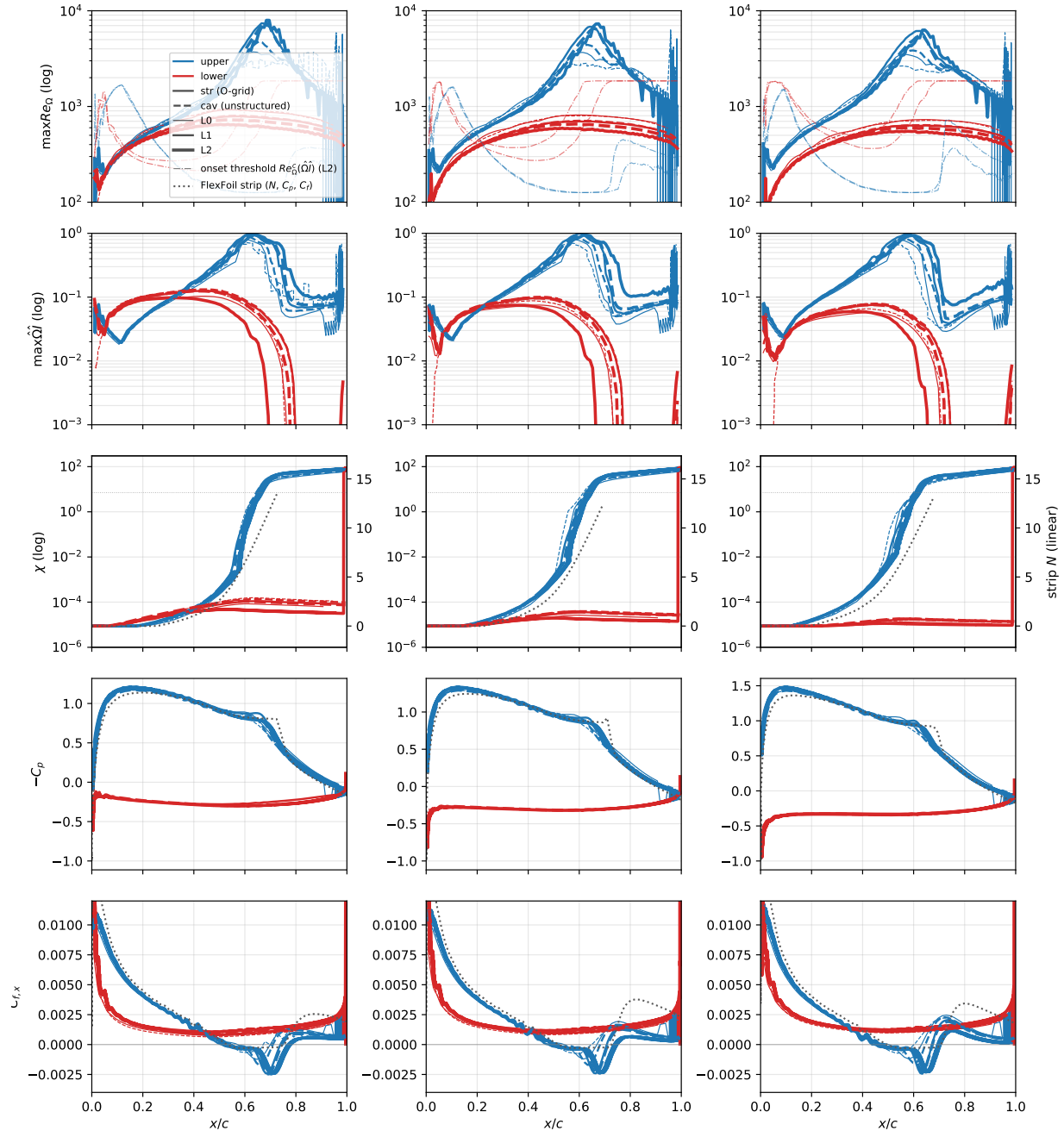
| $\alpha$ | level     | structured O-grid |         | unstructured |         |
|----------|-----------|-------------------|---------|--------------|---------|
|          |           | $C_L$             | $C_D$   | $C_L$        | $C_D$   |
| 4°       | L0        | 1.0244            | 0.01958 | 1.0156       | 0.02134 |
|          | L1        | 1.0279            | 0.01946 | 1.0232       | 0.01941 |
|          | L2        | 1.0284            | 0.01969 | 1.0219       | 0.01976 |
|          | AVL+XFOIL | 1.0284            | 0.02125 |              |         |
| 5°       | L0        | 1.1254            | 0.02219 | 1.1145       | 0.02427 |
|          | L1        | 1.1307            | 0.02197 | 1.1248       | 0.02201 |
|          | L2        | 1.1290            | 0.02212 | 1.1236       | 0.02223 |
|          | AVL+XFOIL | 1.1290            | 0.02315 |              |         |
| 6°       | L0        | 1.2262            | 0.02505 | 1.2171       | 0.02713 |
|          | L1        | 1.2321            | 0.02465 | 1.2256       | 0.02481 |
|          | L2        | 1.2282            | 0.02474 | 1.2238       | 0.02491 |
|          | AVL+XFOIL | 1.2282            | 0.02530 |              |         |











## A Proposed two-branch kernel (for discussion)

Rate  $a = \text{clip}(\text{softmax}_2(a_{\text{inv}}\langle\hat{\Omega}\hat{I}\rangle_+, a_{\text{visc}}\langle\hat{\Omega}\hat{Z}\rangle_+), 0, a_{\text{inv}})$ ,  $\text{softmax}_2(x, y) = \sqrt{x^2 + y^2}$ . Gate  $Re_{\Omega}^c = \text{softmin}(A + B/P_I^2, A + B_c/P_{\text{curv}}^2)$ ,  $P_I = \langle\hat{\Omega}\hat{I}\rangle_+$ ,  $P_{\text{curv}} = \langle\hat{\Omega}\hat{Z}\rangle_+$ ,  $\hat{Z} = -Z/R$ ,  $\langle x \rangle_+ = \max(x, 0)$ . Constants:  $a_{\text{inv}} = 0.19$ ,  $a_{\text{visc}} = 0.0276$ ,  $A = 124.6$ ,  $B = 1.424$ ,  $B_c = 130$ . (env-gated *AI\_A\_VISC*, *AI\_REOMC\_BC*; default 0 = canon.)

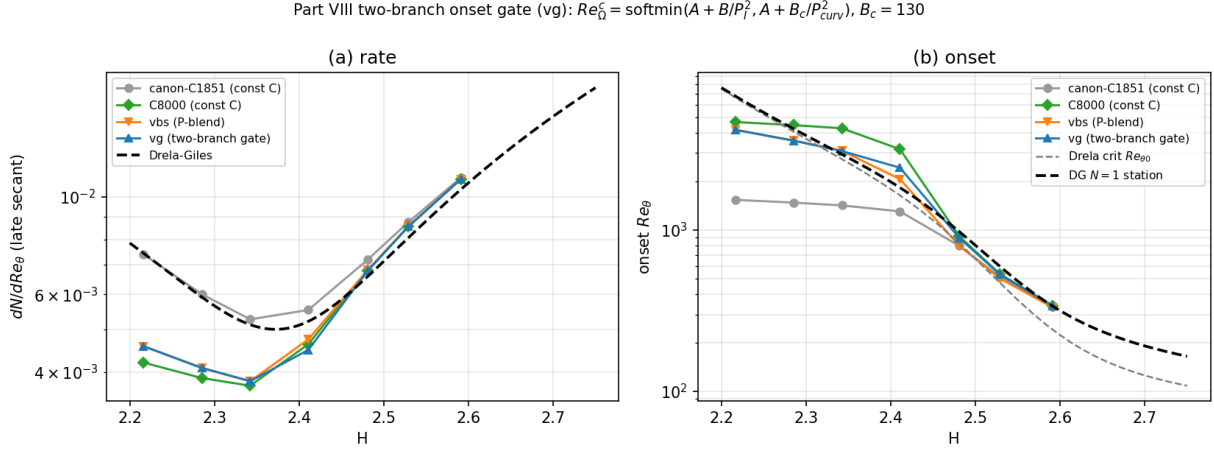


Figure 25: Curves: canon ( $C=1851$ ),  $C=8000$ , vbs, vg. vg both-panel max  $1.63 \times$  DG; (a) rate; (b) onset on the DG  $N=1$  station to  $H=2.2$ ; low- $H$  rate  $0.56 \times$  DG.

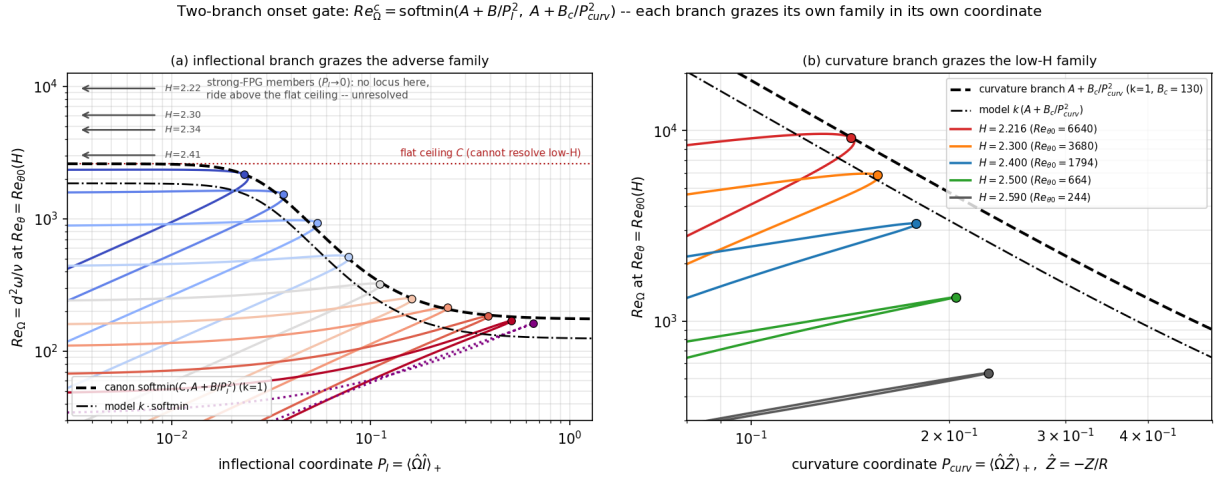


Figure 26: Left  $P_I$  (inflectional/adverse family); right  $P_{\text{curv}}$  (low- $H$  family). Curvature-branch graze ratio by  $H$ :  $2.216 \rightarrow 1.00$ ,  $2.30 \rightarrow 0.76$ ,  $2.40 \rightarrow 0.55$ ,  $2.50 \rightarrow 0.29$ ,  $2.59 \rightarrow 0.15$ . Drela  $Re_{\theta 0}$ : 6640/3680/1794/664/244.

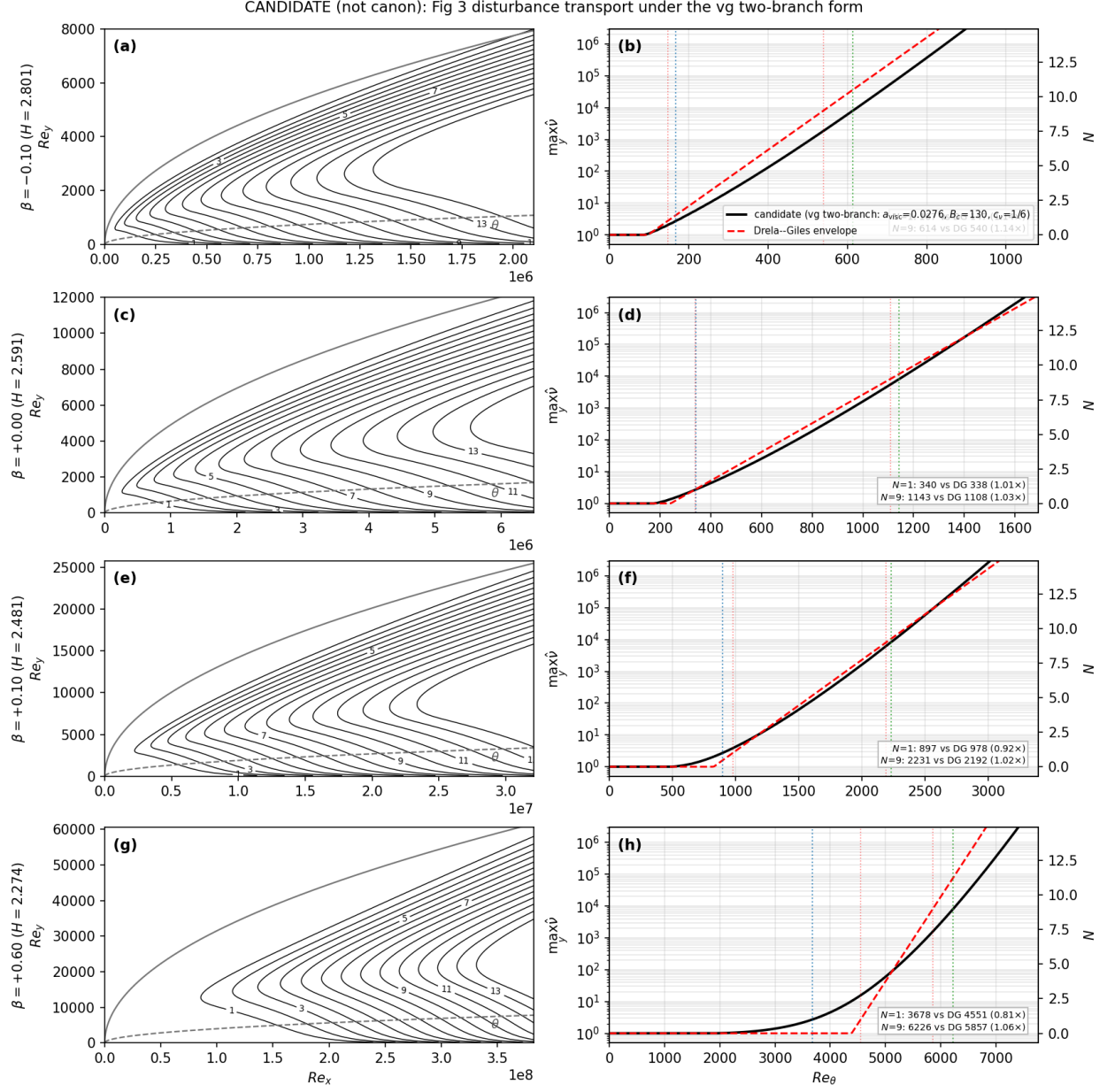


Figure 27: Rows  $(\beta, H, \max \hat{\Omega} \hat{I})$ :  $(-0.10, 2.80, 0.162)$ ,  $(0, 2.59, 0.078)$ ,  $(+0.10, 2.48, 0.037)$ ,  $(+0.60, 2.27, 6.5 \times 10^{-4})$ .  $N=1/N=9$  vs DG: 1.14/1.14; 1.01/1.03; 0.92/1.02; 0.81/1.06.



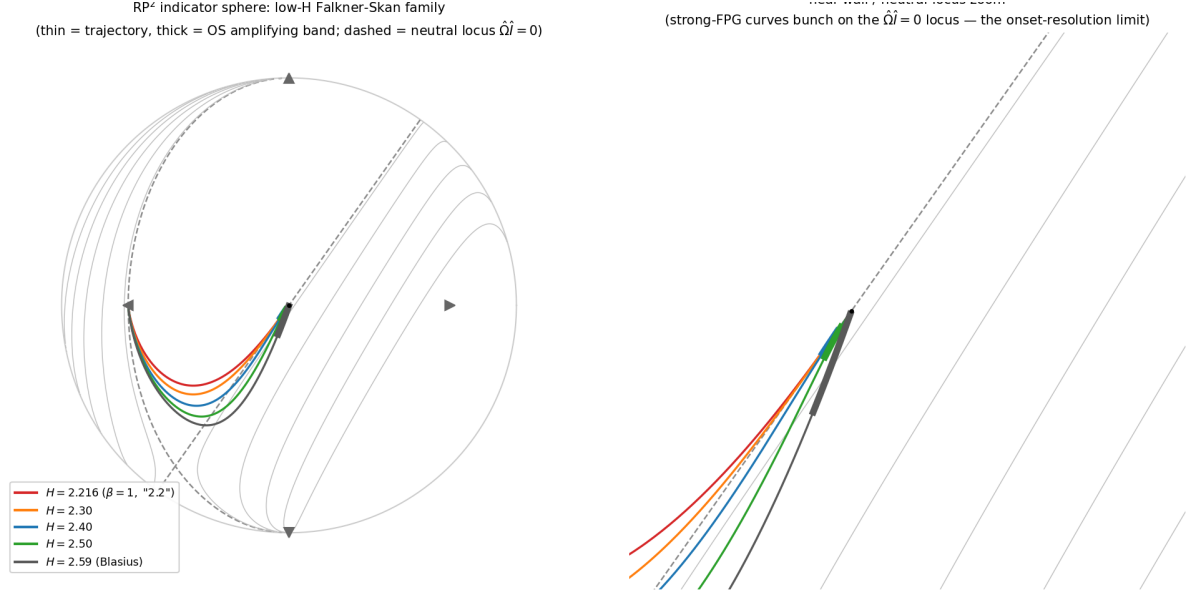


Figure 28: Low- $H$  Falkner-Skan on  $\mathbb{RP}^2$ .  $H = 2.216$  and  $2.30$ :  $\max \hat{\Omega}\hat{I} = 1\text{--}3 \times 10^{-4}$  ( $\Delta = 1.6 \times 10^{-4}$ ), Drela  $Re_{\theta 0}$  6640 vs 3680 ( $1.81\times$ ).

In-solver (vg on, env-gated; canon untouched):  $Re_D = 2 \times 10^6\text{--}2 \times 10^7$  vg vs canon  $\Delta\theta_{tr} \leq 2.5^\circ$ ,  $\Delta C_d \leq 6\%$ ;  $C_d(10^9) = 0.209$  (canon 0.206),  $C_d(10^{10}) = 0.199$  (canon 0.189); flat-plate  $Re_{x,tr}$   $-7.5\%$  ( $Tu$  0.16%) /  $-9\%$  ( $Tu$  0.08%); Eppler-387  $Re = 2 \times 10^5$ ,  $\alpha = 2^\circ$  rooftop unchanged.

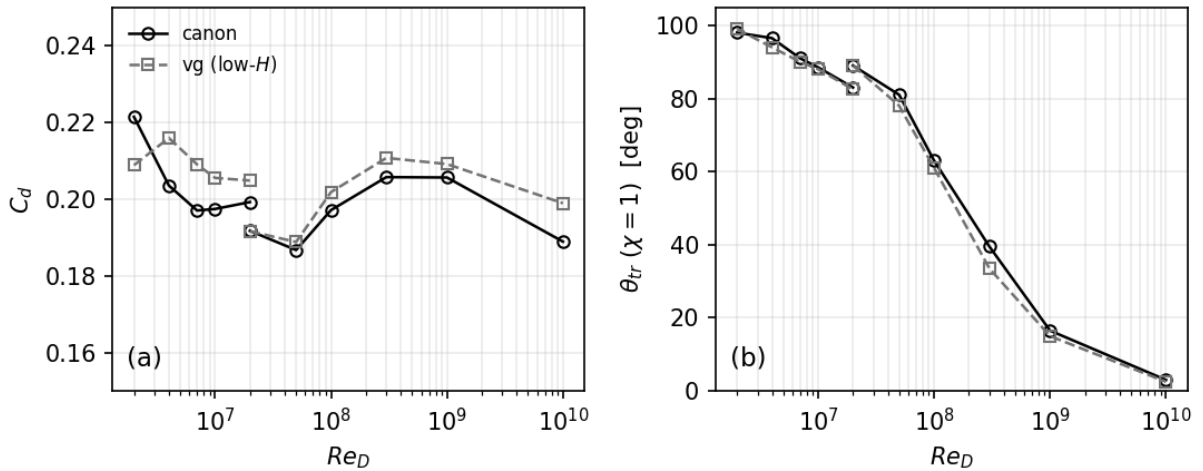


Figure 29: vg (low- $H$ ) vs canon,  $Tu = 0.2\%$  up-ladder,  $Re_D = 2 \times 10^6\text{--}10^{10}$  (highre mesh  $\leq 2 \times 10^7$ , ultra  $\geq 2 \times 10^7$ ). (a)  $C_d$ : both 0.19–0.22, vg +3–6%; neither reaches the transcritical 0.5–0.7. (b)  $\theta_{tr}$  ( $\chi = 1$ ): front advances  $98^\circ \rightarrow 2.5^\circ$ ; vg tracks canon within  $6^\circ$ .